Results — revised submission — original style

1 Datasets

#	Name	N	P	K	MeanIR	CVIR
1	GpositivePseAAC	519	440	4	3.86	1.15
2	CHD_49	555	49	6	5.77	1.60
3	Emotions	593	72	6	1.48	0.16
4	Scene	2407	294	6	1.25	0.11
5	PlantPseAAC	978	440	12	6.69	0.68
6	Water-quality	1060	16	14	1.77	0.29
7	Yeast	2417	103	14	7.20	1.82
8	HumanPseAAC	3106	440	14	15.29	1.05
9	enron	1702	1001	53	10.45	0.80

Table 1: (Table 3) Datasets used in the revised experiments. N, P, K are the number of instances, features, labels. MeanIR and CVIR report per-label imbalance (higher = more imbalanced).

Pending follow-ups (to fill before final submission)

- **LGBM run on `enron`.** Sequentially training (2–3h per task due to K=53 pairwise classifiers); will be merged into `full_enron_split_lgbm_summary/` when complete and the LGBM aggregate panels regenerated.
- **Extended PartialAbstention metrics.** The evaluator has been extended with 11 new PA metrics (`jaccard_pa`, `{example,macro,micro}_{precision,recall,f1}_pa`, `mfrd_pa`, `afrd_pa`). Existing runs were summarised with the previous metric set; the new columns will populate automatically on the next end-to-end training+evaluation pass.

2 Method encoding

For every panel that follows, the x-axis encodes the classifier as one of 12 methods, grouped into three blocks:

Block A — Order-based predictors (ours, indices 1–8). Each name has the form <order>-<target>-<heig
order $\in$ {PA = partial order, PR = pre-order}; *target* $\in$ {H = Hamming, S = Subset 0/1}; *height* $\in$ {2, ∞ (omitted in the label)}.

- 1 = **PA-H-2**: partial order, Hamming target, height ≤ 2.

- $2 =$ **PA-H**: partial order, Hamming target, unrestricted height.
- $3 =$ **PA-S-2**: partial order, Subset 0/1 target, height ≤ 2.
- $4 =$ **PA-S**: partial order, Subset 0/1, unrestricted.
- $5 =$ **PR-H-2**: pre-order, Hamming, height ≤ 2.
- $6 =$ **PR-H**: pre-order, Hamming, unrestricted.
- $7 =$ **PR-S-2**: pre-order, Subset 0/1, height ≤ 2.
- $8 =$ **PR-S**: pre-order, Subset 0/1, unrestricted.

Block B — Standard MLC baselines from the original paper (indices 9–11).

- $9 =$ **BR** (Binary Relevance): K independent binary classifiers, one per label.
- $10 =$ **CC** (Classifier Chain): sequential chain that uses previous labels as features.
- $11 =$ **CLR** (Calibrated Label Ranking): pairwise label ranking with a calibration label that separates relevant from irrelevant.

Block C — Extended baseline added for the revision (index 12).

- $12 =$ **ECC** (Ensemble of Classifier Chains): averages over multiple CCs with random chain orders.

Reading the panels. Each panel plots one dotted line with markers per noise level, colored as in the original paper's Table 5: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$, $\alpha = 0.3$. *PartialAbstention* and *ScoreVector* panels show only indices 1–8 (the remaining baselines do not export a partial-abstention prediction nor a probabilistic score vector). Vertical dashed lines at $x = 4.5$, $x = 8.5$, and $x = 11.5$ separate the four method groups: PA (1–4), PR (5–8), standard baselines BR/CC/CLR (9–11), and ECC (12). Missing methods show as gaps in the line (with a small grey "x" on the axis when the method has no data at all for that panel).

3 Paper results (RF base learner)

Reproduces the layout of the original paper's main experiment tables (Tables 5–7): 3 datasets across, 4 metrics down, one table per prediction type (binary vector, partial abstention, score vector). Per the revision, $f^{\text{ham}}_{\text{MLC}}$ and $f^{\text{sub}}_{\text{MLC}}$ are dropped in favour of $f^{\text{Jacc}}_{\text{MLC}}$.

#	Name	N	P	K	MeanIR	CVIR
1	GpositivePseAAC	519	440	4	3.86	1.15
2	CHD_49	555	49	6	5.77	1.60
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8	HumanPseAAC	3106	440	14	15.29	1.05
9	enron	1702	1001	53	10.45	0.80

Table 2: (Table 3) Datasets used in the revised experiments. N, P, K are the number of instances, features, labels. MeanIR and CVIR report per-label imbalance (higher = more imbalanced).

3.1 Standard MLC predictions (BinaryVector)

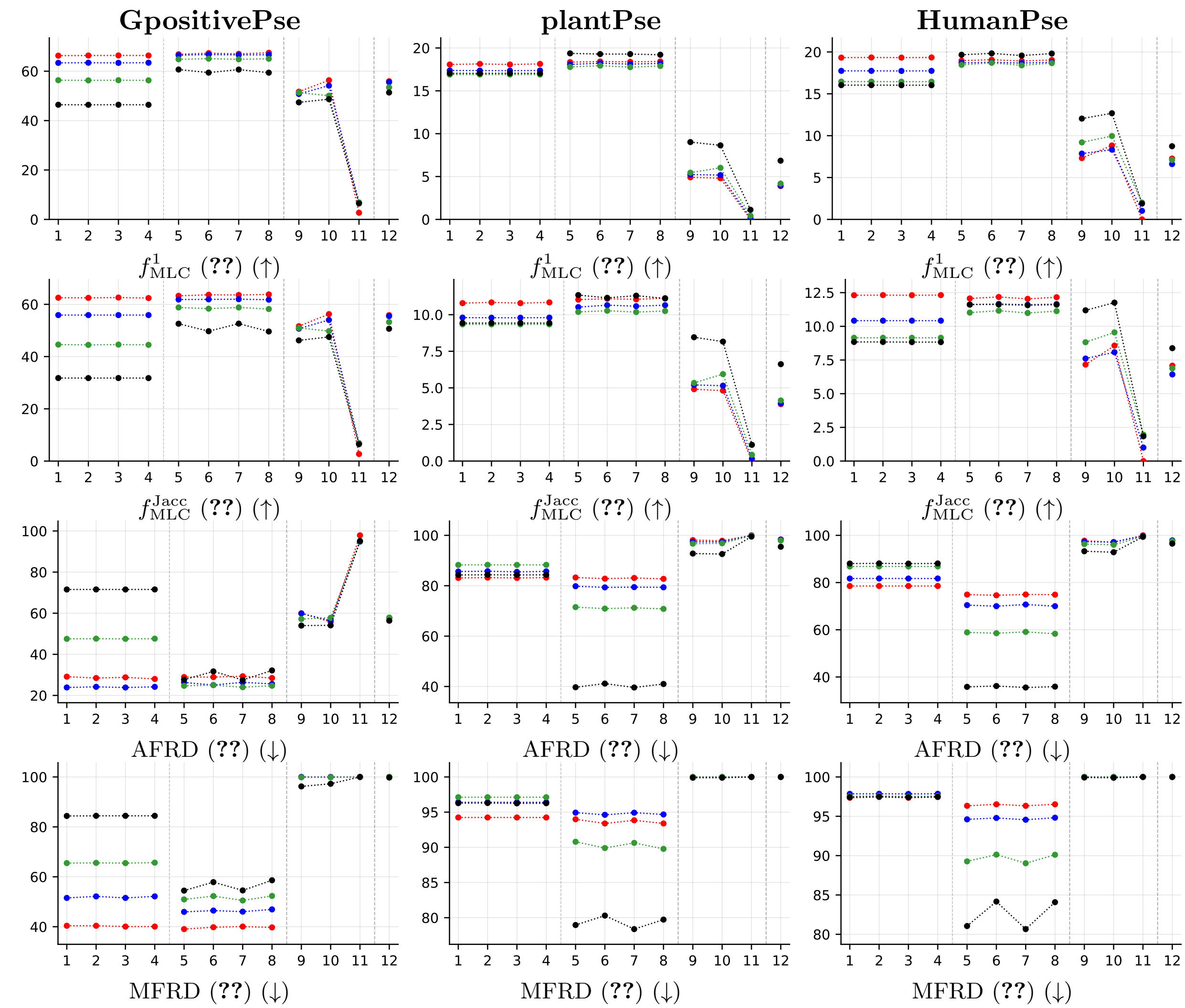


Table 3: (Table 4) Average scores (in %, y-axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x-axis): (PA-H-2, PA-H, PA-S-2, PA-S), (PR-H-2, PR-H, PR-S-2, PR-S), and (BR, CC, CLR) are encoded as (1, 2, 3, 4), (5, 6, 7, 8), and (9, 10, 11), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

3.2 Partial abstention (PartialAbstention)

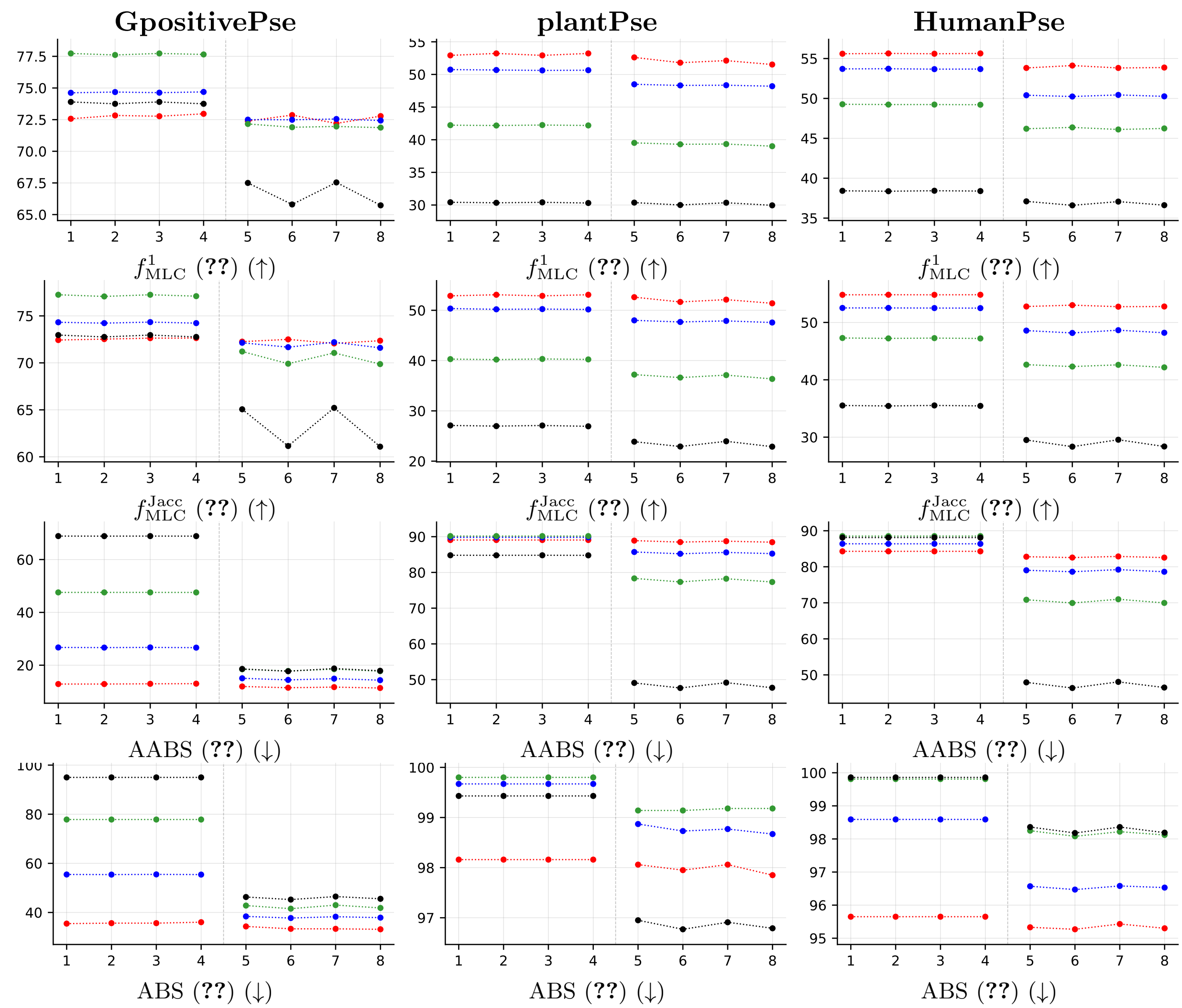


Table 4: (Table 5) Average scores (in %, y-axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x-axis): (PA-H-2, PA-H, PA-S-2, PA-S) and (PR-H-2, PR-H, PR-S-2, PR-S) are encoded as (1, 2, 3, 4) and (5, 6, 7, 8), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

3.3 Experimental results — appendix tables (RF base learner)

Appendix counterpart of the main Paper results section. Tables G2–G5 mirror the original paper's appendix exactly (same metric sets, with the 6 non-main datasets split into imbalanced vs. balanced buckets following the original layout). Tables G6–G7 are revision additions reporting the new metrics ($f^{\mathrm{Jacc}}_{\mathrm{MLC}}$, macro-$f_1$, micro-$f_1$).

3.3.1 Standard MLC predictions (BinaryVector)

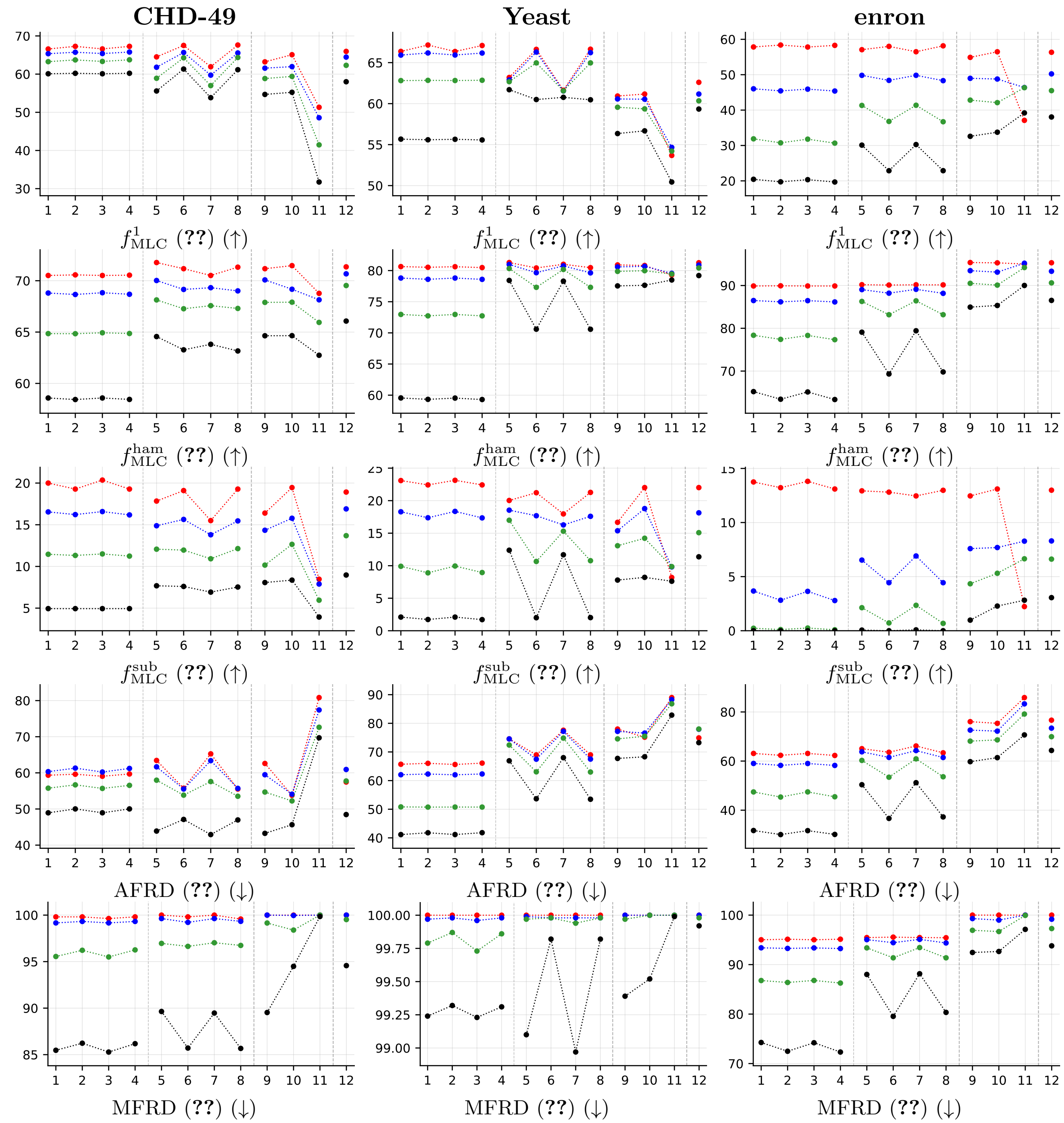


Table 5: (Table G2) Average scores (in %, y-axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x-axis): (PA-H-2, PA-H, PA-S-2, PA-S), (PR-H-2, PR-H, PR-S-2, PR-S), and (BR, CC, CLR) are encoded as (1, 2, 3, 4), (5, 6, 7, 8), and (9, 10, 11), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

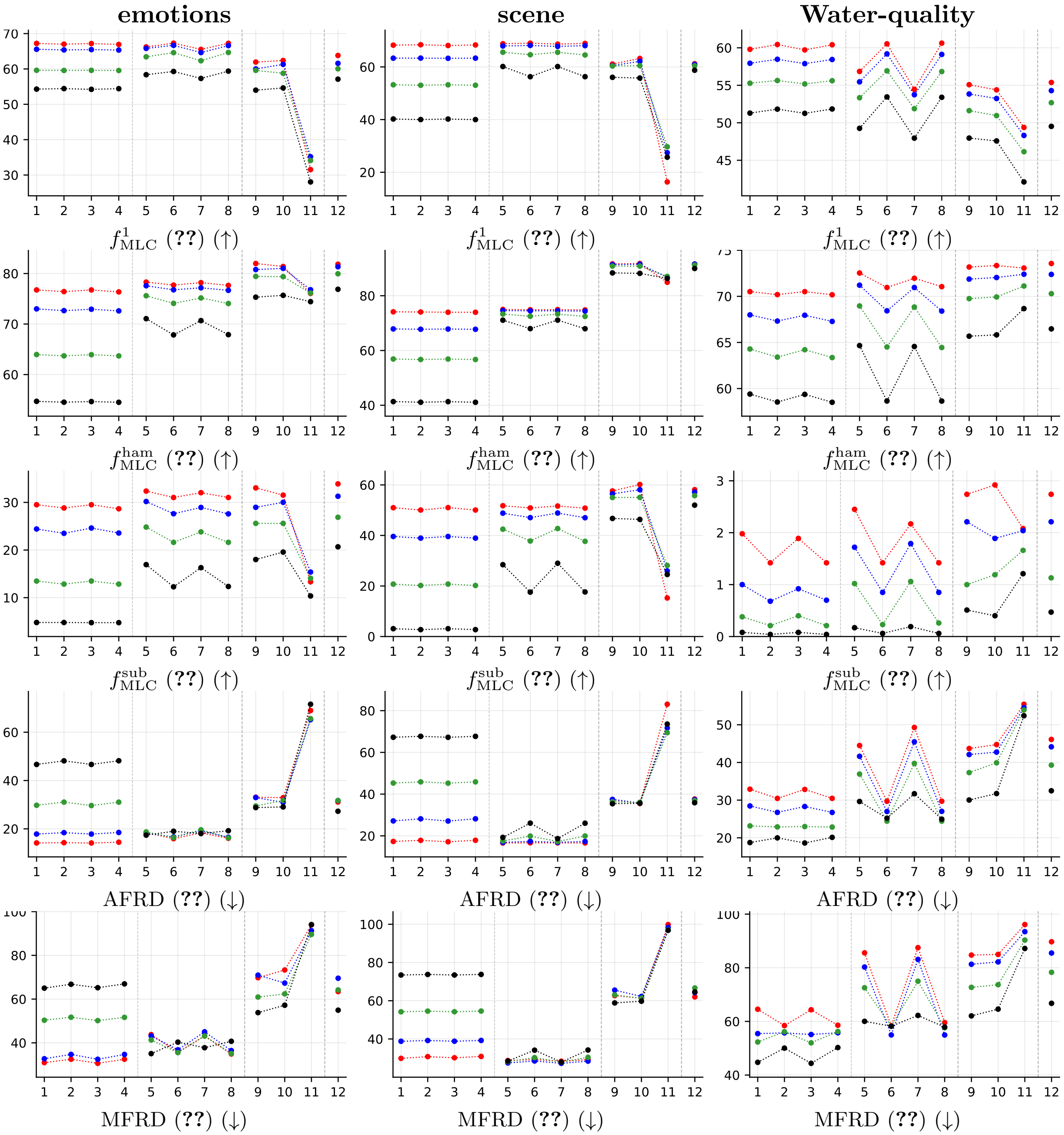


Table 6: (Table G3) Average scores (in %, y-axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x-axis): (PA-H-2, PA-H, PA-S-2, PA-S), (PR-H-2, PR-H, PR-S-2, PR-S), and (BR, CC, CLR) are encoded as (1, 2, 3, 4), (5, 6, 7, 8), and (9, 10, 11), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

3.3.2 Partial abstention (PartialAbstention)

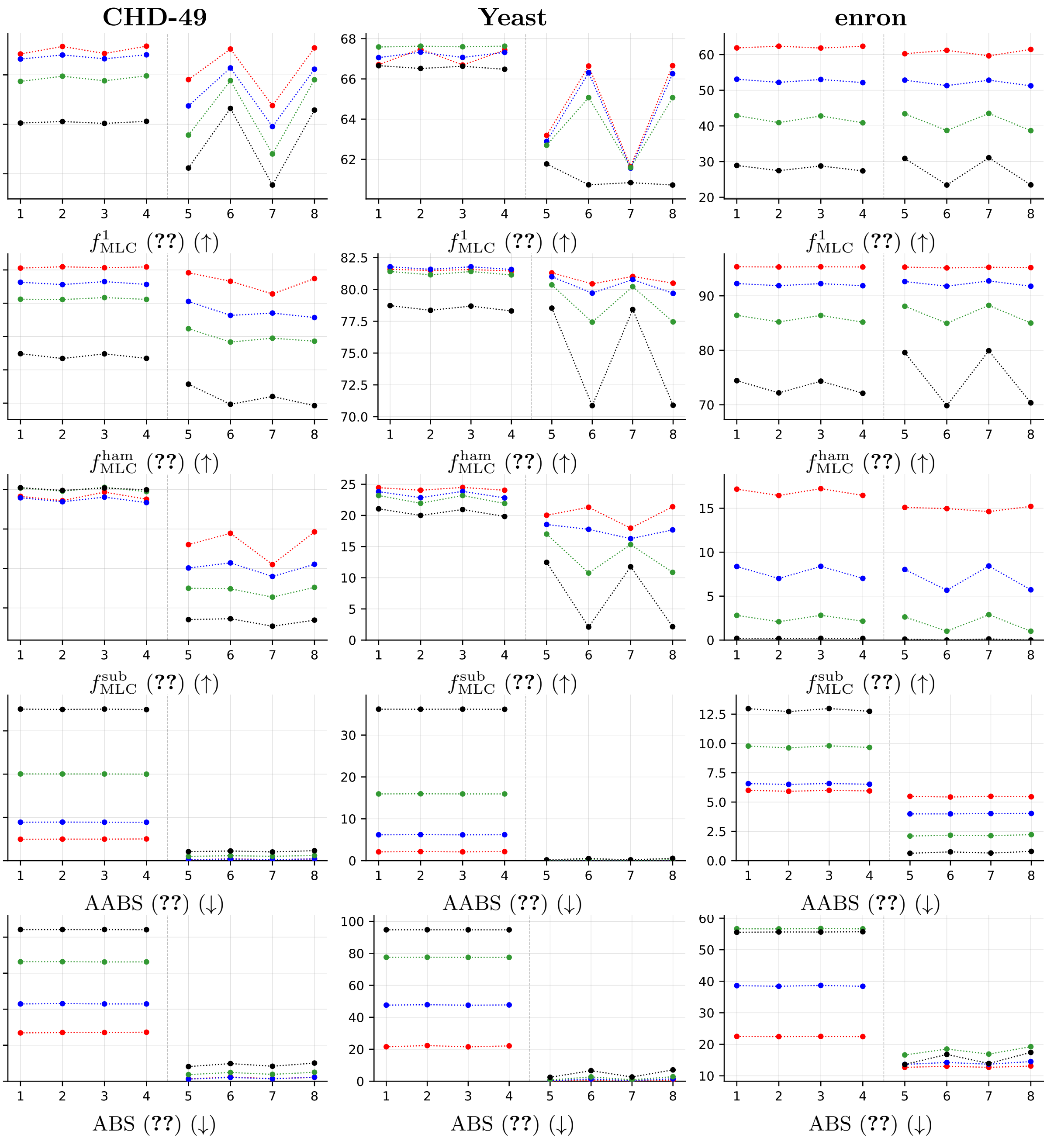


Table 7: (Table G4) Average scores (in %, y-axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x-axis): (PA-H-2, PA-H, PA-S-2, PA-S) and (PR-H-2, PR-H, PR-S-2, PR-S) are encoded as (1, 2, 3, 4) and (5, 6, 7, 8), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

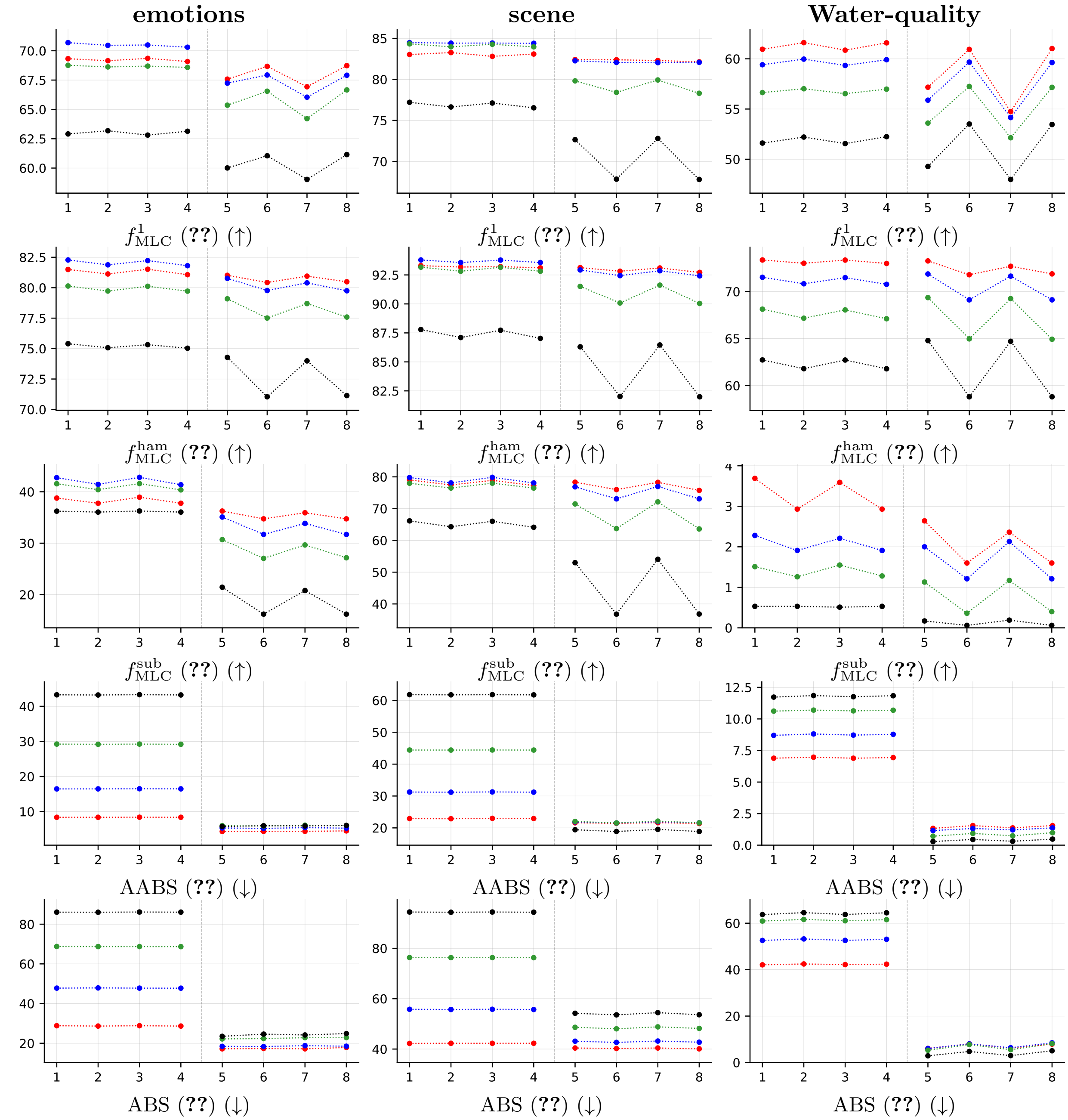


Table 8: (Table G5) Average scores (in %, y-axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x-axis): (PA-H-2, PA-H, PA-S-2, PA-S) and (PR-H-2, PR-H, PR-S-2, PR-S) are encoded as (1, 2, 3, 4) and (5, 6, 7, 8), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

3.3.3 Standard MLC predictions (BinaryVector)

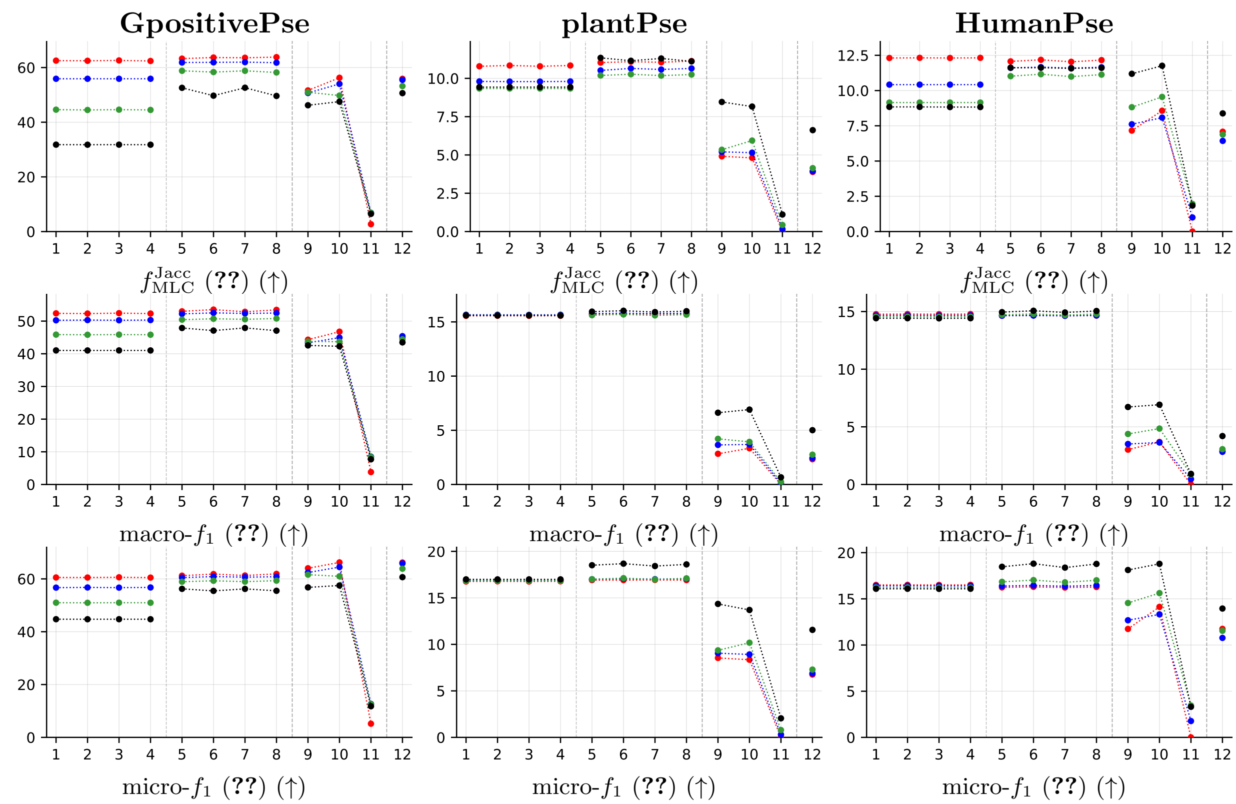


Table 9: (Table G6, new metrics — main datasets) Average scores (in %, y-axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x-axis): (PA-H-2, PA-H, PA-S-2, PA-S), (PR-H-2, PR-H, PR-S-2, PR-S), and (BR, CC, CLR) are encoded as (1, 2, 3, 4), (5, 6, 7, 8), and (9, 10, 11), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

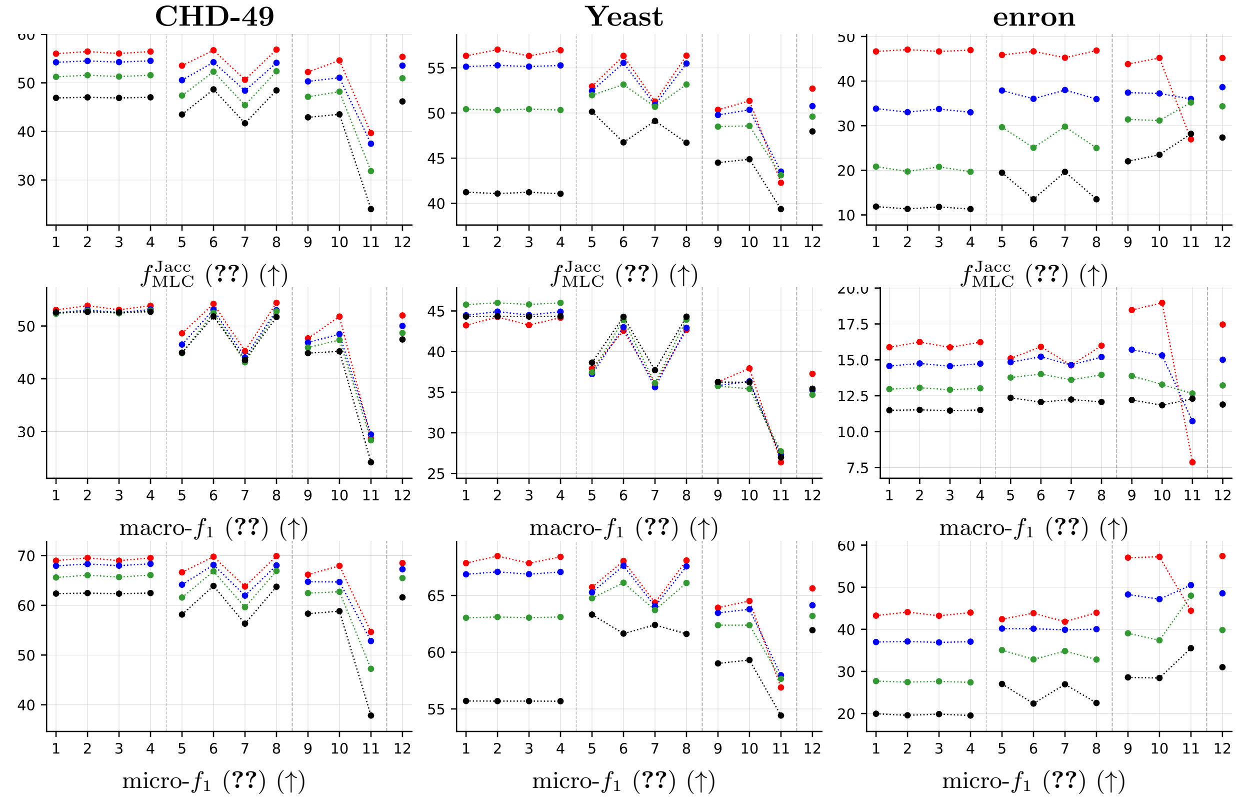


Table 10: (Table G6, new metrics — imbalanced) Average scores (in %, y-axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x-axis): (PA-H-2, PA-H, PA-S-2, PA-S), (PR-H-2, PR-H, PR-S-2, PR-S), and (BR, CC, CLR) are encoded as (1, 2, 3, 4), (5, 6, 7, 8), and (9, 10, 11), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

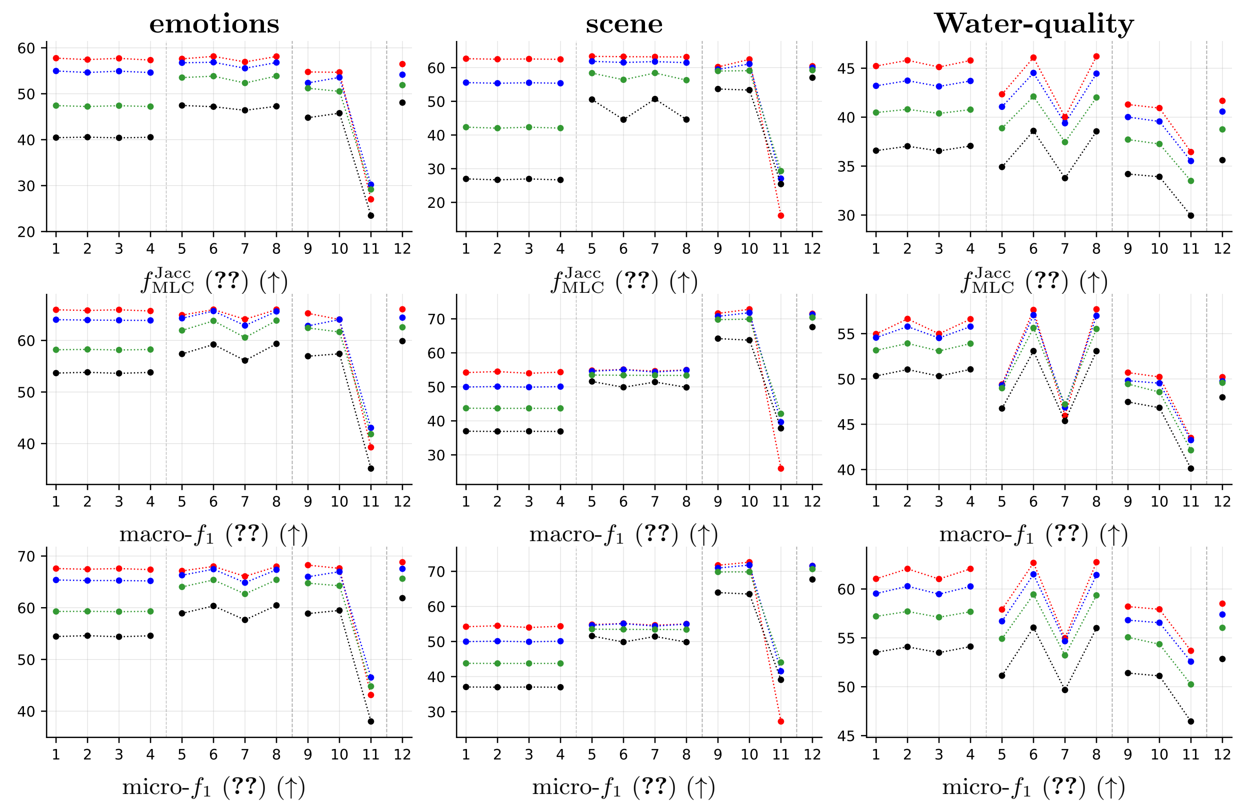


Table 11: (Table G6, new metrics — balanced) Average scores (in %, y-axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x-axis): (PA-H-2, PA-H, PA-S-2, PA-S), (PR-H-2, PR-H, PR-S-2, PR-S), and (BR, CC, CLR) are encoded as (1, 2, 3, 4), (5, 6, 7, 8), and (9, 10, 11), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

3.3.4 Partial abstention (PartialAbstention)

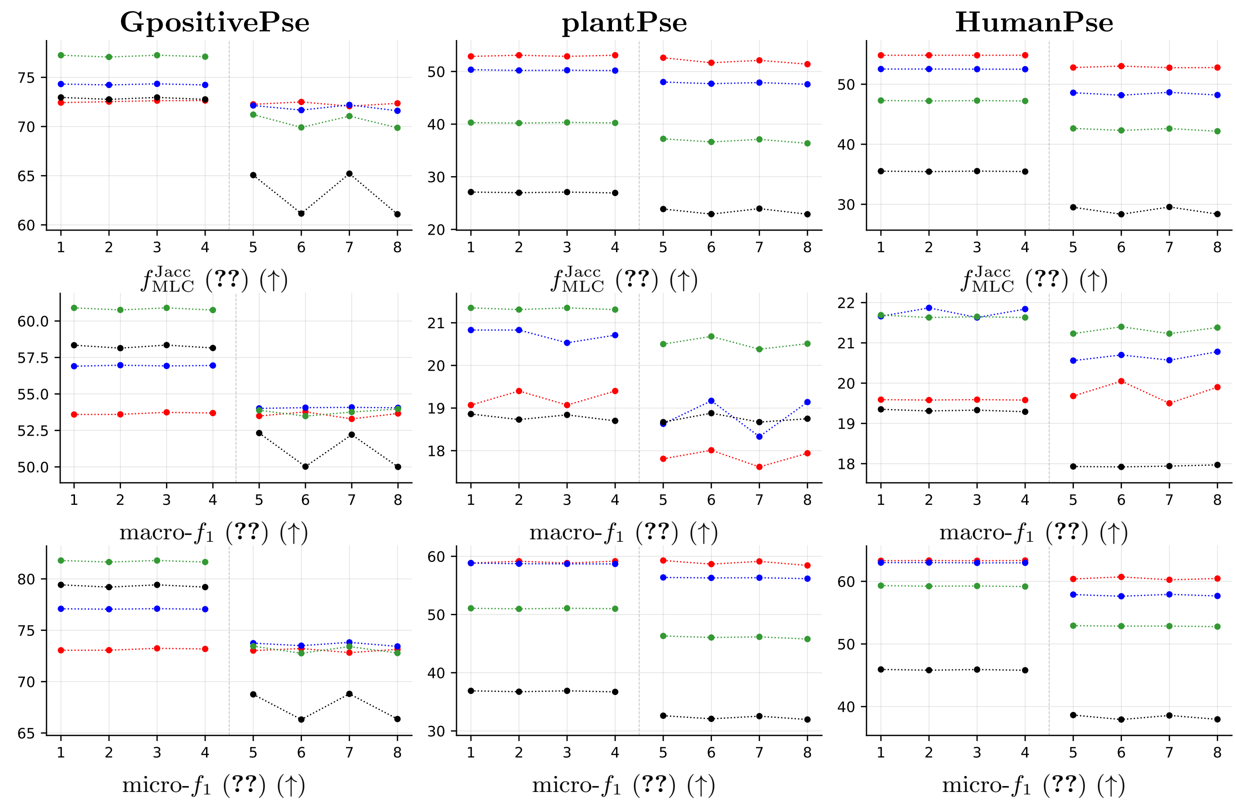


Table 12: (Table G7, new metrics — main datasets) Average scores (in %, y-axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x-axis): (PA-H-2, PA-H, PA-S-2, PA-S) and (PR-H-2, PR-H, PR-S-2, PR-S) are encoded as (1, 2, 3, 4) and (5, 6, 7, 8), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

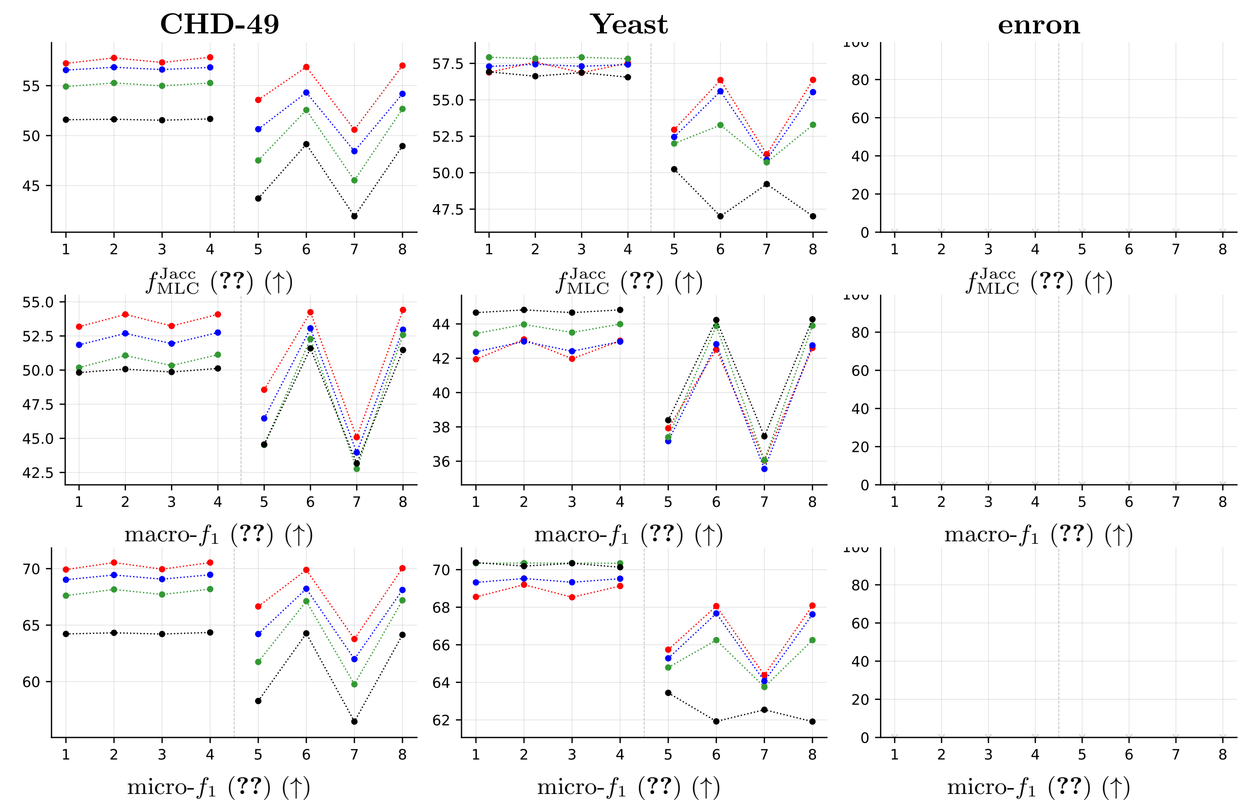


Table 13: (Table G7, new metrics — imbalanced) Average scores (in %, y-axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x-axis): (PA-H-2, PA-H, PA-S-2, PA-S) and (PR-H-2, PR-H, PR-S-2, PR-S) are encoded as (1, 2, 3, 4) and (5, 6, 7, 8), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

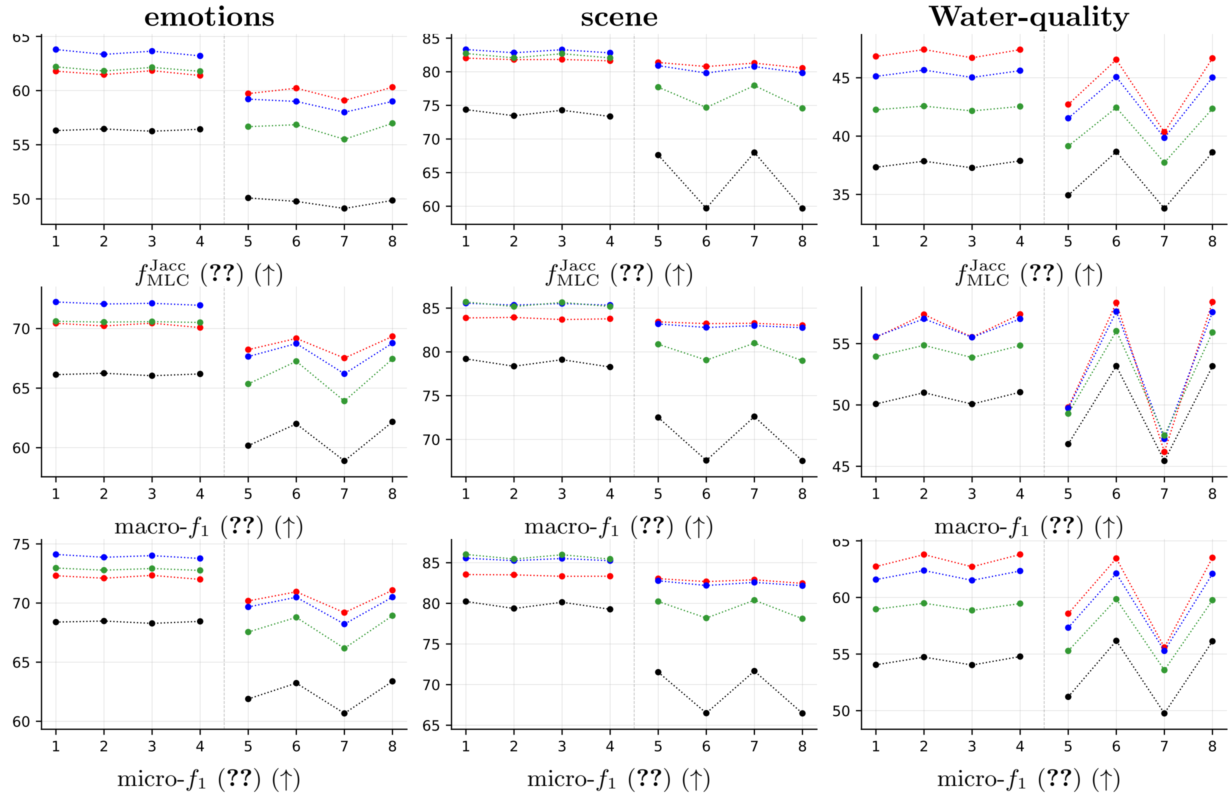


Table 14: (Table G7, new metrics — balanced) Average scores (in %, y-axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x-axis): (PA-H-2, PA-H, PA-S-2, PA-S) and (PR-H-2, PR-H, PR-S-2, PR-S) are encoded as (1, 2, 3, 4) and (5, 6, 7, 8), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

3.4 Abstention overview: aggregate trends

Purpose. This compact section shows the headline abstention-vs-standard-MLC trend across noise levels, averaged over all datasets, for both base learners (RF, LGBM) and both retained quality metrics (f_1, Jaccard). Use it as a quick visual summary before diving into the detailed per-method / per-dataset figures in Section 5.

How to read. In each panel: blue bar = standard MLC (no abstention) on its native quality metric; coloured bar = with-abstention quality on retained labels. Green "$+x.y$" label = score-point gain of abstention over the standard-MLC bar of the same method. The four sub-panels per figure correspond to $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$.

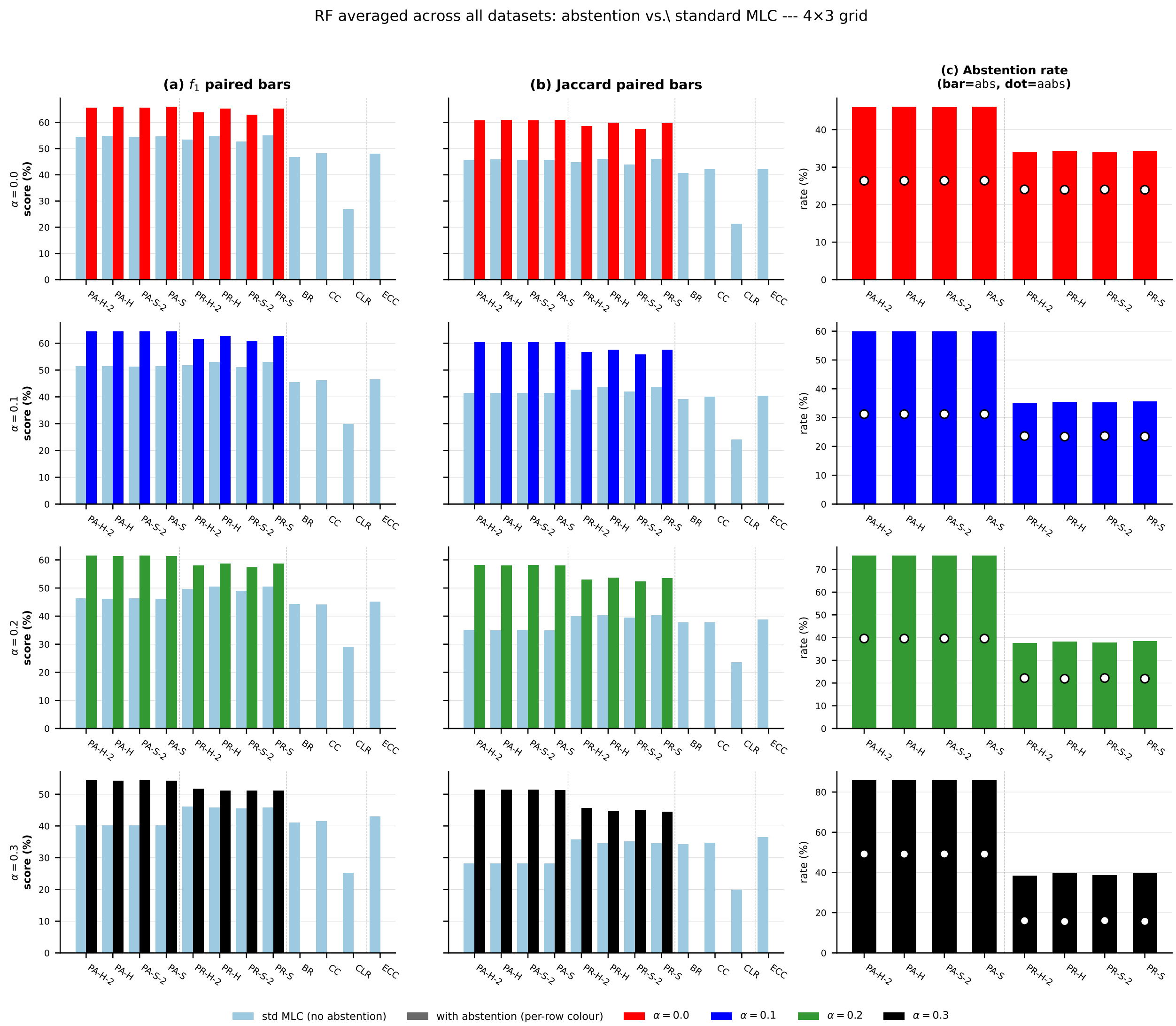


Figure 1: Figure 1: RF averaged across all datasets — table-shaped 4×3 grid: rows = $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$, columns = ((a) f_1 paired bars, (b) Jaccard paired bars, (c) abstention rate). Light blue = standard MLC, α-coloured = with-abstention; in (c), bar = `abs` (instance-level), dot = `aabs` (per-cell). Color = $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$.

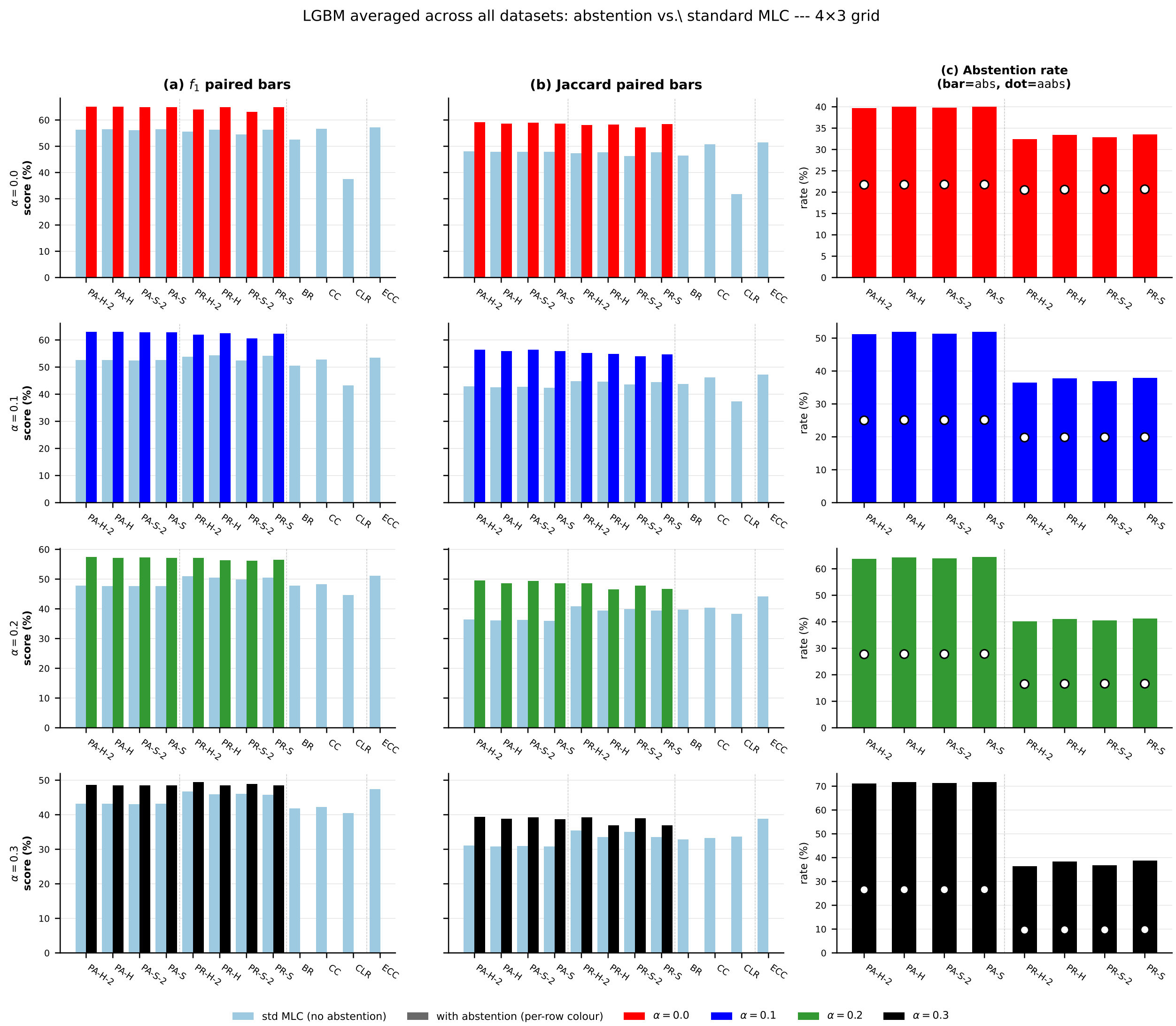


Figure 2: Figure 1: LGBM averaged across all datasets — table-shaped 4×3 grid: rows = $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$, columns = ((a) f_1 paired bars, (b) Jaccard paired bars, (c) abstention rate). Light blue = standard MLC, α-coloured = with-abstention; in (c), bar = `abs` (instance-level), dot = `aabs` (per-cell). Color = $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$.

4 Main results: RF base learner, aggregated

RF • Aggregate • BinaryVector — Standard MLC predictions

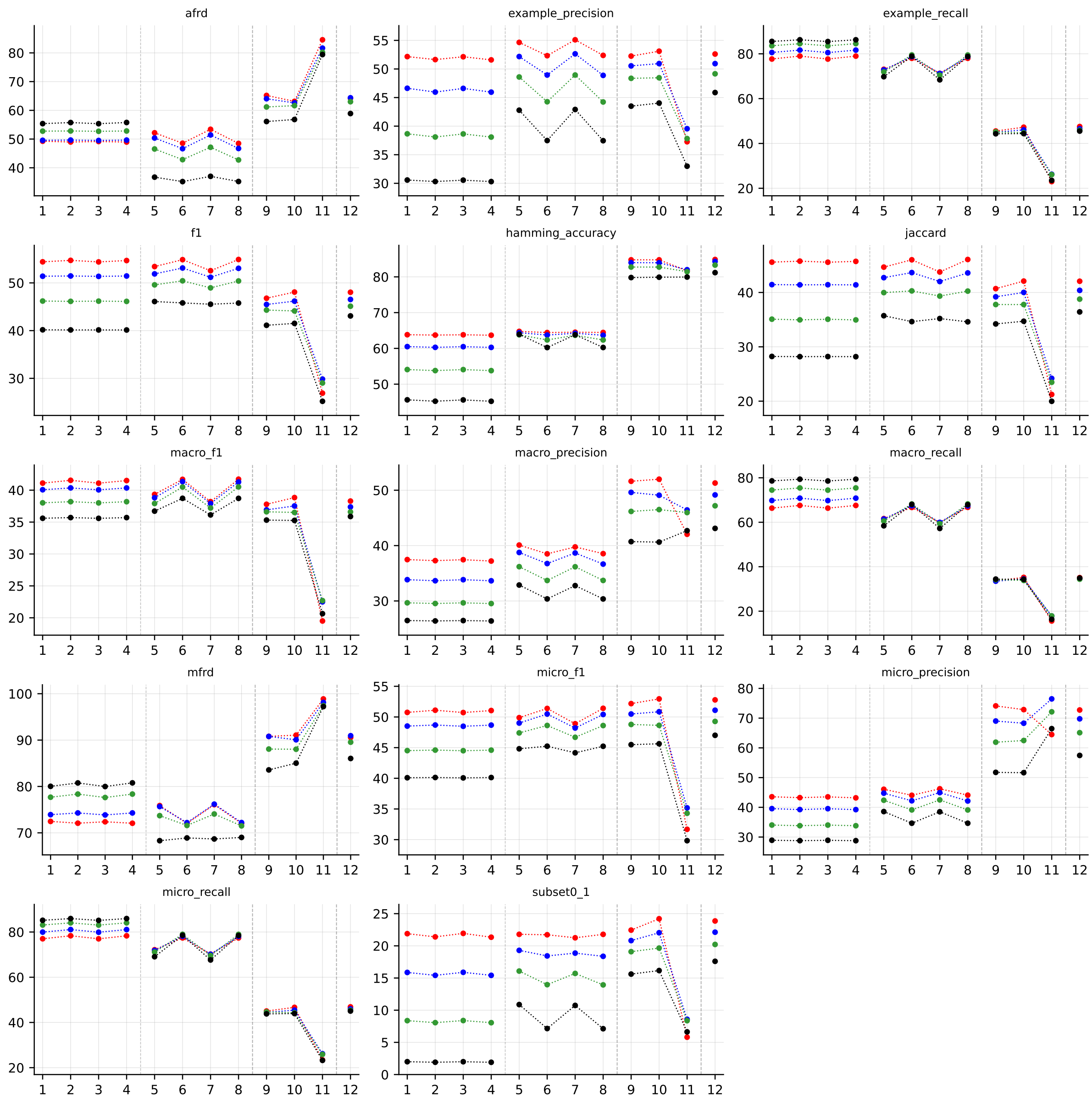


Table 15: Standard MLC predictions (BinaryVector), RF base learner, averaged across datasets. Method indices and line colors follow the encoding paragraph above.

RF • Aggregate • PartialAbstention — Partial abstention

RF • Aggregate • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

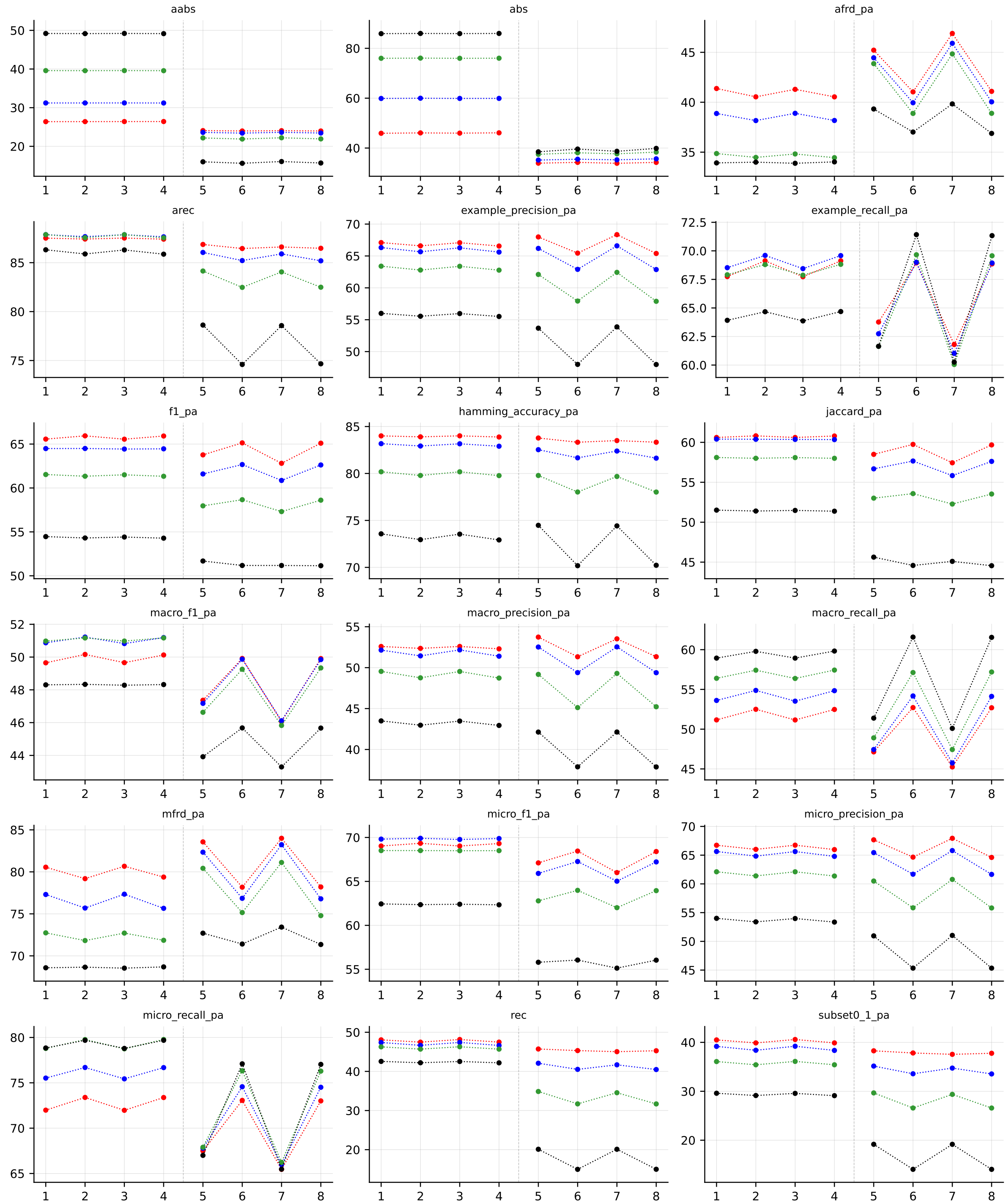


Table 16: Partial abstention (PartialAbstention), RF base learner, averaged across datasets. Method indices and line colors follow the encoding paragraph above.

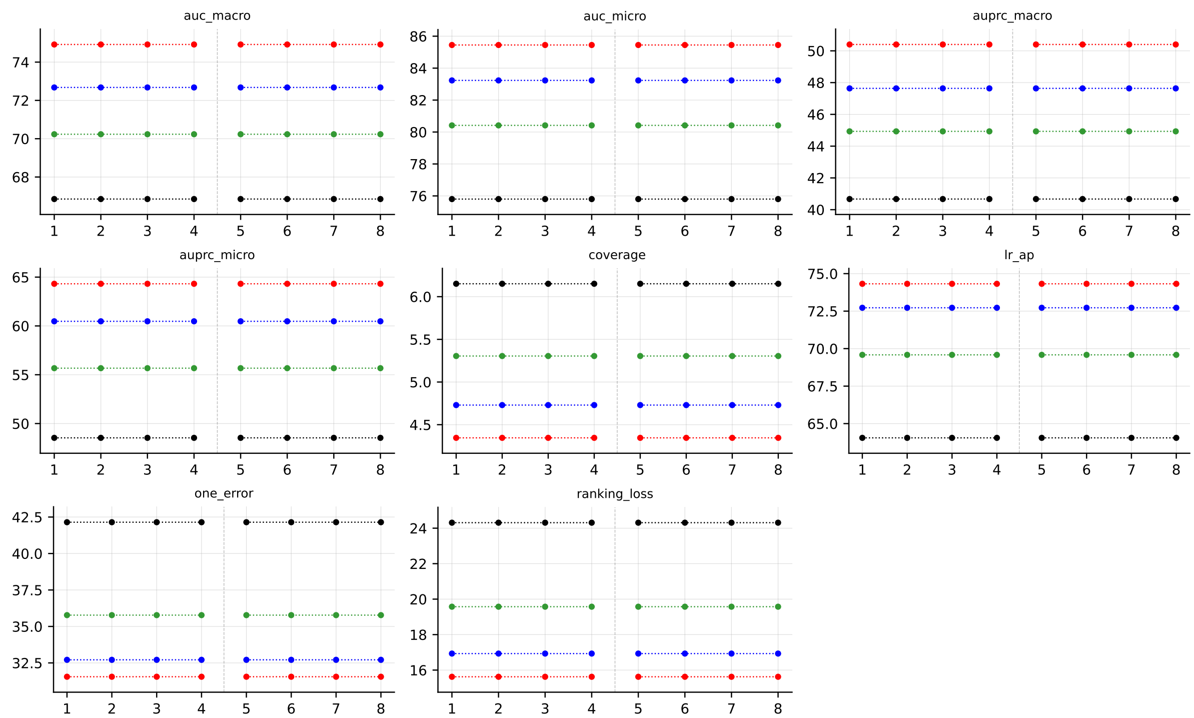


Table 17: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector), RF base learner, averaged across datasets. Method indices and line colors follow the encoding paragraph above.

5 Abstention vs standard MLC

Why this section exists. The paper's central claim is that an order-based predictor can *abstain* on labels it is unsure about and, on the labels it does predict, achieve higher quality than a standard MLC method that is forced to predict every label. The two chart families below make that claim visible.

Terminology. '*Standard MLC*' (a.k.a. `BinaryVector` / BV) means each method outputs a 0/1 prediction for every one of the K labels. '*With abstention*' (a.k.a. `PartialAbstention` / PA) means an order-based predictor may return $\perp$ (abstain) on some labels; the PA quality metrics (`f1_pa`, `jaccard_pa`, ...) are computed only over the non-abstained labels. Only the 8 PA/PR methods (indices 1–8) can abstain; BR/CC/CLR/ECC cannot.

Reading the coverage-risk chart. Each point is one (PA/PR method, noise level α) pair. The x-axis is the abstention rate (fraction of labels the method skipped); the y-axis is the quality metric on the labels that *were* predicted (higher is better). Marker shape encodes method, colour encodes noise level. The dashed grey line is the best standard-MLC baseline (BR / CC / CLR / ECC) on the BV quality metric — it is the bar that "no abstention" has to clear. **Any point above the dashed line is direct evidence that abstaining improved quality beyond what any standard MLC method achieves.** Points further right traded coverage for that gain.

Reading the paired-bars chart. Same data, different view. The four sub-panels correspond to the four noise levels. Inside each sub-panel: 12 method positions, two bars per position — blue is the standard-MLC quality (`f1`, no abstention), warm-colour is the with-abstention quality on retained labels. Only the 8 PA/PR methods (1–8) have a warm bar; the standard MLC baselines (9–12) only have a blue bar by construction. The green "$+x.y$" label above each warm bar is the gain in score-points over that method's own BV bar.

5.1 RF, metric `f1_pa` — aggregate

Figure 3: RF (f1_pa) averaged across all datasets: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

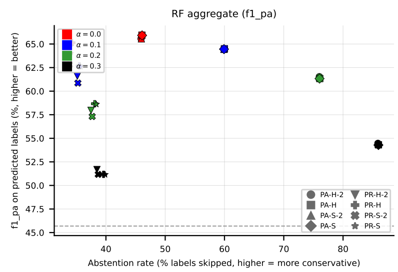


Figure 4: RF (f1_pa) averaged across all datasets: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 5: RF (f1_pa) averaged across all datasets: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $- f_1$) heatmap (right).

Figure 6: RF (f1_pa) averaged across all datasets — robustness view. Left: efficiency-quality scatter (x = coverage = $1 -$ abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

5.2 RF, metric `f1_pa` — per dataset

Figure 7: RF (f1_pa) on `GpositivePseAAC`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

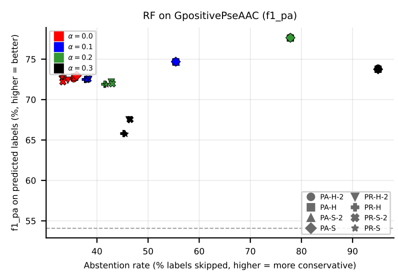
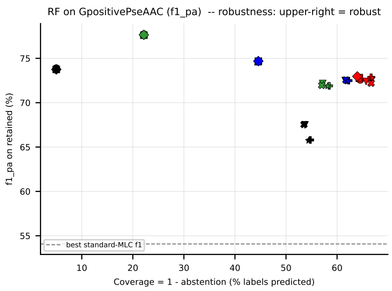


Figure 8: RF (f1_pa) on `GpositivePseAAC`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 9: RF (f1_pa) on `GpositivePseAAC`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $- f_1$) heatmap (right).



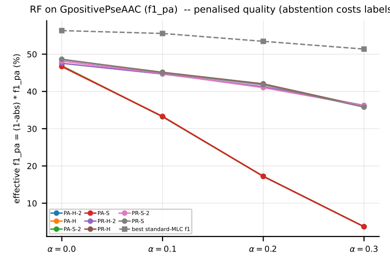
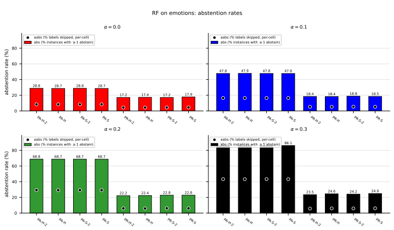


Figure 10: RF (f1_pa) on `GpositivePseAAC` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.



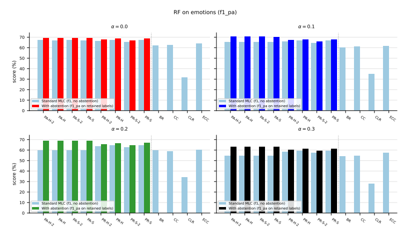


Figure 11: RF (f1_pa) on `emotions`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

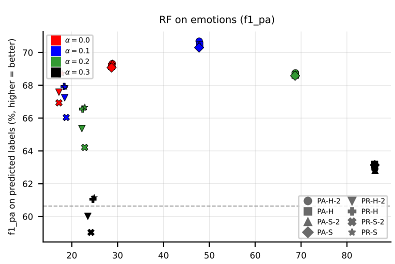


Figure 12: RF (f1_pa) on `emotions`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 13: RF (f1_pa) on `emotions`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $-$ f_1) heatmap (right).

Figure 14: RF (f1_pa) on `emotions` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 15: RF (f1_pa) on `scene`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

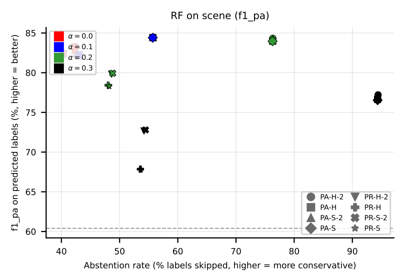


Figure 16: RF (f1_pa) on `scene`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 17: RF (f1_pa) on `scene`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa − f_1) heatmap (right).

Figure 18: RF (f1_pa) on `scene` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 19: RF (f1_pa) on `PlantPseAAC`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

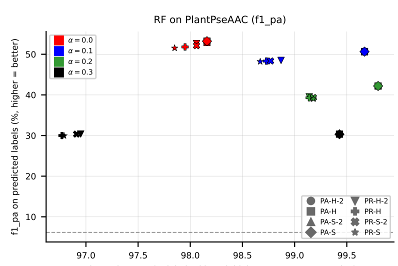


Figure 20: RF (f1_pa) on `PlantPseAAC`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 21: RF (f1_pa) on `PlantPseAAC`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $- f_1$) heatmap (right).

Figure 22: RF (f1_pa) on `PlantPseAAC` — robustness view. Left: efficiency-quality scatter (x = coverage = 1 − abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 23: RF (f1_pa) on `HumanPseAAC`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

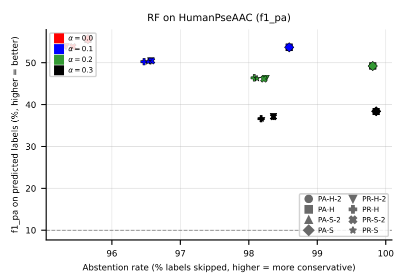


Figure 24: RF (f1_pa) on `HumanPseAAC`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 25: RF (f1_pa) on `HumanPseAAC`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $- f_1$) heatmap (right).

Figure 26: RF (f1_pa) on HumanPseAAC — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa $= (1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 27: RF (f1_pa) on Yeast: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = abs (% of instances with at least one abstained label, instance-level); dot = aabs (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

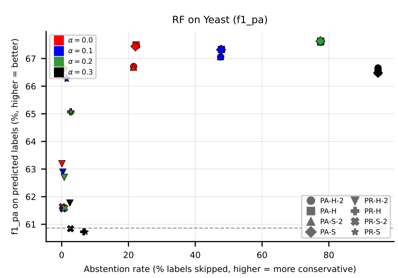


Figure 28: RF (f1_pa) on Yeast: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 29: RF (f1_pa) on `Yeast`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $-$ f_1) heatmap (right).

Figure 30: RF (f1_pa) on `Yeast` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 31: RF (f1_pa) on `CHD_49`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

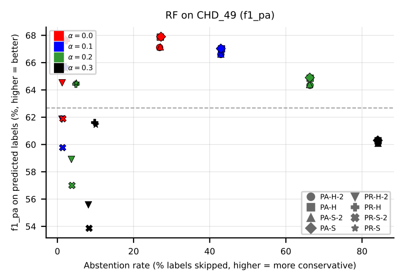


Figure 32: RF (f1_pa) on `CHD_49`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 33: RF (f1_pa) on `CHD_49`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $- f_1$) heatmap (right).

Figure 34: RF (f1_pa) on `CHD_49` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 35: RF (f1_pa) on `Water-quality`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

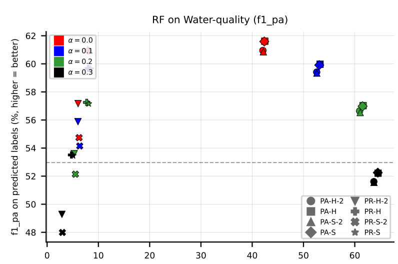


Figure 36: RF (f1_pa) on `Water-quality`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 37: RF (f1_pa) on `Water-quality`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa − f_1) heatmap (right).

Figure 38: RF (f1_pa) on `Water-quality` — robustness view. Left: efficiency-quality scatter (x = coverage = 1 − abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = (1 − abs) · $f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 39: RF (f1_pa) on **enron**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

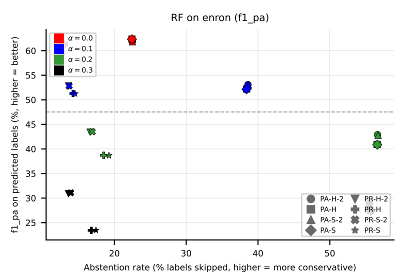
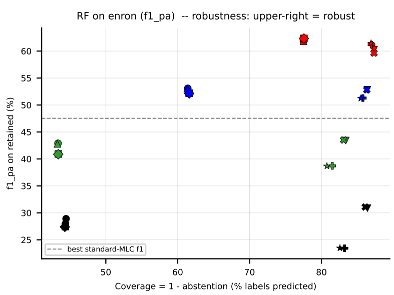


Figure 40: RF (f1_pa) on **enron**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 41: RF (f1_pa) on **enron**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa − f_1) heatmap (right).



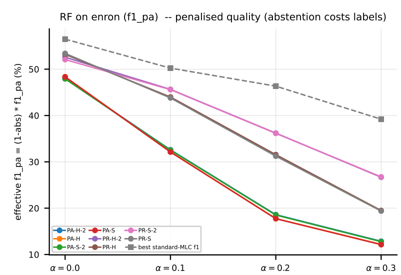


Figure 42: RF (f1_pa) on `enron` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 -$ abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

5.3 RF, metric jaccard_pa — aggregate

Figure 43: RF (jaccard_pa) averaged across all datasets: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

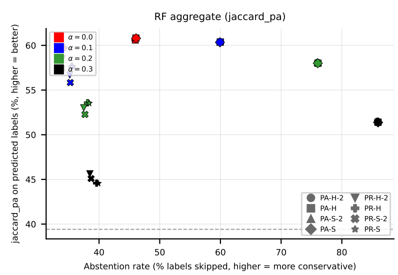


Figure 44: RF (jaccard_pa) averaged across all datasets: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 45: RF (jaccard_pa) averaged across all datasets: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $-$ f_1) heatmap (right).

Figure 46: RF (jaccard_pa) averaged across all datasets — robustness view. Left: efficiency-quality scatter (x = coverage = 1 − abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1-\text{abs}) \cdot \mathit{jaccard\_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

5.4 RF, metric `jaccard_pa` — per dataset

Figure 47: RF (jaccard_pa) on `GpositivePseAAC`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

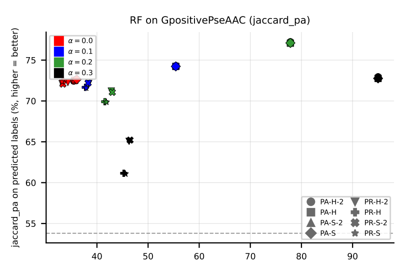
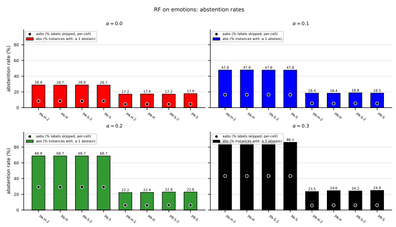


Figure 48: RF (jaccard_pa) on `GpositivePseAAC`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 49: RF (jaccard_pa) on `GpositivePseAAC`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $- f_1$) heatmap (right).

Figure 50: RF (jaccard_pa) on `GpositivePseAAC` — robustness view. Left: efficiency-quality scatter (x = coverage = 1 − abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1-\text{abs}) \cdot jaccard\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.



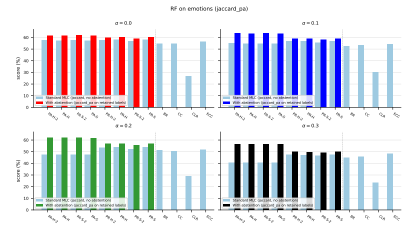


Figure 51: RF (jaccard_pa) on `emotions`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

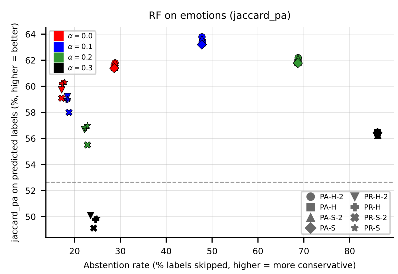
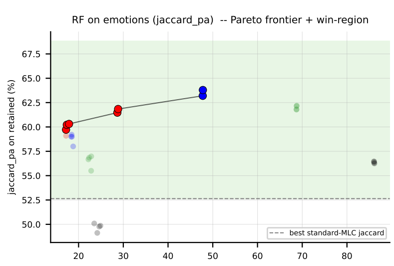


Figure 52: RF (jaccard_pa) on `emotions`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.



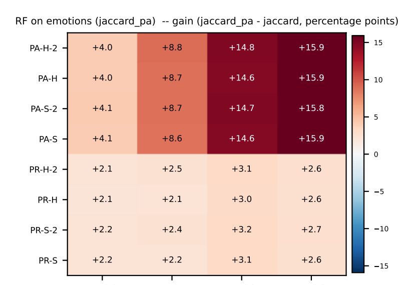


Figure 53: RF (jaccard_pa) on `emotions`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa − f_1) heatmap (right).

Figure 54: RF (jaccard_pa) on `emotions` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot jaccard\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 55: RF (jaccard_pa) on `scene`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

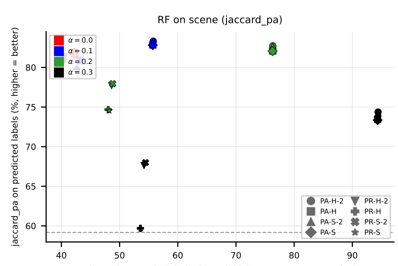


Figure 56: RF (jaccard_pa) on `scene`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 57: RF (jaccard_pa) on `scene`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $- f_1$) heatmap (right).

Figure 58: RF (jaccard_pa) on `scene` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot jaccard\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 59: RF (jaccard_pa) on **PlantPseAAC**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

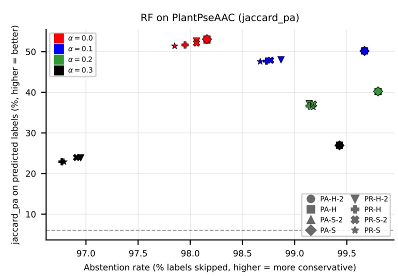


Figure 60: RF (jaccard_pa) on **PlantPseAAC**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 61: RF (jaccard_pa) on **PlantPseAAC**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $-$ f_1) heatmap (right).

Figure 62: RF (jaccard_pa) on `PlantPseAAC` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot jaccard\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 63: RF (jaccard_pa) on `HumanPseAAC`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

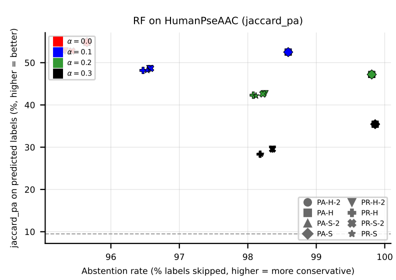
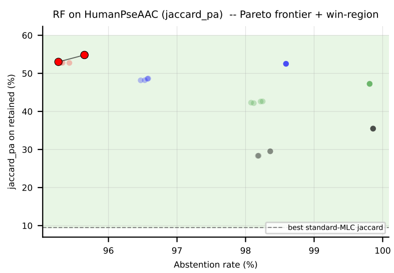
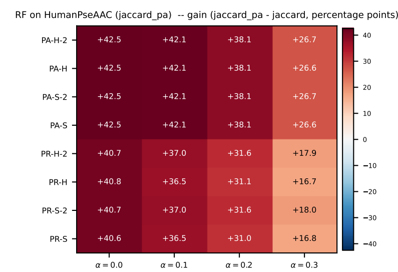


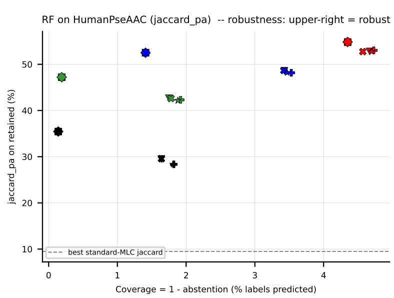
Figure 64: RF (jaccard_pa) on `HumanPseAAC`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.





	$\alpha = 0.0$	$\alpha = 0.1$	$\alpha = 0.2$	$\alpha = 0.3$
PA-H-2	+42.5	+42.1	+38.1	+26.7
PA-H	+42.5	+42.1	+38.1	+26.6
PA-S-2	+42.5	+42.1	+38.1	+26.7
PA-S	+42.5	+42.1	+38.1	+26.6
PR-H-2	+40.7	+37.0	+31.6	+17.9
PR-H	+40.8	+36.5	+31.1	+16.7
PR-S-2	+40.7	+37.0	+31.6	+18.0
PR-S	+40.6	+36.5	+31.0	+16.8

Figure 65: RF (jaccard_pa) on `HumanPseAAC`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $- f_1$) heatmap (right).



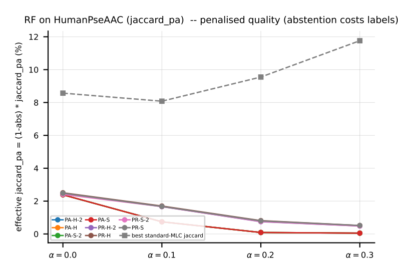
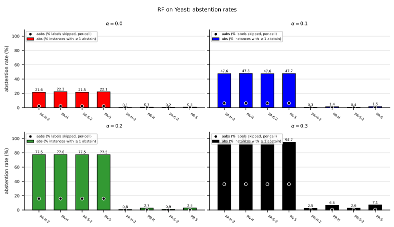


Figure 66: RF (jaccard_pa) on `HumanPseAAC` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot jaccard\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.



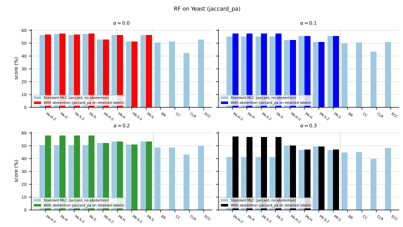


Figure 67: RF (jaccard_pa) on `Yeast`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

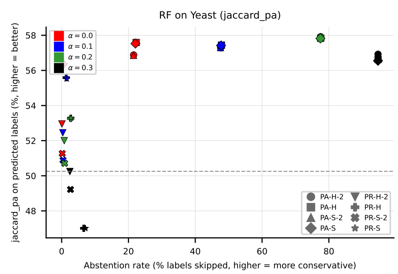
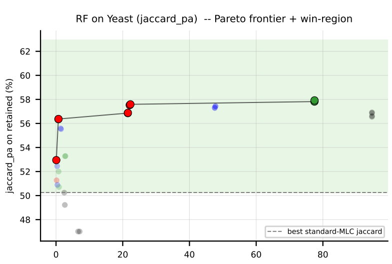


Figure 68: RF (jaccard_pa) on `Yeast`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.



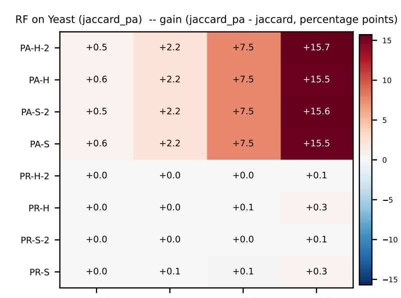


Figure 69: RF (jaccard_pa) on `Yeast`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $-$ f_1) heatmap (right).

Figure 70: RF (jaccard_pa) on `Yeast` — robustness view. Left: efficiency-quality scatter (x = coverage $= 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa $= (1 - \text{abs}) \cdot \mathit{jaccard\_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 71: RF (jaccard_pa) on `CHD_49`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

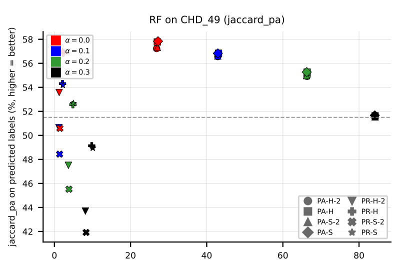
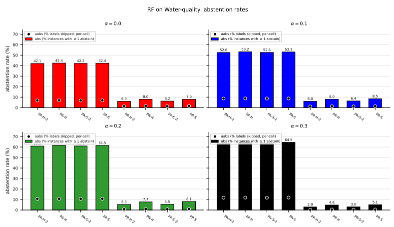


Figure 72: RF (jaccard_pa) on `CHD_49`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 73: RF (jaccard_pa) on CHD_49: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $- f_1$) heatmap (right).

Figure 74: RF (jaccard_pa) on CHD_49 — robustness view. Left: efficiency-quality scatter (x = coverage = $1 -$ abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa $= (1 - \text{abs}) \cdot jaccard\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.



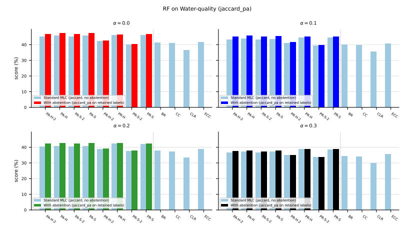


Figure 75: RF (jaccard_pa) on `Water-quality`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

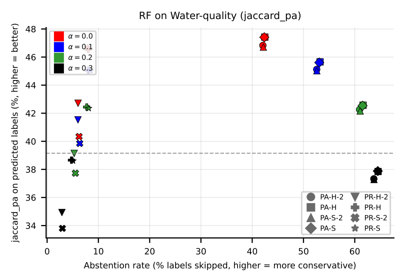
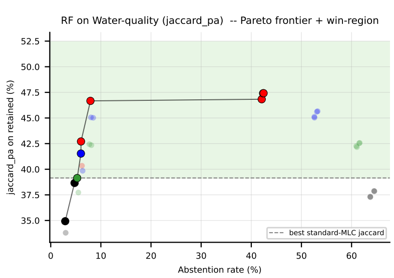


Figure 76: RF (jaccard_pa) on `Water-quality`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.



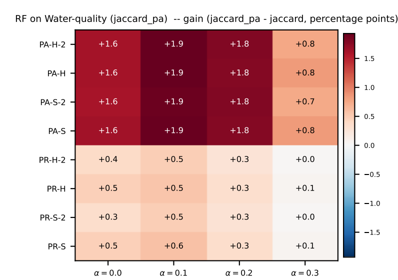


Figure 77: RF (jaccard_pa) on `Water-quality`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $-$ f_1) heatmap (right).

Figure 78: RF (jaccard_pa) on `Water-quality` — robustness view. Left: efficiency-quality scatter (x = coverage = 1−abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot \mathit{jaccard\_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 79: RF (jaccard_pa) on `enron`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

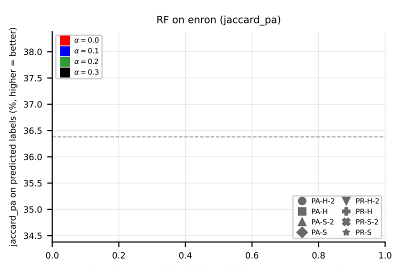


Figure 80: RF (jaccard_pa) on `enron`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 81: RF (jaccard_pa) on `enron`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $-$ f_1) heatmap (right).

Figure 82: RF (jaccard_pa) on `enron` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa $= (1 - \text{abs}) \cdot jaccard\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

5.5 LGBM, metric `f1_pa` — aggregate

Figure 83: LGBM (f1_pa) averaged across all datasets: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

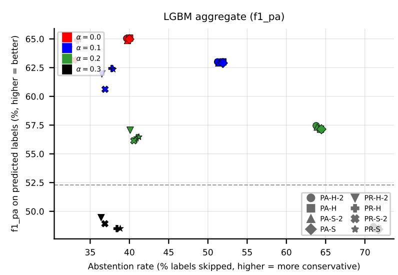


Figure 84: LGBM (f1_pa) averaged across all datasets: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 85: LGBM (f1_pa) averaged across all datasets: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $- f_1$) heatmap (right).

Figure 86: LGBM (f1_pa) averaged across all datasets — robustness view. Left: efficiency-quality scatter (x = coverage = 1 − abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1-\text{abs})\cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

5.6 LGBM, metric `f1_pa` — per dataset

Figure 87: LGBM (f1_pa) on `GpositivePseAAC`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

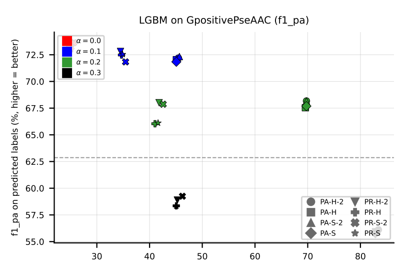


Figure 88: LGBM (f1_pa) on `GpositivePseAAC`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 89: LGBM (f1_pa) on `GpositivePseAAC`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $-$ f_1) heatmap (right).

Figure 90: LGBM (f1_pa) on `GpositivePseAAC` — robustness view. Left: efficiency-quality scatter (x = coverage = 1 $-$ abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 91: LGBM (f1_pa) on `emotions`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

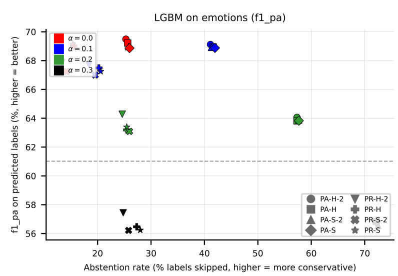


Figure 92: LGBM (f1_pa) on `emotions`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 93: LGBM (f1_pa) on `emotions`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $- f_1$) heatmap (right).

Figure 94: LGBM (f1_pa) on `emotions` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 95: LGBM (f1_pa) on `scene`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

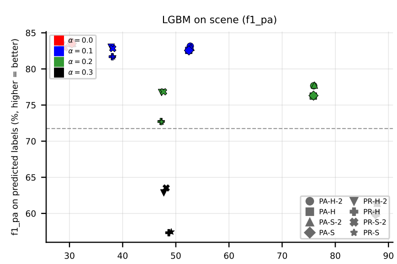


Figure 96: LGBM (f1_pa) on `scene`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 97: LGBM (f1_pa) on `scene`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $-$ f_1) heatmap (right).

Figure 98: LGBM (f1_pa) on `scene` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 99: LGBM (f1_pa) on `PlantPseAAC`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

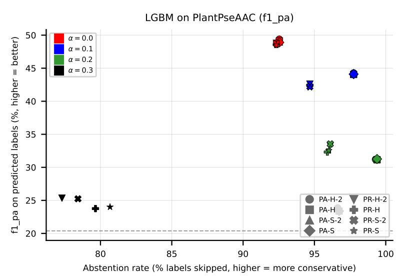


Figure 100: LGBM (f1_pa) on `PlantPseAAC`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 101: LGBM (f1_pa) on `PlantPseAAC`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $- f_1$) heatmap (right).

Figure 102: LGBM (f1_pa) on `PlantPseAAC` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 103: LGBM (f1_pa) on `HumanPseAAC`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

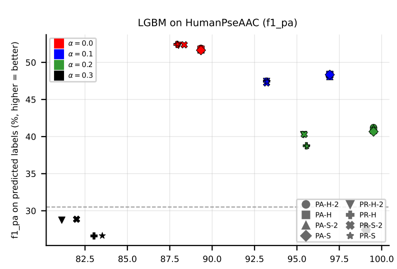
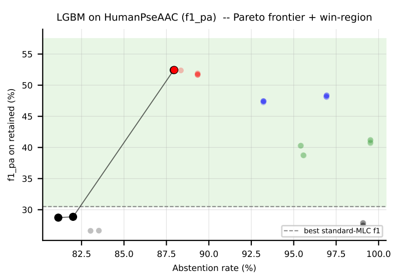


Figure 104: LGBM (f1_pa) on `HumanPseAAC`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.



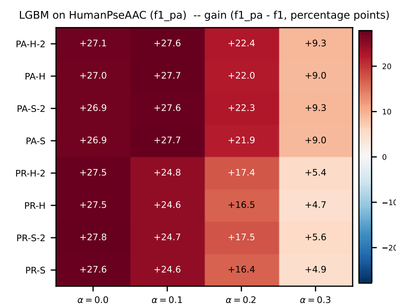
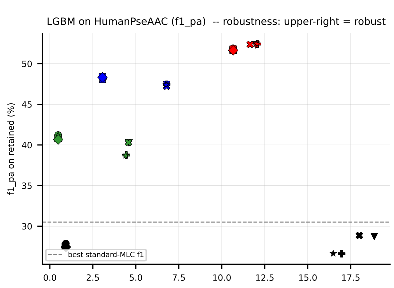


Figure 105: LGBM (f1_pa) on `HumanPseAAC`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $- f_1$) heatmap (right).



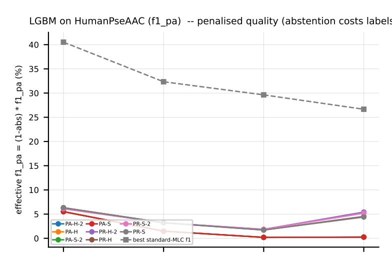


Figure 106: LGBM (f1_pa) on `HumanPseAAC` — robustness view. Left: efficiency-quality scatter (x = coverage = $1-$abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1-\text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 107: LGBM (f1_pa) on `Yeast`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

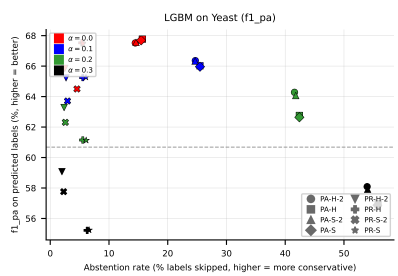


Figure 108: LGBM (f1_pa) on `Yeast`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 109: LGBM (f1_pa) on `Yeast`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $- f_1$) heatmap (right).

Figure 110: LGBM (f1_pa) on `Yeast` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 111: LGBM (f1_pa) on `CHD_49`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

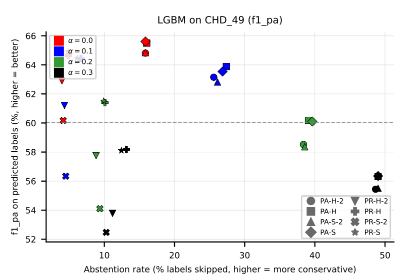


Figure 112: LGBM (f1_pa) on `CHD_49`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 113: LGBM (f1_pa) on `CHD_49`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $-$ f_1) heatmap (right).

Figure 114: LGBM (f1_pa) on `CHD_49` — robustness view. Left: efficiency-quality scatter (x = coverage = 1 $-$ abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 115: LGBM (f1_pa) on `Water-quality`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

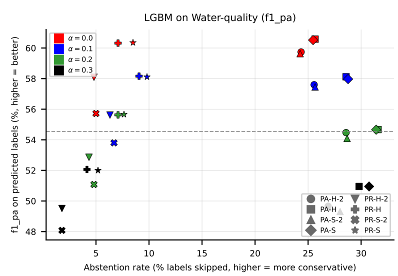


Figure 116: LGBM (f1_pa) on `Water-quality`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 117: LGBM (f1_pa) on `Water-quality`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $- f_1$) heatmap (right).

Figure 118: LGBM (f1_pa) on `Water-quality` — robustness view. Left: efficiency-quality scatter (x = coverage = 1 − abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

5.7 LGBM, metric `jaccard_pa` — aggregate

Figure 119: LGBM (jaccard_pa) averaged across all datasets: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

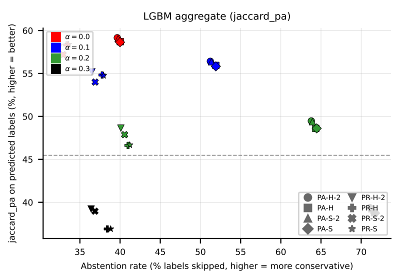


Figure 120: LGBM (jaccard_pa) averaged across all datasets: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 121: LGBM (jaccard_pa) averaged across all datasets: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $-$ f_1) heatmap (right).

Figure 122: LGBM (jaccard_pa) averaged across all datasets — robustness view. Left: efficiency-quality scatter (x = coverage = 1 − abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1-\text{abs}) \cdot jaccard\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

5.8 LGBM, metric `jaccard_pa` — per dataset

Figure 123: LGBM (jaccard_pa) on `GpositivePseAAC`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

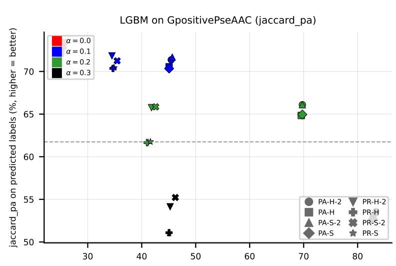


Figure 124: LGBM (jaccard_pa) on `GpositivePseAAC`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 125: LGBM (jaccard_pa) on `GpositivePseAAC`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $-$ f_1) heatmap (right).

Figure 126: LGBM (jaccard_pa) on `GpositivePseAAC` — robustness view. Left: efficiency-quality scatter (x = coverage = 1 − abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot jaccard\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 127: LGBM (jaccard_pa) on `emotions`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

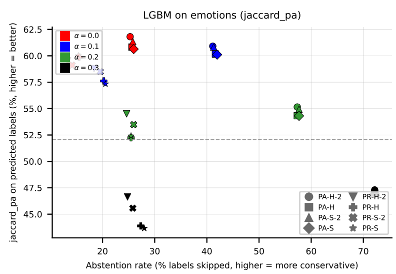


Figure 128: LGBM (jaccard_pa) on `emotions`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 129: LGBM (jaccard_pa) on `emotions`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa − f_1) heatmap (right).

Figure 130: LGBM (jaccard_pa) on `emotions` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot jaccard\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 131: LGBM (jaccard_pa) on `scene`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

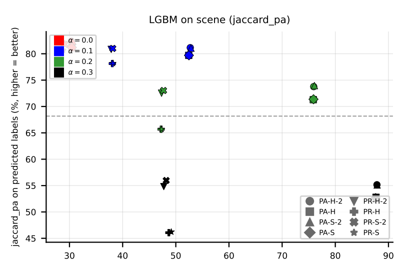
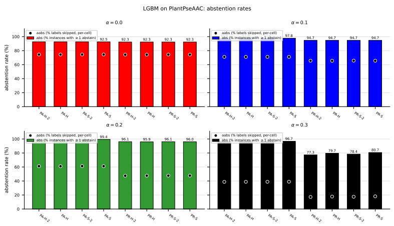


Figure 132: LGBM (jaccard_pa) on `scene`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 133: LGBM (jaccard_pa) on `scene`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $- f_1$) heatmap (right).

Figure 134: LGBM (jaccard_pa) on `scene` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa $= (1 - \text{abs}) \cdot jaccard\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.



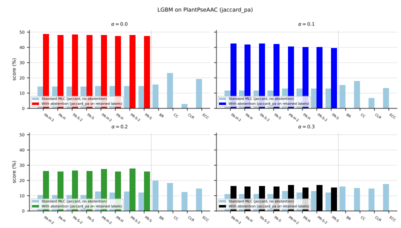


Figure 135: LGBM (jaccard_pa) on `PlantPseAAC`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

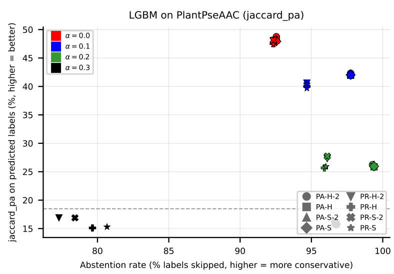
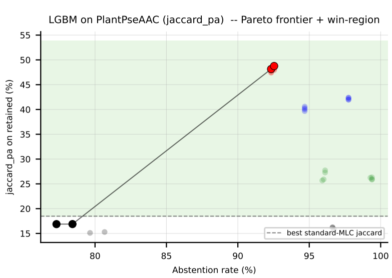


Figure 136: LGBM (jaccard_pa) on `PlantPseAAC`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.



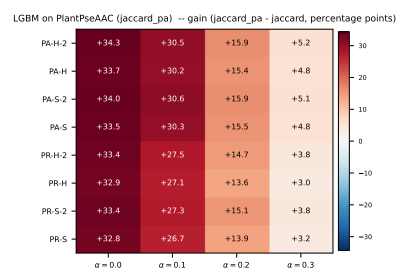


Figure 137: LGBM (jaccard_pa) on `PlantPseAAC`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa − f_1) heatmap (right).

Figure 138: LGBM (jaccard_pa) on `PlantPseAAC` — robustness view. Left: efficiency-quality scatter (x = coverage = 1 − abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1-\text{abs}) \cdot jaccard\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 139: LGBM (jaccard_pa) on `HumanPseAAC`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

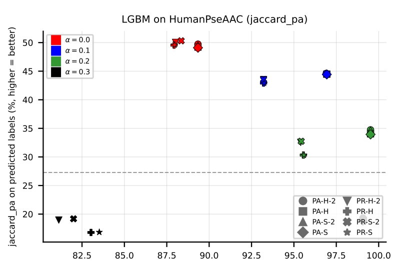


Figure 140: LGBM (jaccard_pa) on `HumanPseAAC`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 141: LGBM (jaccard_pa) on `HumanPseAAC`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $-$ f_1) heatmap (right).

Figure 142: LGBM (jaccard_pa) on `HumanPseAAC` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 -$ abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1-\text{abs})\cdot \textit{jaccard\_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 143: LGBM (jaccard_pa) on `Yeast`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

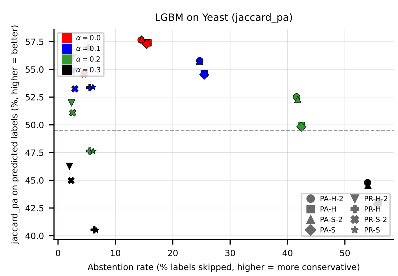


Figure 144: LGBM (jaccard_pa) on `Yeast`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 145: LGBM (jaccard_pa) on `Yeast`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $- f_1$) heatmap (right).

Figure 146: LGBM (jaccard_pa) on `Yeast` — robustness view. Left: efficiency-quality scatter (x = coverage = 1 − abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot \mathit{jaccard\_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 147: LGBM (jaccard_pa) on `CHD_49`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

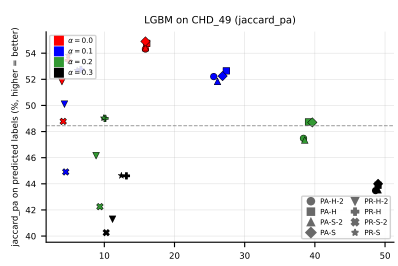


Figure 148: LGBM (jaccard_pa) on `CHD_49`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 149: LGBM (jaccard_pa) on `CHD_49`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $-$ f_1) heatmap (right).

Figure 150: LGBM (jaccard_pa) on `CHD_49` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot jaccard\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 151: LGBM (jaccard_pa) on `Water-quality`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

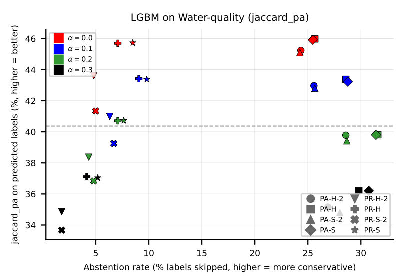


Figure 152: LGBM (jaccard_pa) on `Water-quality`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 153: LGBM (jaccard_pa) on `Water-quality`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $- f_1$) heatmap (right).

Figure 154: LGBM (jaccard_pa) on `Water-quality` — robustness view. Left: efficiency-quality scatter (x = coverage = 1 − abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1-\text{abs})\cdot jaccard\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

A Experimental results — appendix tables (RF base learner)

Appendix counterpart of the main Paper results section. Tables G2–G5 mirror the original paper's appendix exactly (same metric sets, with the 6 non-main datasets split into imbalanced vs. balanced buckets following the original layout). Tables G6–G7 are revision additions reporting the new metrics ($f_{\mathrm{MLC}}^{\mathrm{Jacc}}$, macro-$f_1$, micro-$f_1$).

A.1 Standard MLC predictions (BinaryVector)

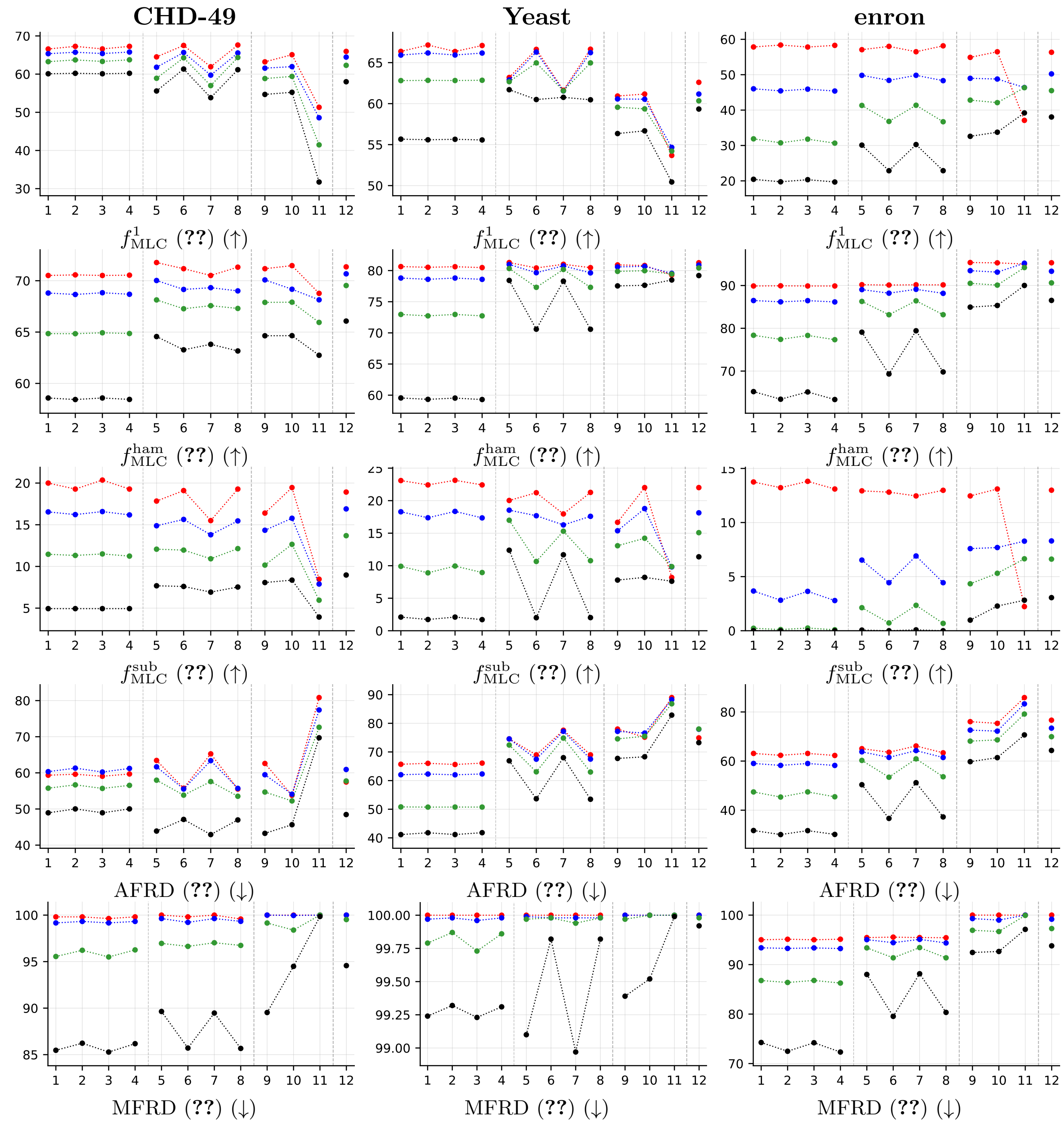


Table 18: (Table G2) Average scores (in %, y-axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x-axis): (PA-H-2, PA-H, PA-S-2, PA-S), (PR-H-2, PR-H, PR-S-2, PR-S), and (BR, CC, CLR) are encoded as (1, 2, 3, 4), (5, 6, 7, 8), and (9, 10, 11), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

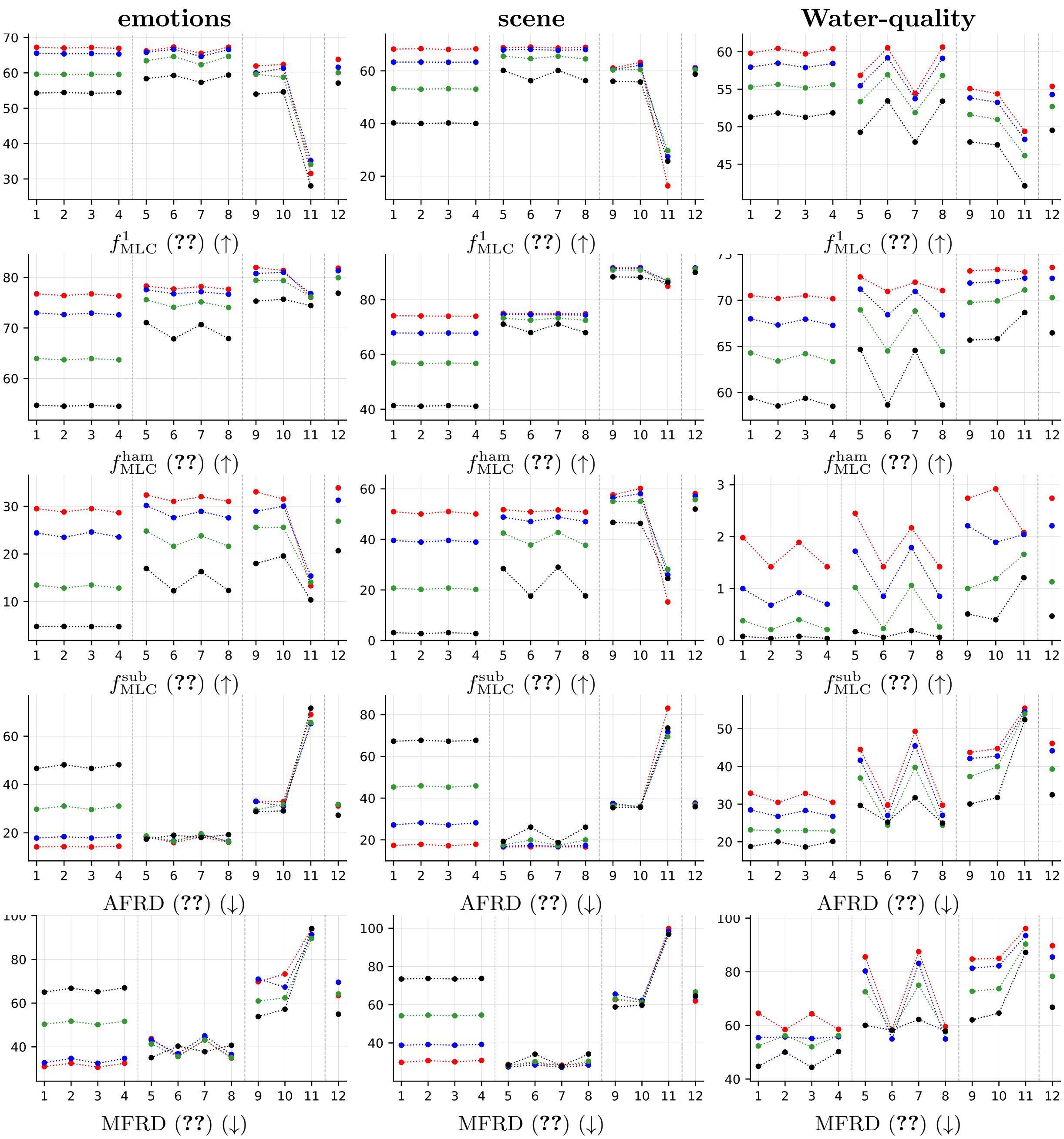


Table 19: (Table G3) Average scores (in %, y-axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x-axis): (PA-H-2, PA-H, PA-S-2, PA-S), (PR-H-2, PR-H, PR-S-2, PR-S), and (BR, CC, CLR) are encoded as (1, 2, 3, 4), (5, 6, 7, 8), and (9, 10, 11), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

A.2 Partial abstention (PartialAbstention)

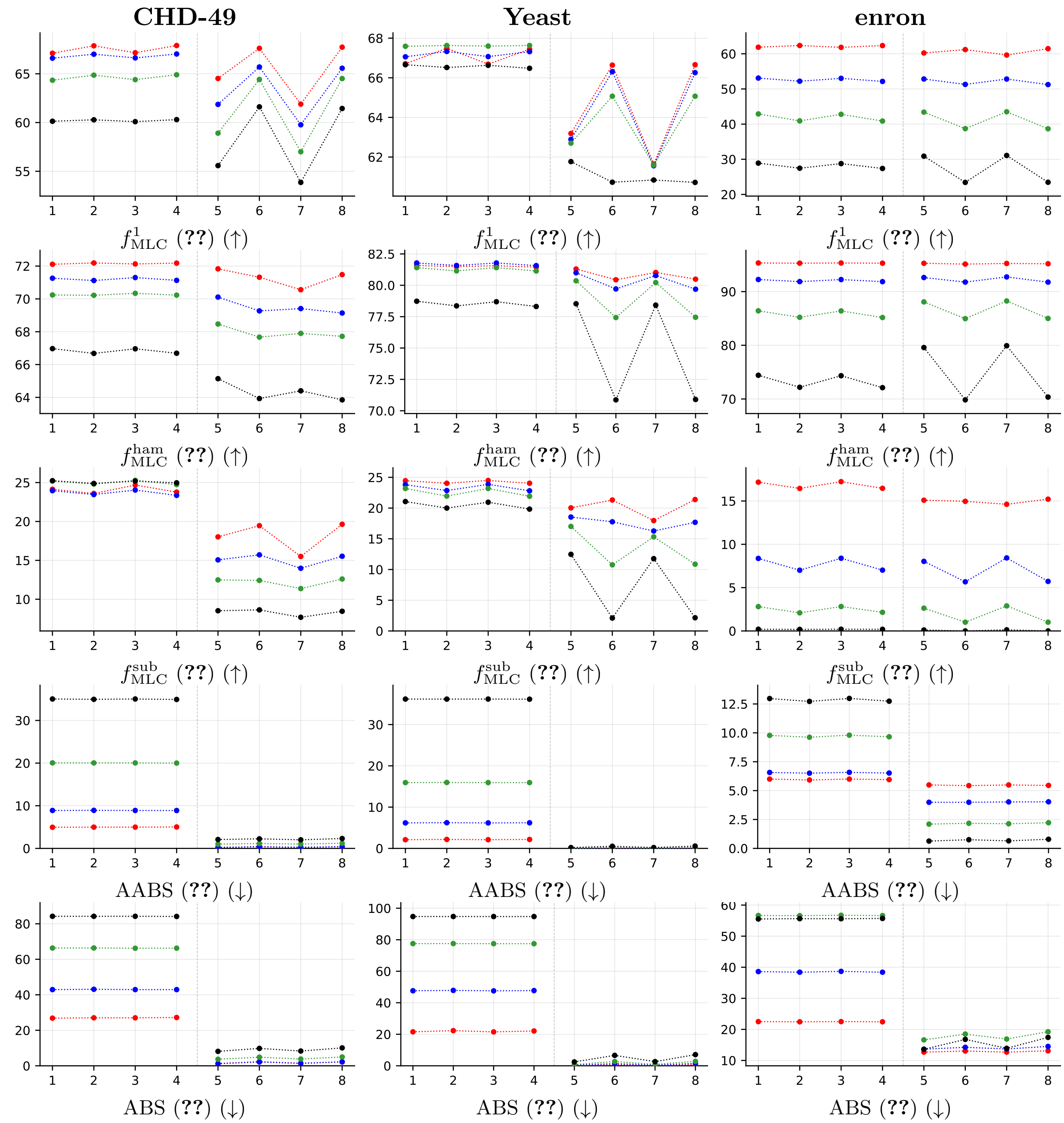


Table 20: (Table G4) Average scores (in %, y-axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x-axis): (PA-H-2, PA-H, PA-S-2, PA-S) and (PR-H-2, PR-H, PR-S-2, PR-S) are encoded as (1, 2, 3, 4) and (5, 6, 7, 8), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

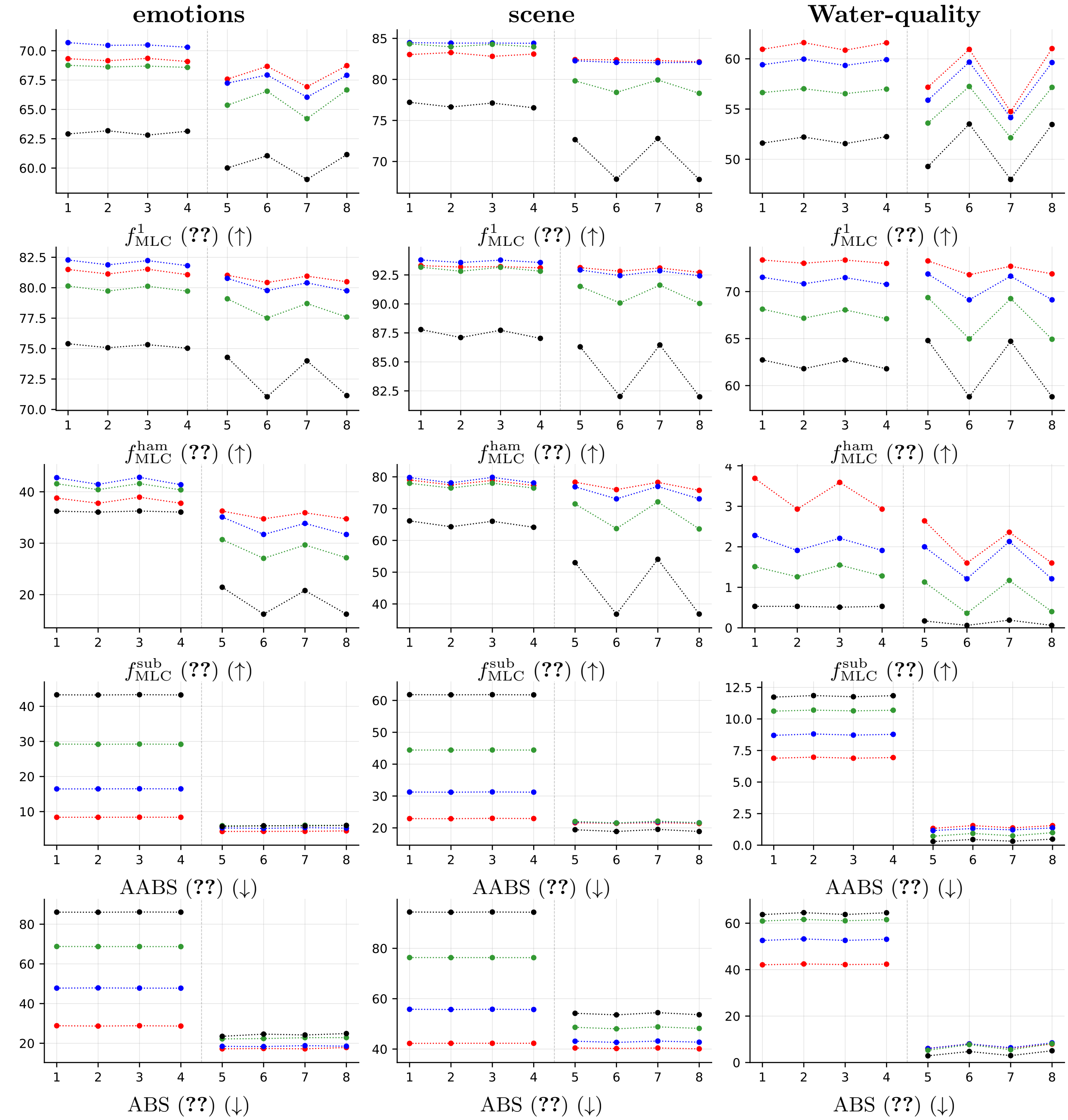


Table 21: (Table G5) Average scores (in %, y-axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x-axis): (PA-H-2, PA-H, PA-S-2, PA-S) and (PR-H-2, PR-H, PR-S-2, PR-S) are encoded as (1, 2, 3, 4) and (5, 6, 7, 8), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

A.3 Standard MLC predictions (BinaryVector)

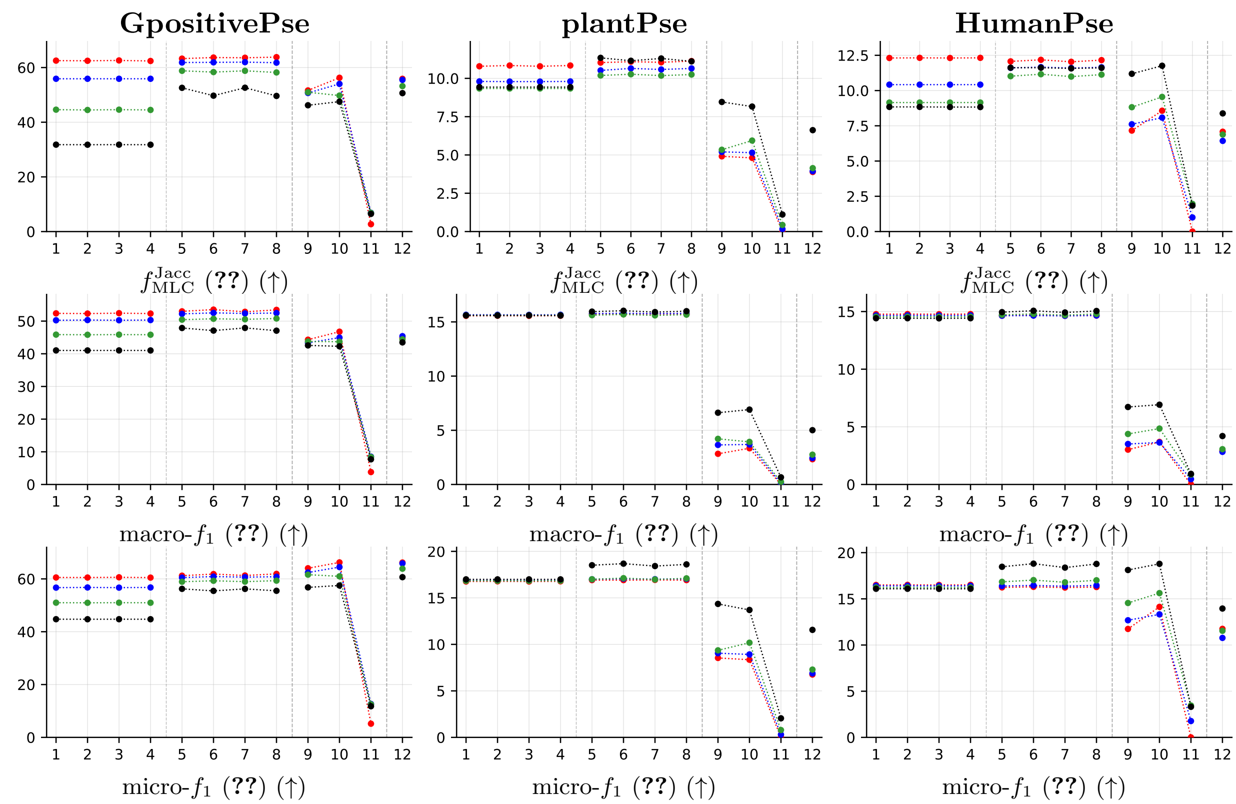


Table 22: (Table G6, new metrics — main datasets) Average scores (in %, y-axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x-axis): (PA-H-2, PA-H, PA-S-2, PA-S), (PR-H-2, PR-H, PR-S-2, PR-S), and (BR, CC, CLR) are encoded as (1, 2, 3, 4), (5, 6, 7, 8), and (9, 10, 11), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

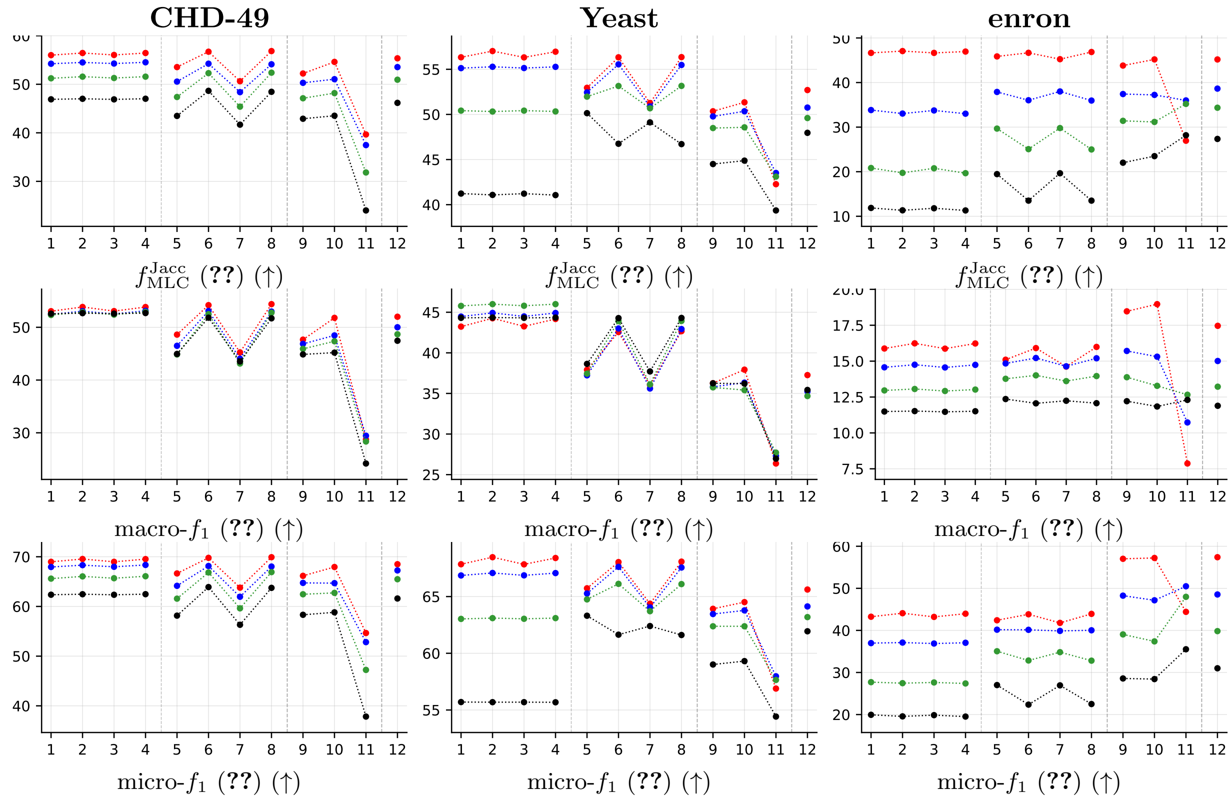


Table 23: (Table G6, new metrics — imbalanced) Average scores (in %, y-axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x-axis): (PA-H-2, PA-H, PA-S-2, PA-S), (PR-H-2, PR-H, PR-S-2, PR-S), and (BR, CC, CLR) are encoded as (1, 2, 3, 4), (5, 6, 7, 8), and (9, 10, 11), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

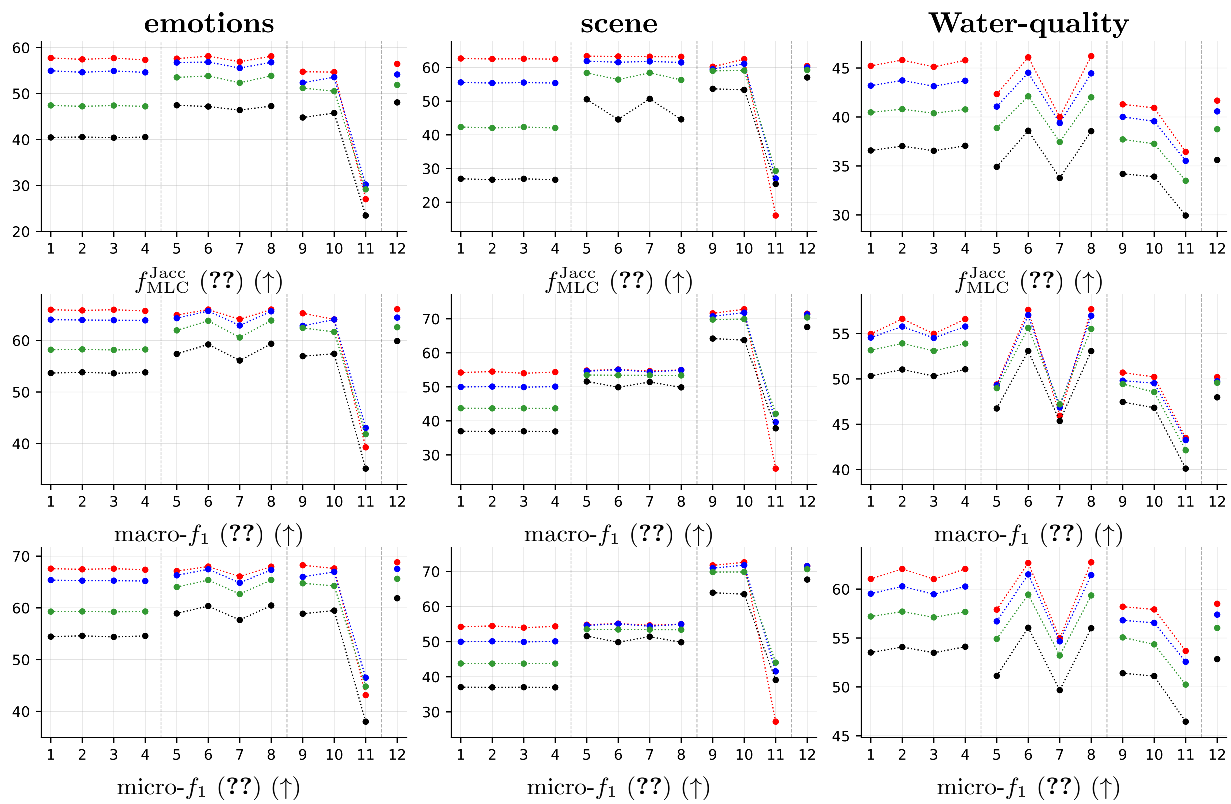


Table 24: (Table G6, new metrics — balanced) Average scores (in %, y-axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x-axis): (PA-H-2, PA-H, PA-S-2, PA-S), (PR-H-2, PR-H, PR-S-2, PR-S), and (BR, CC, CLR) are encoded as (1, 2, 3, 4), (5, 6, 7, 8), and (9, 10, 11), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

A.4 Partial abstention (PartialAbstention)

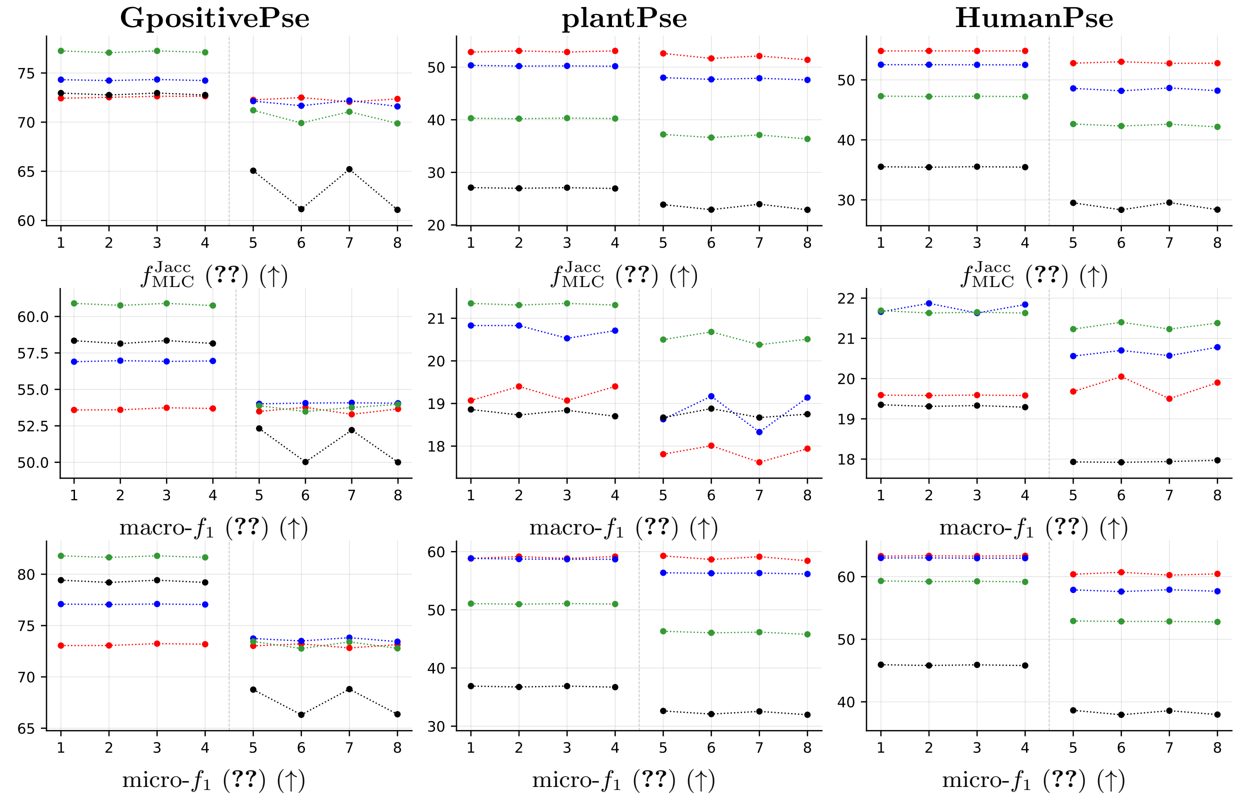


Table 25: (Table G7, new metrics — main datasets) Average scores (in %, y-axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x-axis): (PA-H-2, PA-H, PA-S-2, PA-S) and (PR-H-2, PR-H, PR-S-2, PR-S) are encoded as (1, 2, 3, 4) and (5, 6, 7, 8), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

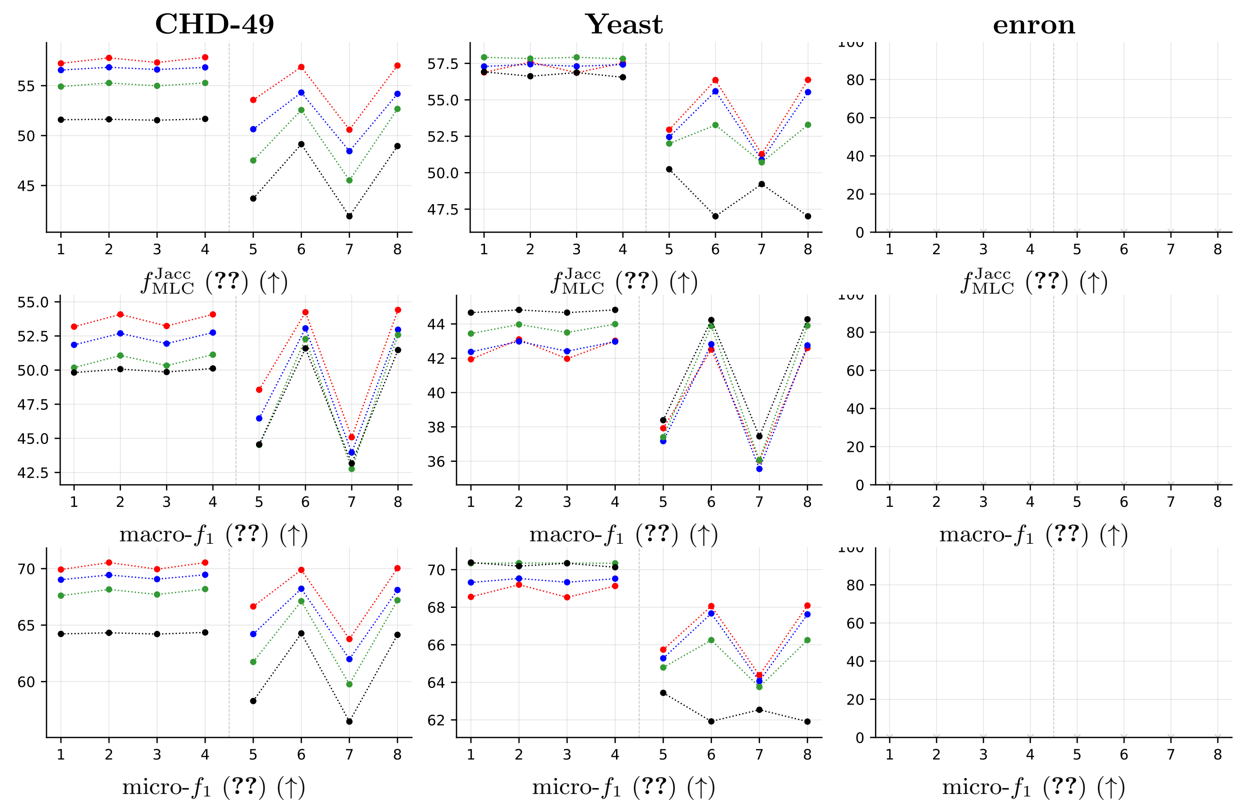


Table 26: (Table G7, new metrics — imbalanced) Average scores (in %, y-axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x-axis): (PA-H-2, PA-H, PA-S-2, PA-S) and (PR-H-2, PR-H, PR-S-2, PR-S) are encoded as (1, 2, 3, 4) and (5, 6, 7, 8), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

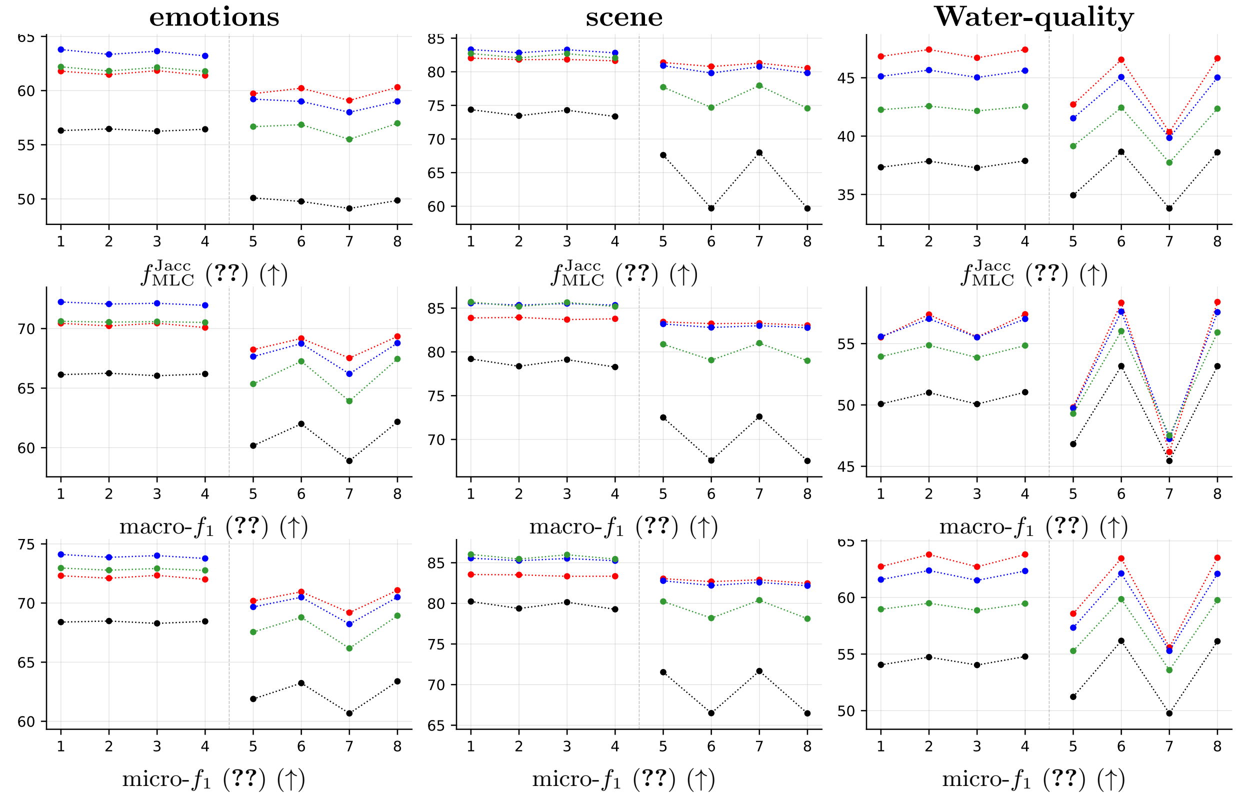


Table 27: (Table G7, new metrics — balanced) Average scores (in %, y-axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x-axis): (PA-H-2, PA-H, PA-S-2, PA-S) and (PR-H-2, PR-H, PR-S-2, PR-S) are encoded as (1, 2, 3, 4) and (5, 6, 7, 8), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

B Per-dataset results (RF base learner)

B.1 GpositivePseAAC ($K = 4$)

RF • `GpositivePseAAC` • BinaryVector — Standard MLC predictions

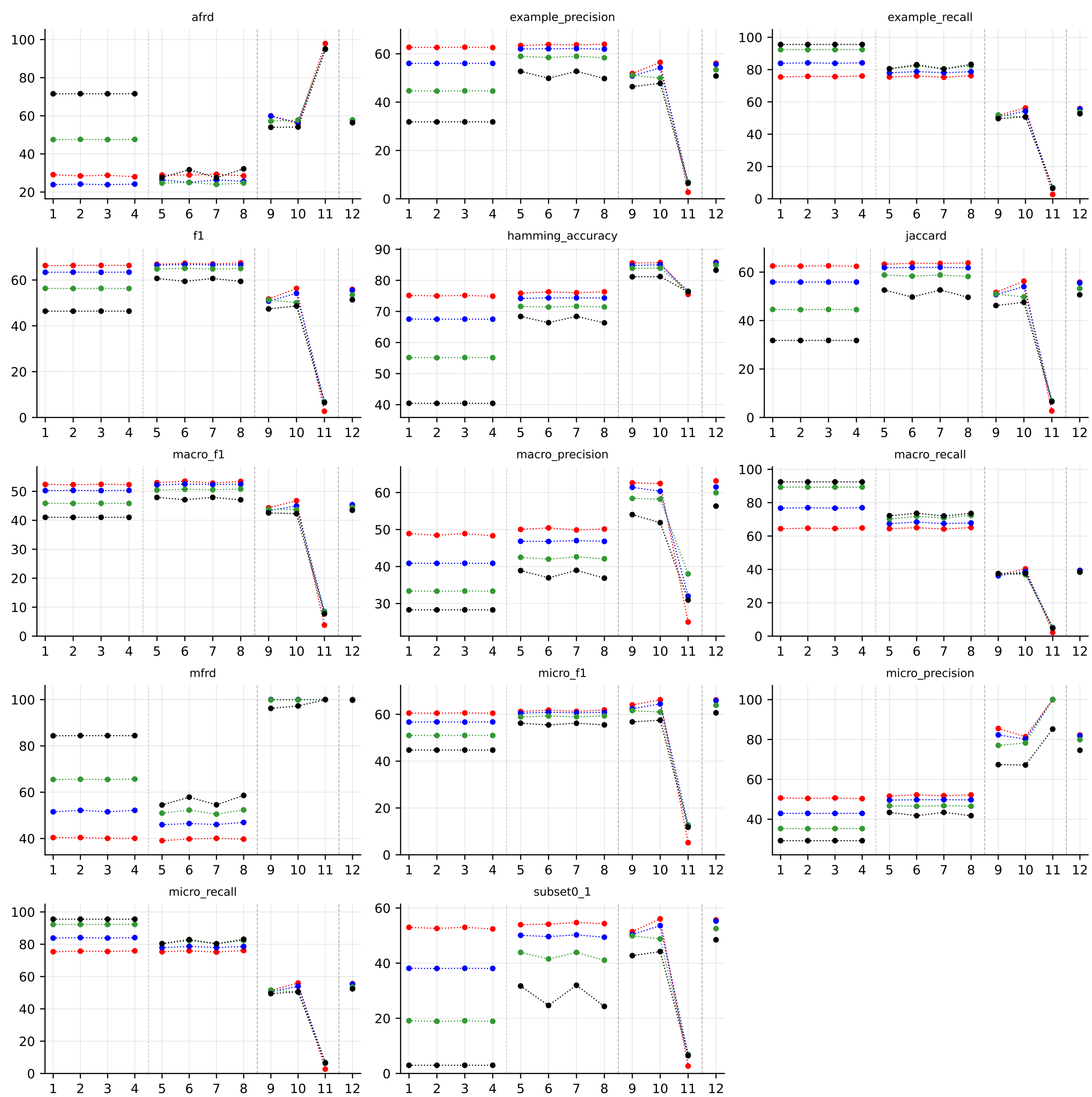


Table 28: Standard MLC predictions (BinaryVector) on `GpositivePseAAC` (RF base learner).

RF • `GpositivePseAAC` • PartialAbstention — Partial abstention

RF • `GpositivePseAAC` • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

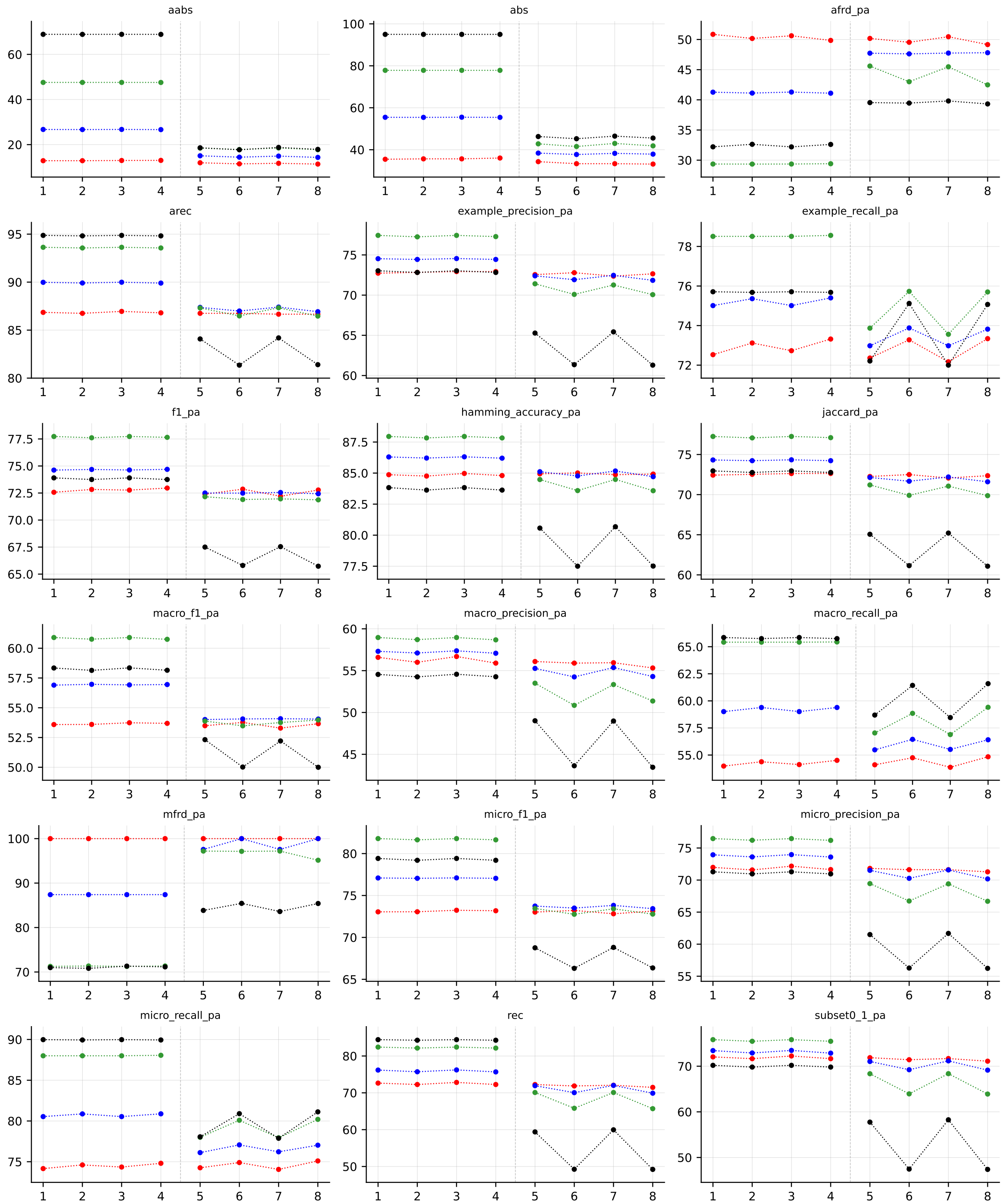


Table 29: Partial abstention (PartialAbstention) on `GpositivePseAAC` (RF base learner).

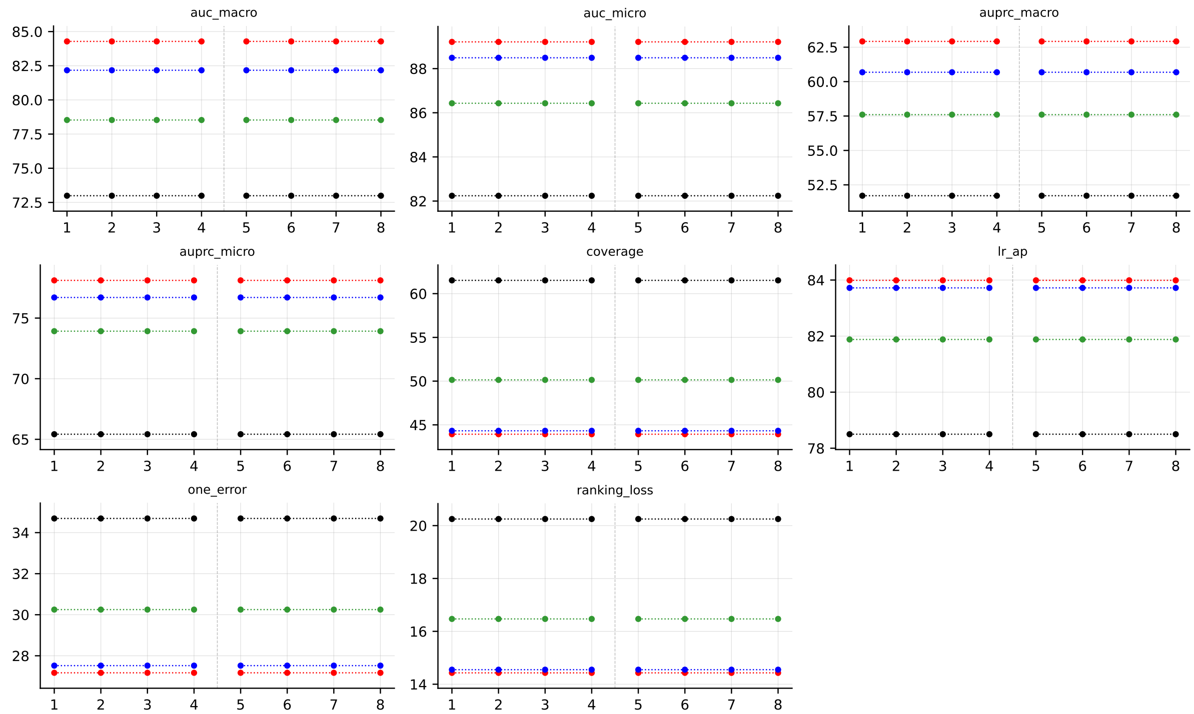


Table 30: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `GpositivePseAAC` (RF base learner).

B.2 Emotions ($K = 6$)

RF • `emotions` • BinaryVector — Standard MLC predictions

RF • `emotions` • PartialAbstention — Partial abstention

RF • `emotions` • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

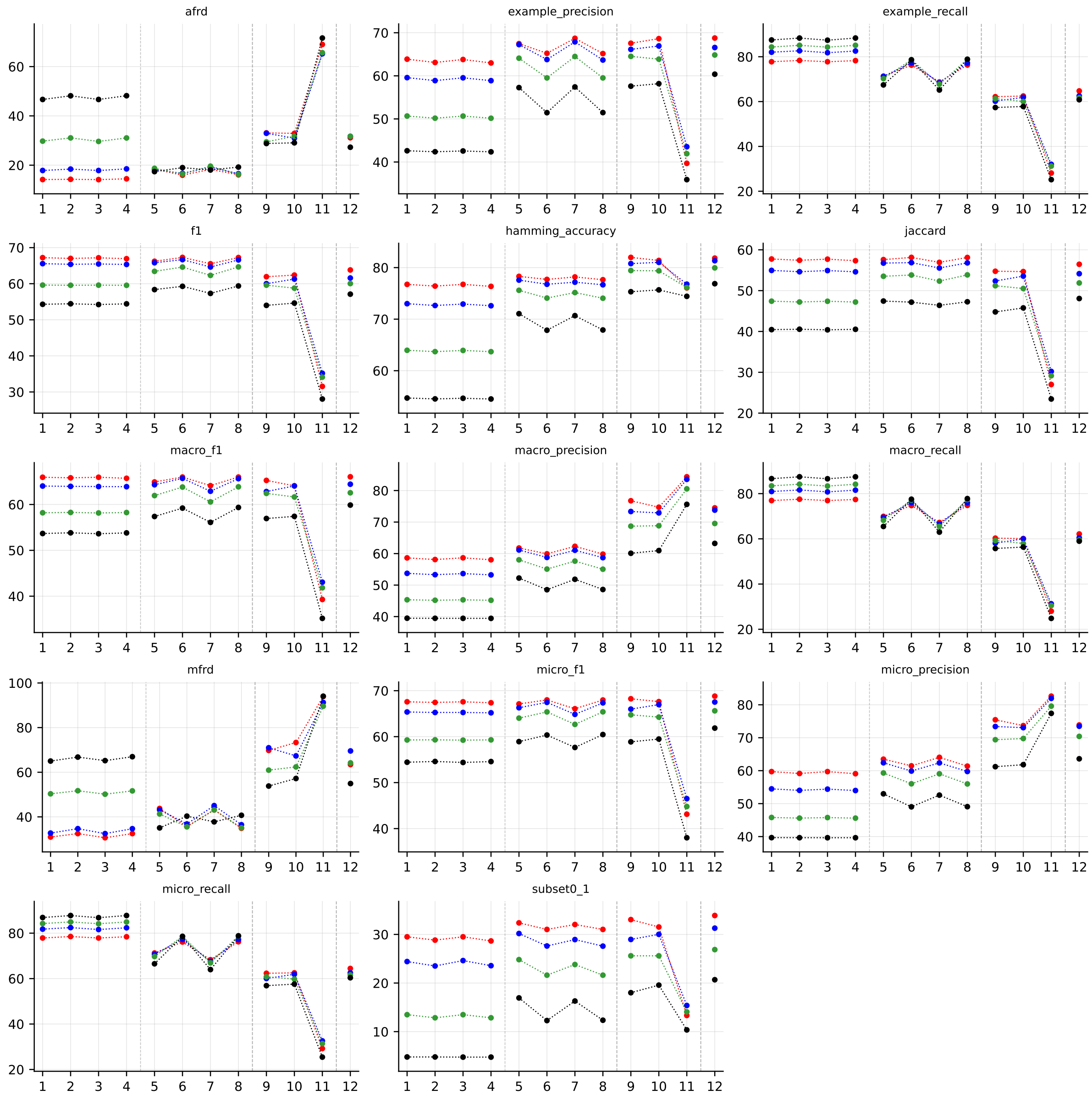


Table 31: Standard MLC predictions (BinaryVector) on `emotions` (RF base learner).

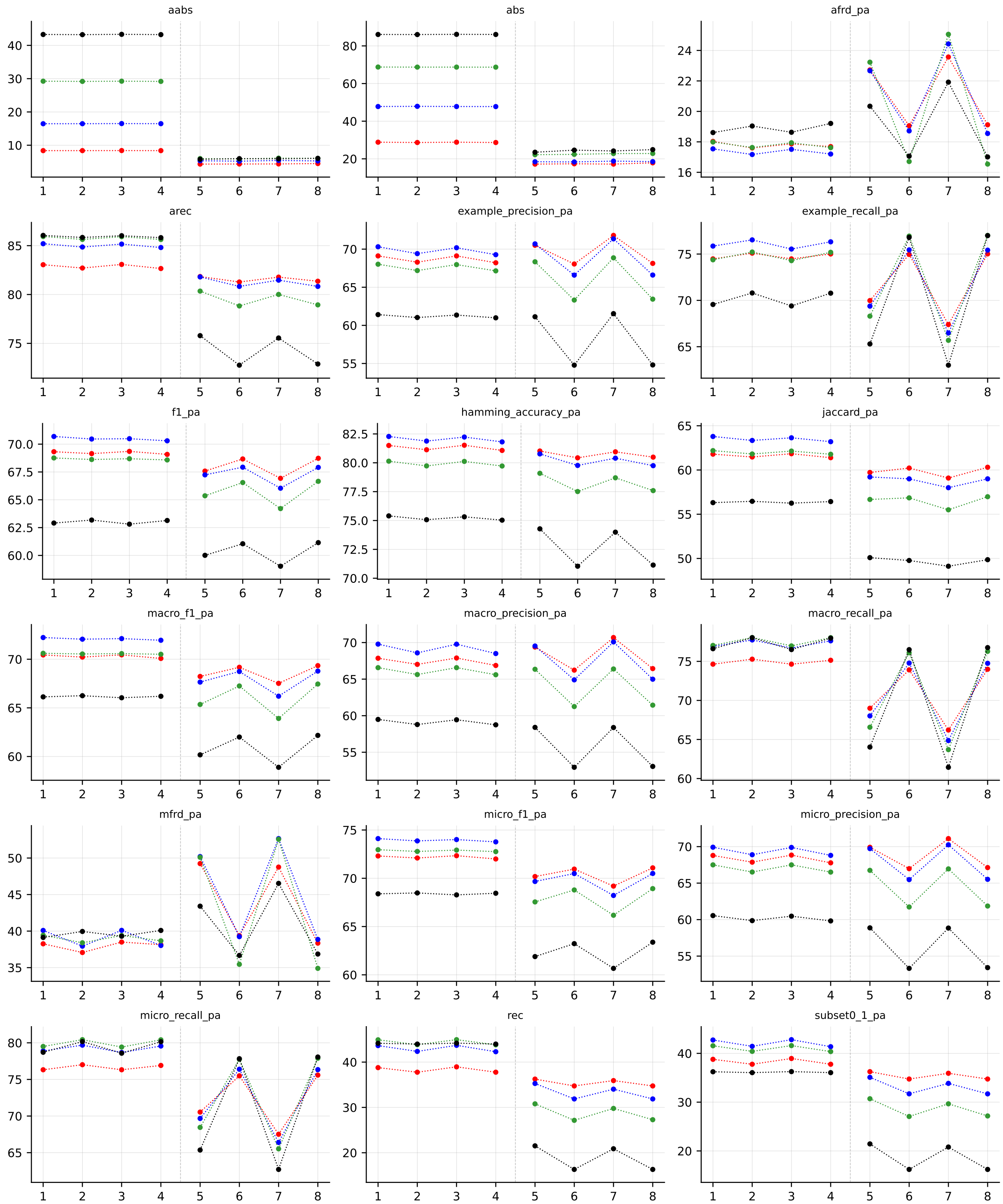


Table 32: Partial abstention (PartialAbstention) on `emotions` (RF base learner).

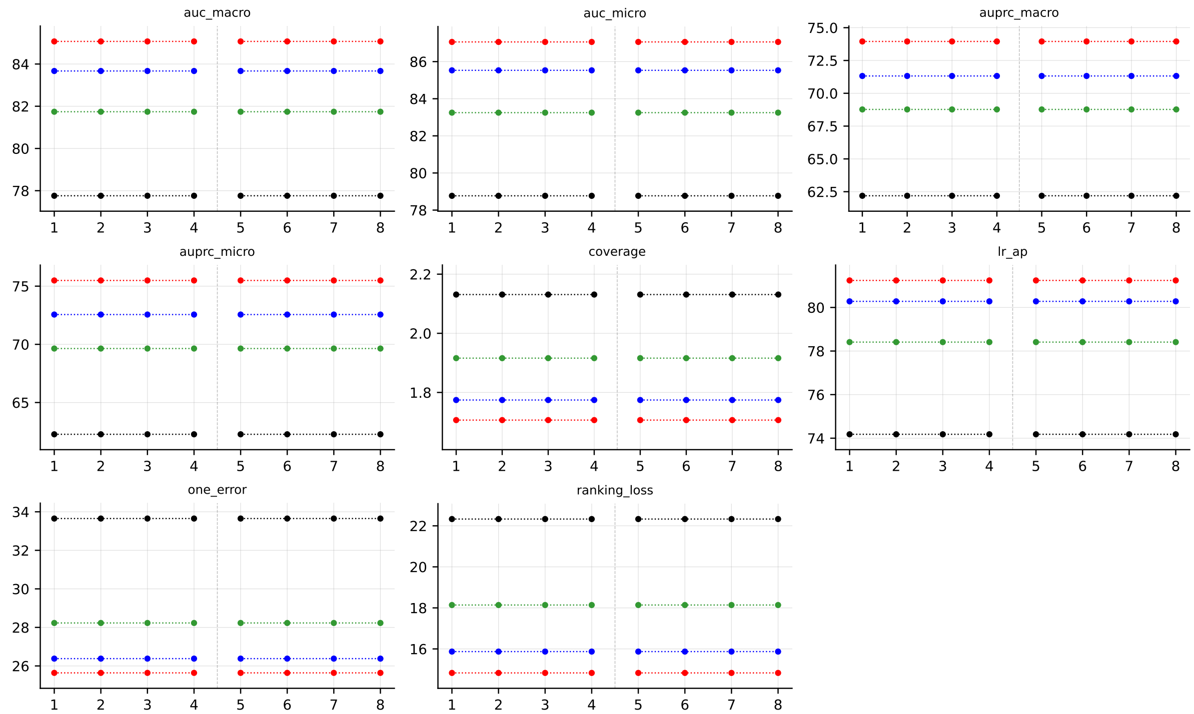


Table 33: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `emotions` (RF base learner).

B.3 Scene ($K = 6$)

RF • `scene` • BinaryVector — Standard MLC predictions

RF • `scene` • PartialAbstention — Partial abstention

RF • `scene` • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

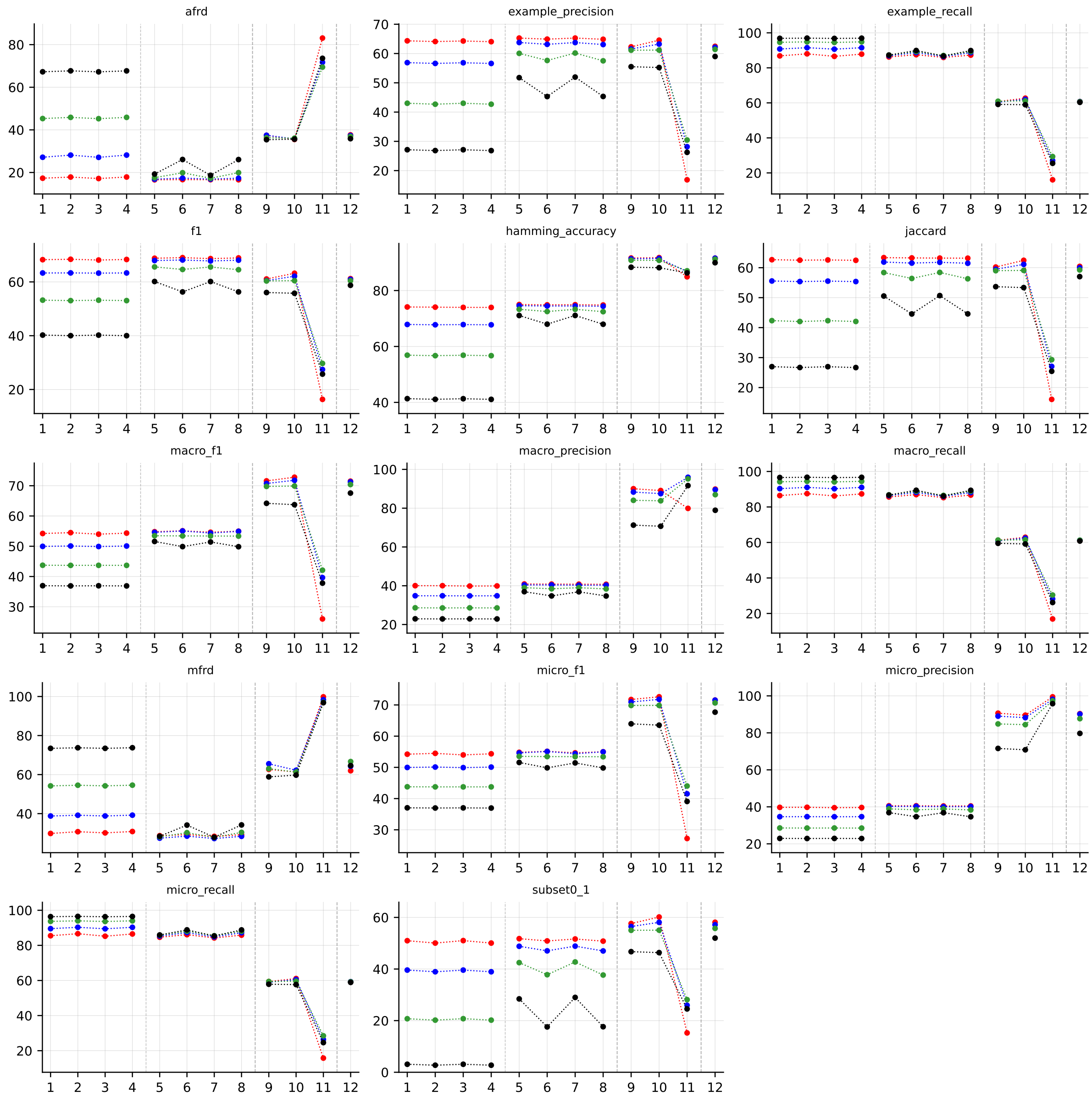


Table 34: Standard MLC predictions (BinaryVector) on `scene` (RF base learner).

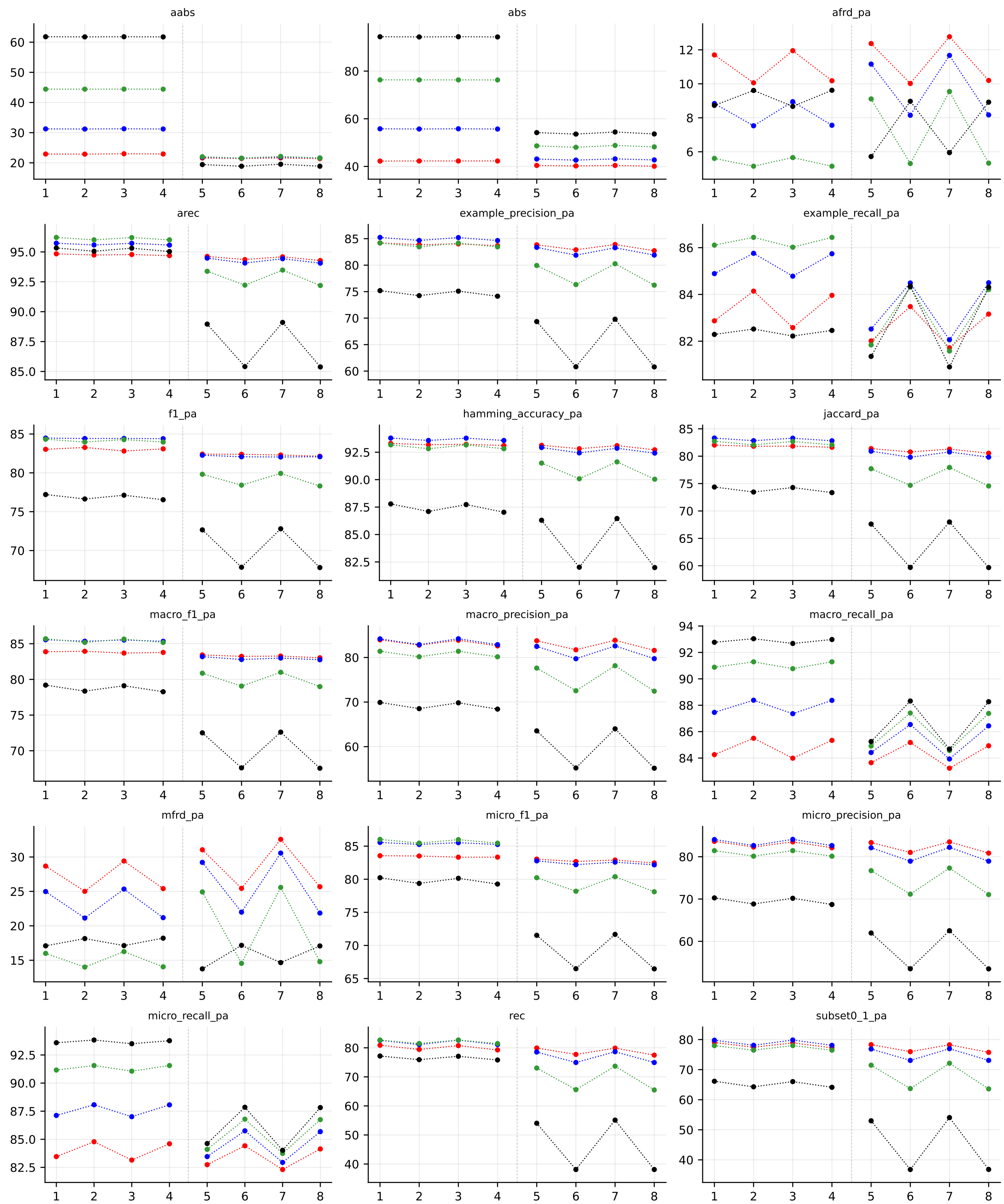


Table 35: Partial abstention (PartialAbstention) on `scene` (RF base learner).

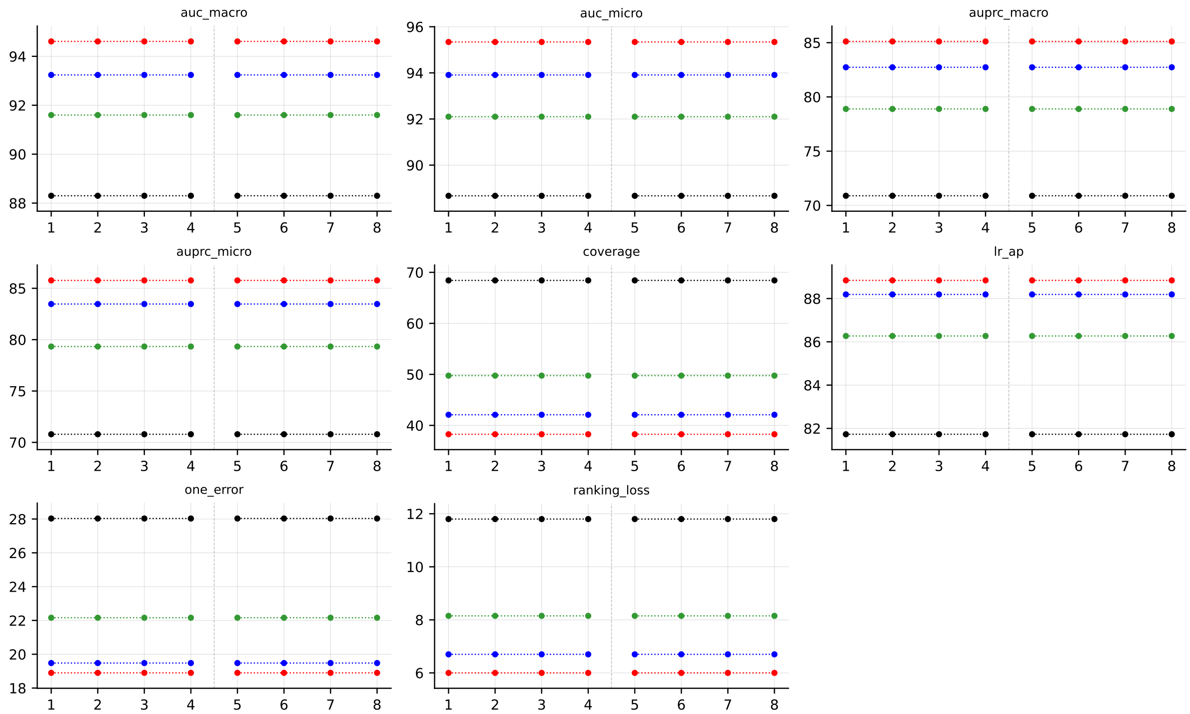


Table 36: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `scene` (RF base learner).

B.4 PlantPseAAC ($K = 12$)

RF • `PlantPseAAC` • BinaryVector — Standard MLC predictions

RF • `PlantPseAAC` • PartialAbstention — Partial abstention

RF • `PlantPseAAC` • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

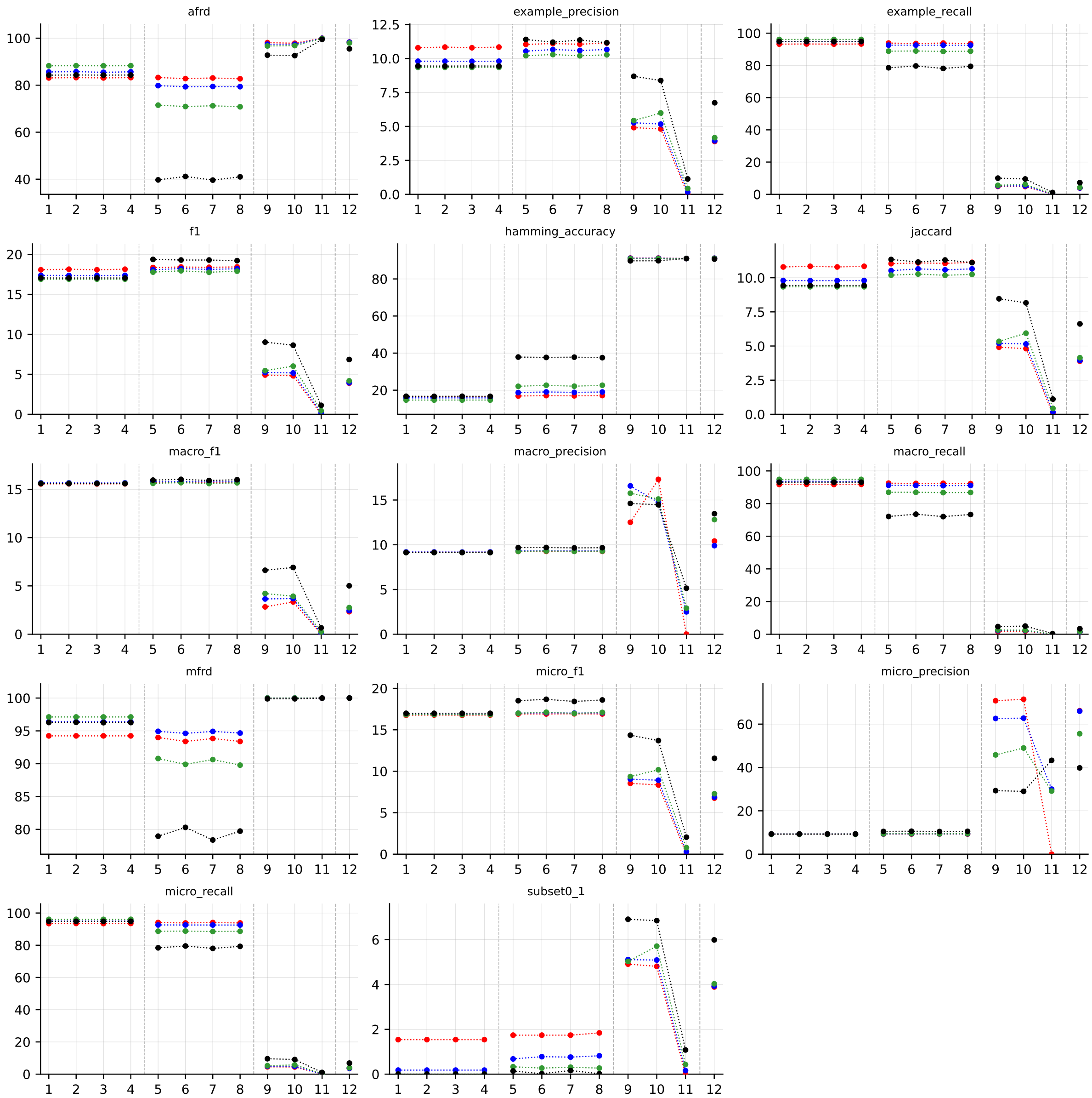


Table 37: Standard MLC predictions (BinaryVector) on `PlantPseAAC` (RF base learner).

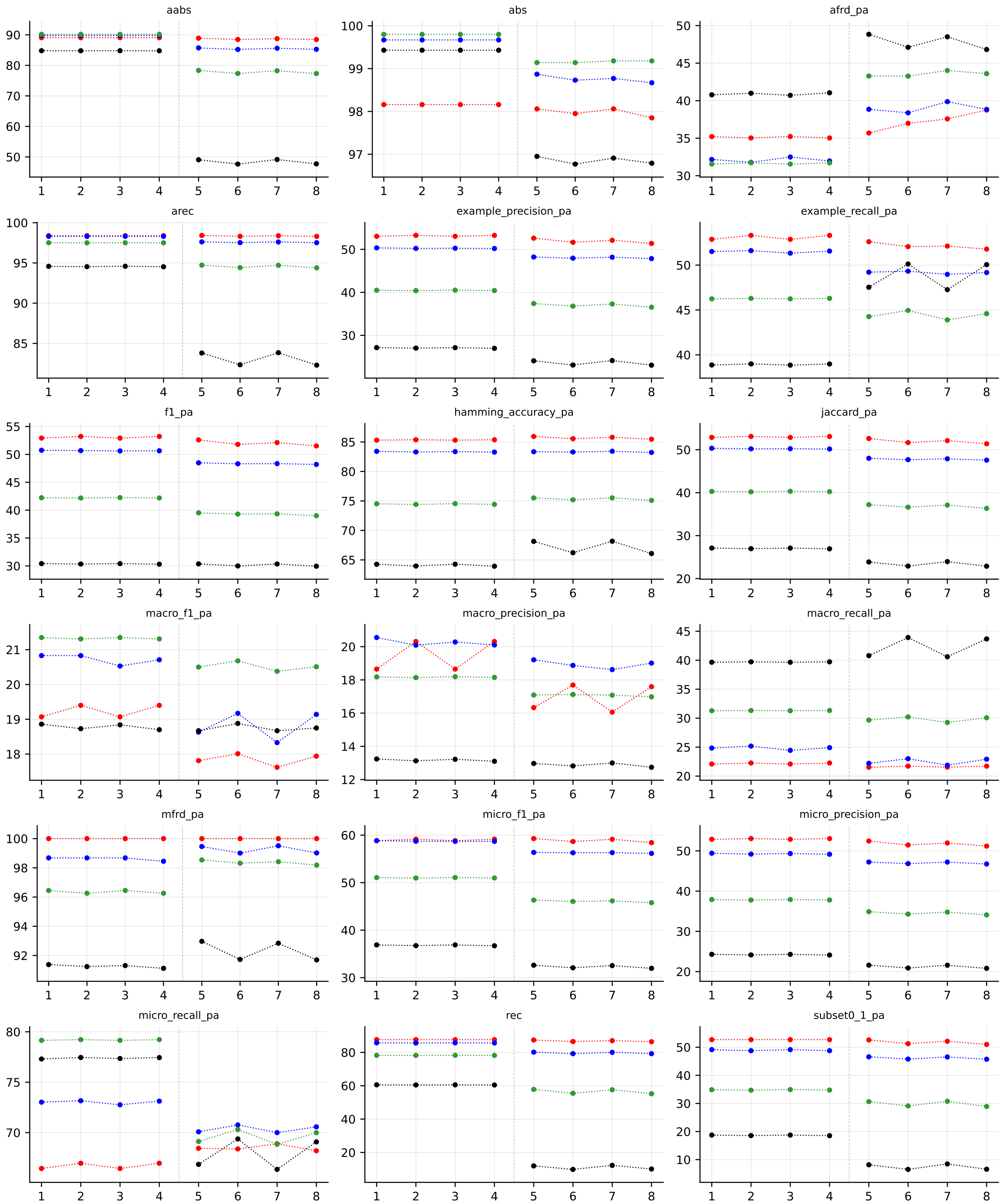


Table 38: Partial abstention (PartialAbstention) on `PlantPseAAC` (RF base learner).

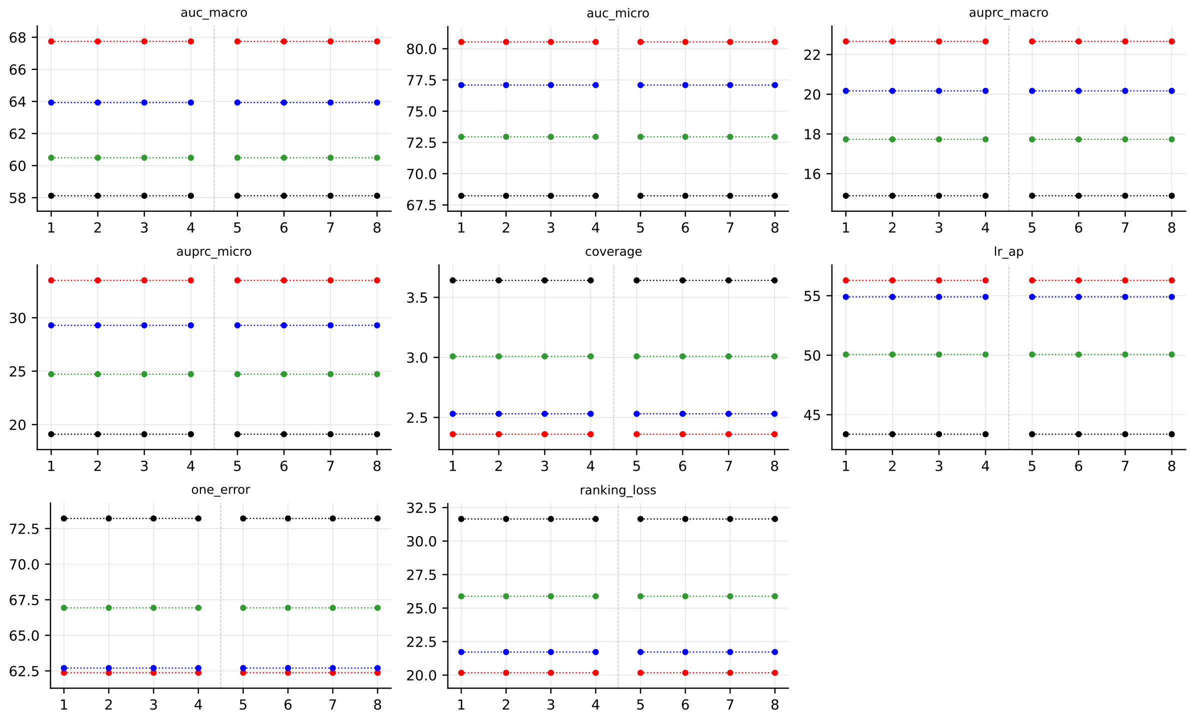


Table 39: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `PlantPseAAC` (RF base learner).

B.5 HumanPseAAC ($K = 14$)

RF • `HumanPseAAC` • BinaryVector — Standard MLC predictions

RF • `HumanPseAAC` • PartialAbstention — Partial abstention

RF • `HumanPseAAC` • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

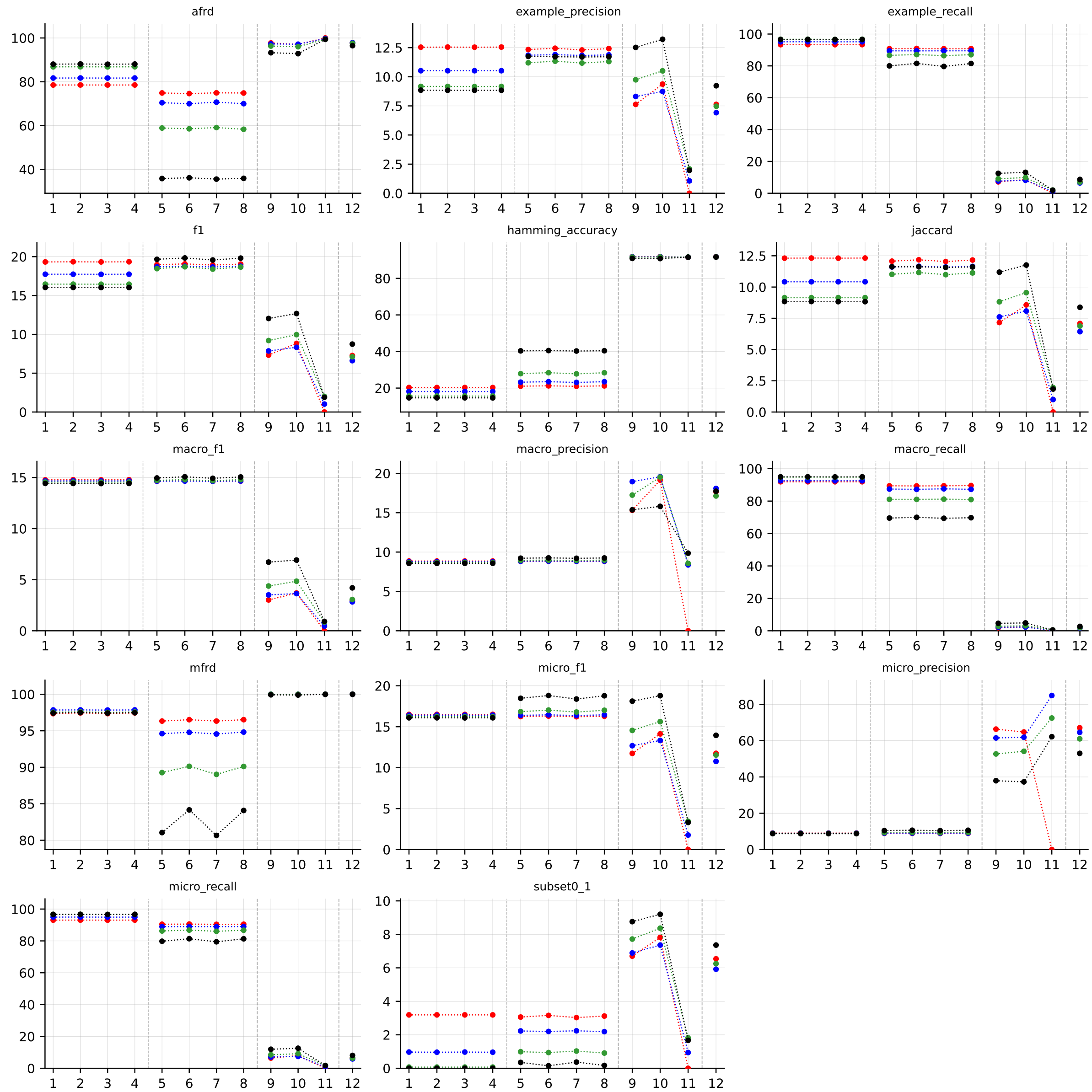


Table 40: Standard MLC predictions (BinaryVector) on `HumanPseAAC` (RF base learner).

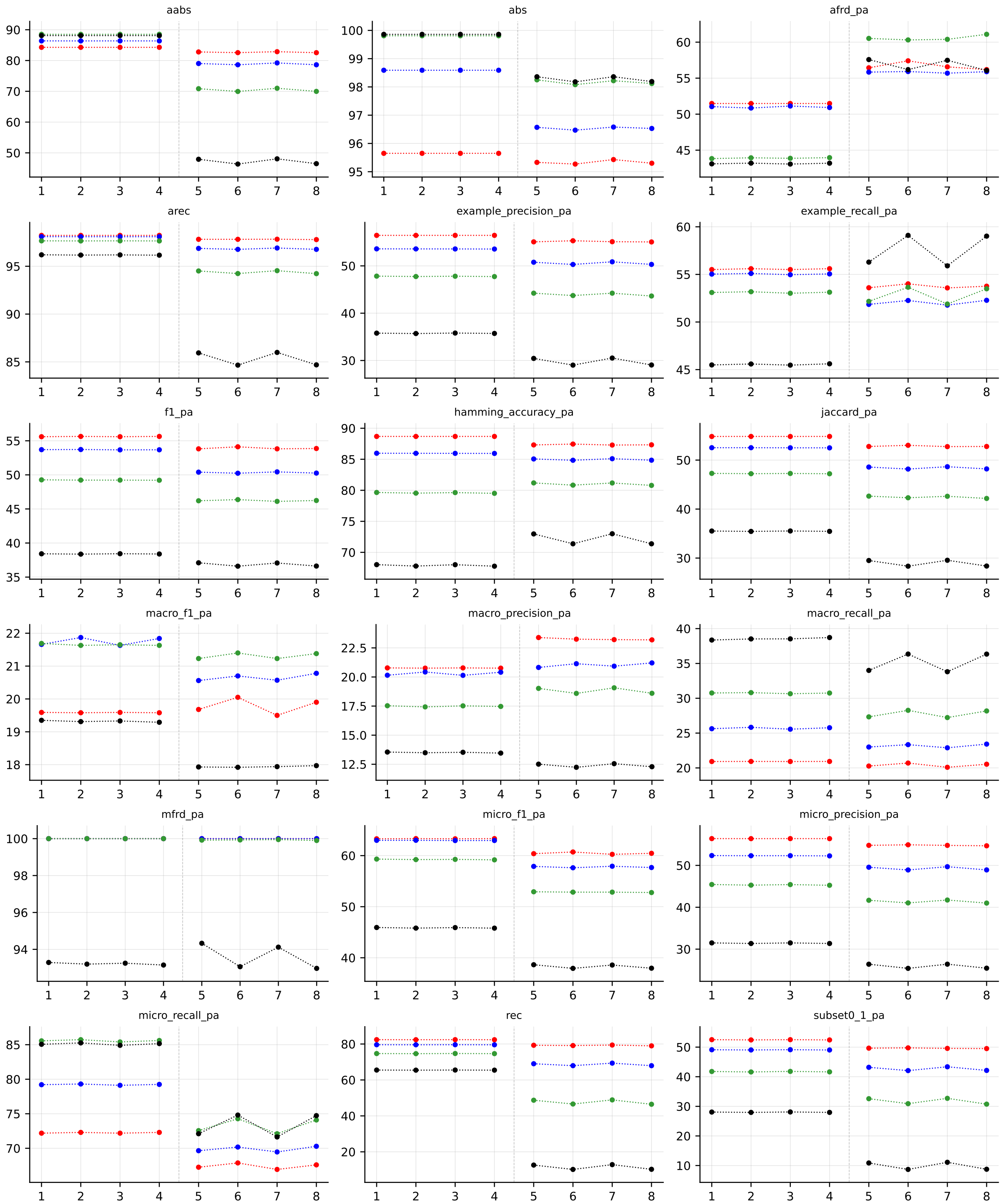


Table 41: Partial abstention (PartialAbstention) on `HumanPseAAC` (RF base learner).

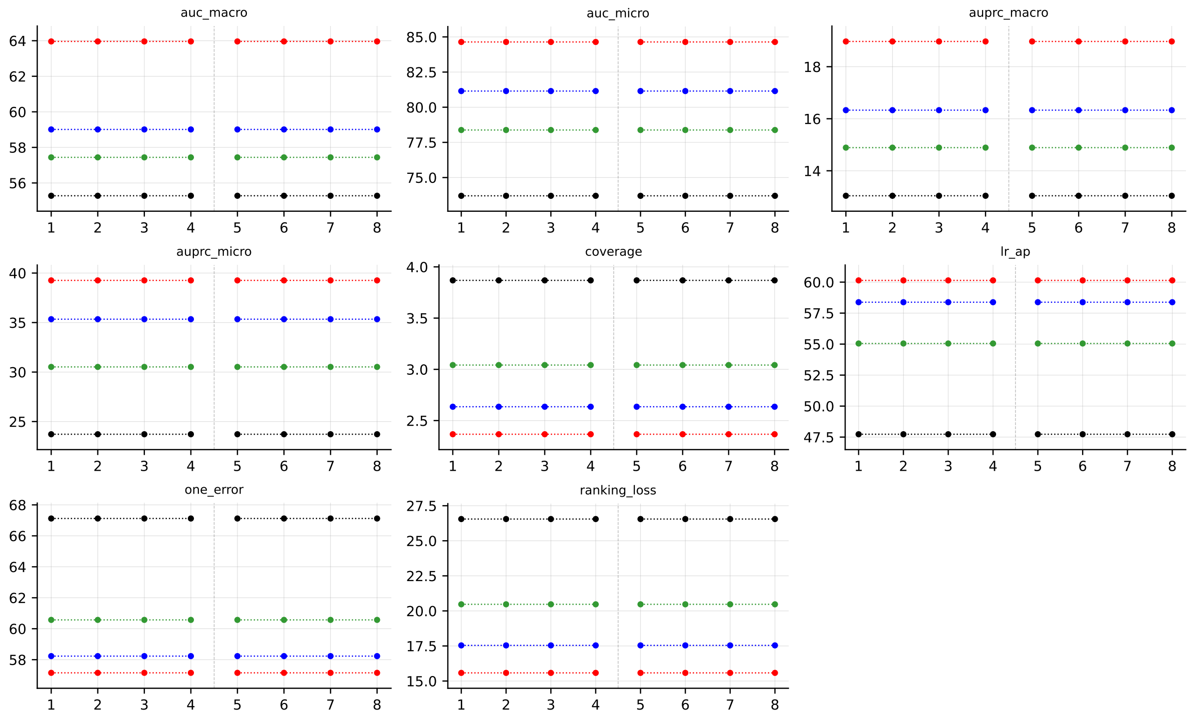


Table 42: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `HumanPseAAC` (RF base learner).

B.6 Yeast ($K = 14$)

RF • `Yeast` • BinaryVector — Standard MLC predictions

RF • `Yeast` • PartialAbstention — Partial abstention

RF • `Yeast` • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

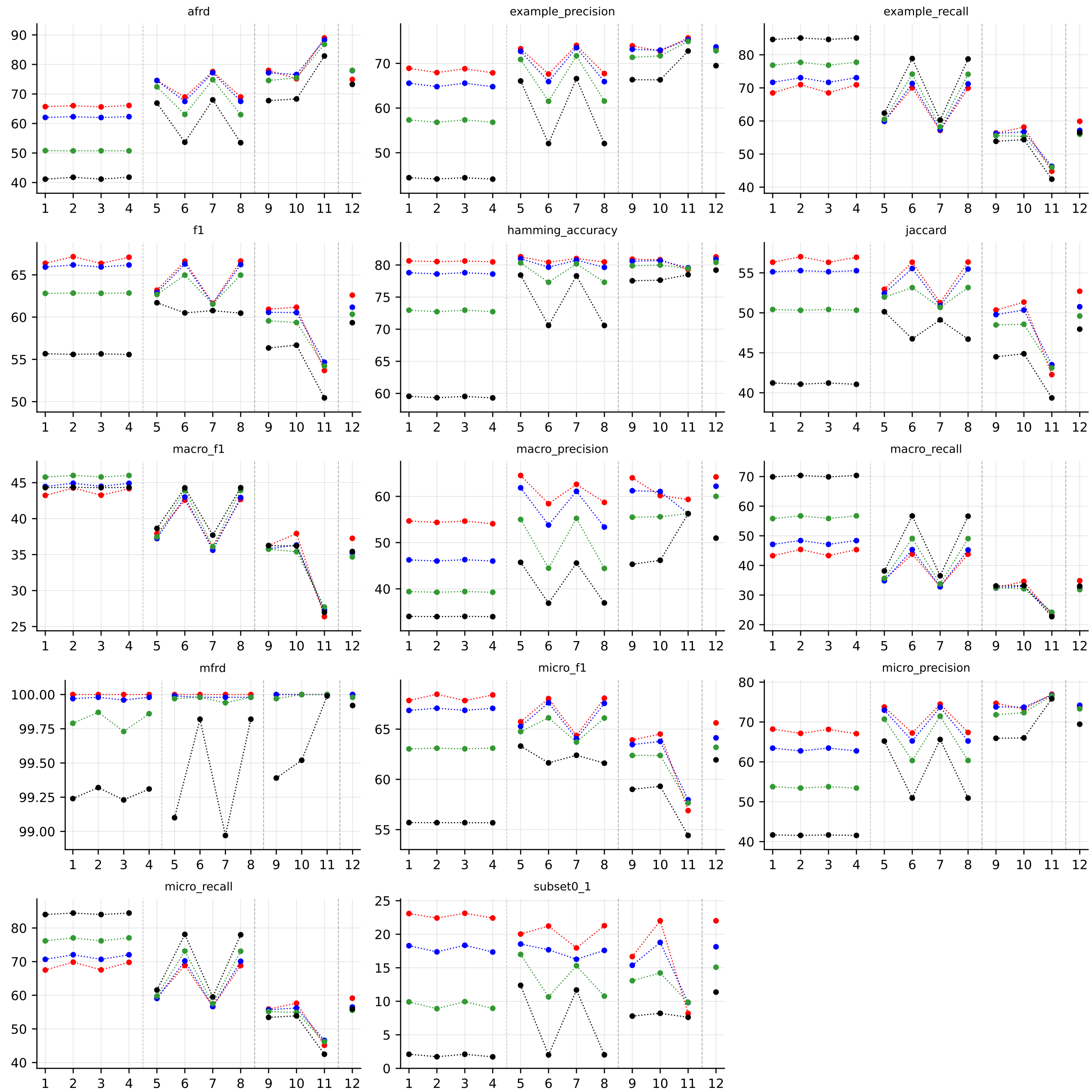


Table 43: Standard MLC predictions (BinaryVector) on `Yeast` (RF base learner).

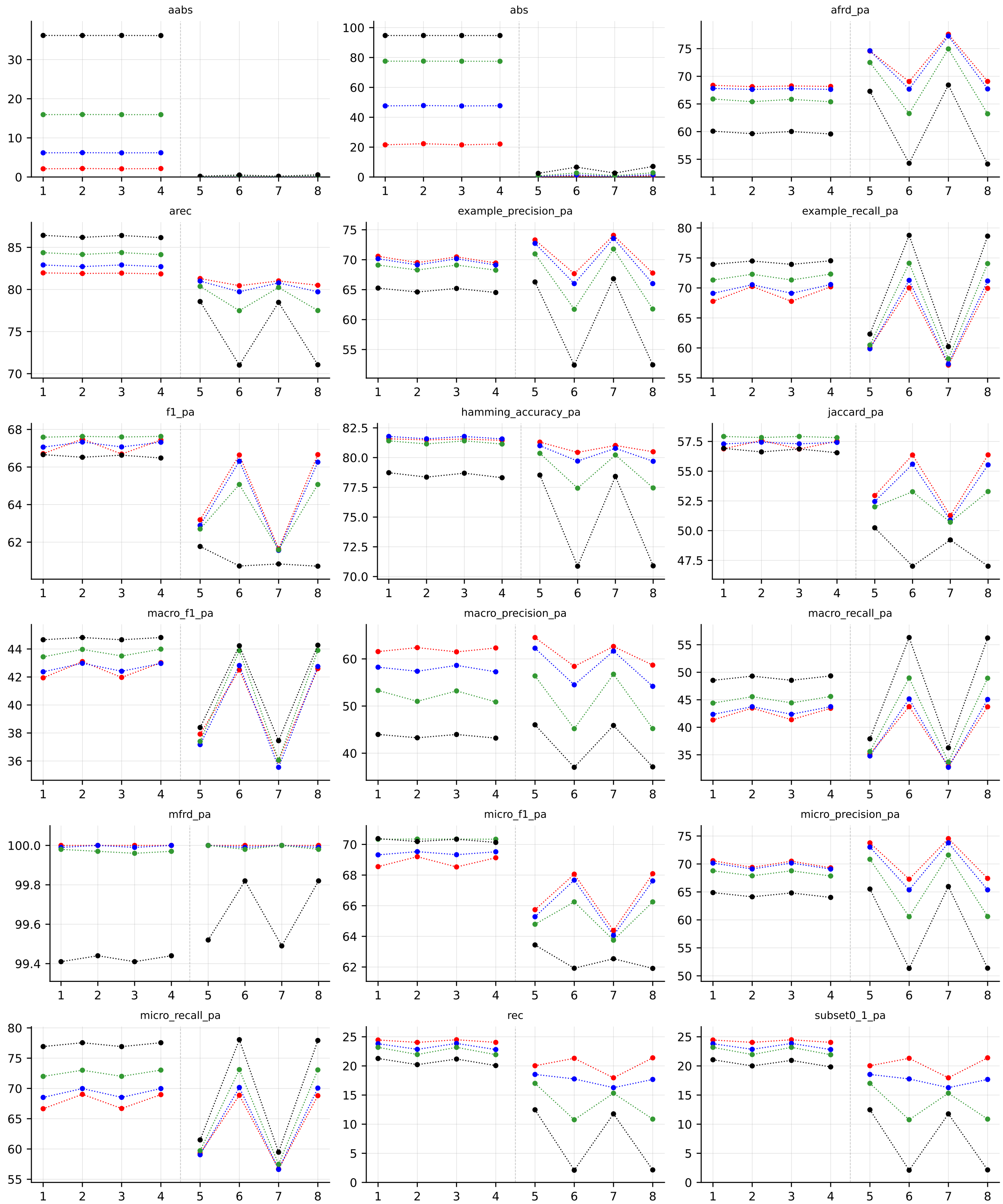


Table 44: Partial abstention (PartialAbstention) on `Yeast` (RF base learner).

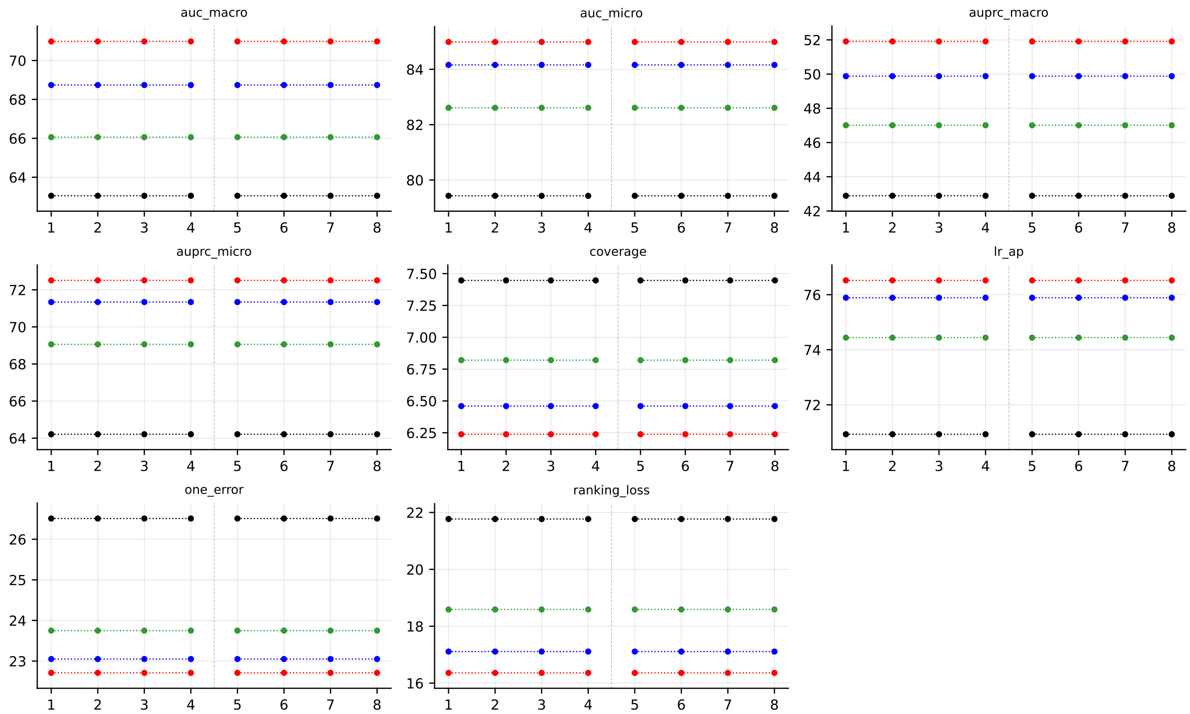


Table 45: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `Yeast` (RF base learner).

B.7 CHD_49 ($K = 6$)

RF • `CHD_49` • BinaryVector — Standard MLC predictions

RF • `CHD_49` • PartialAbstention — Partial abstention

RF • `CHD_49` • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

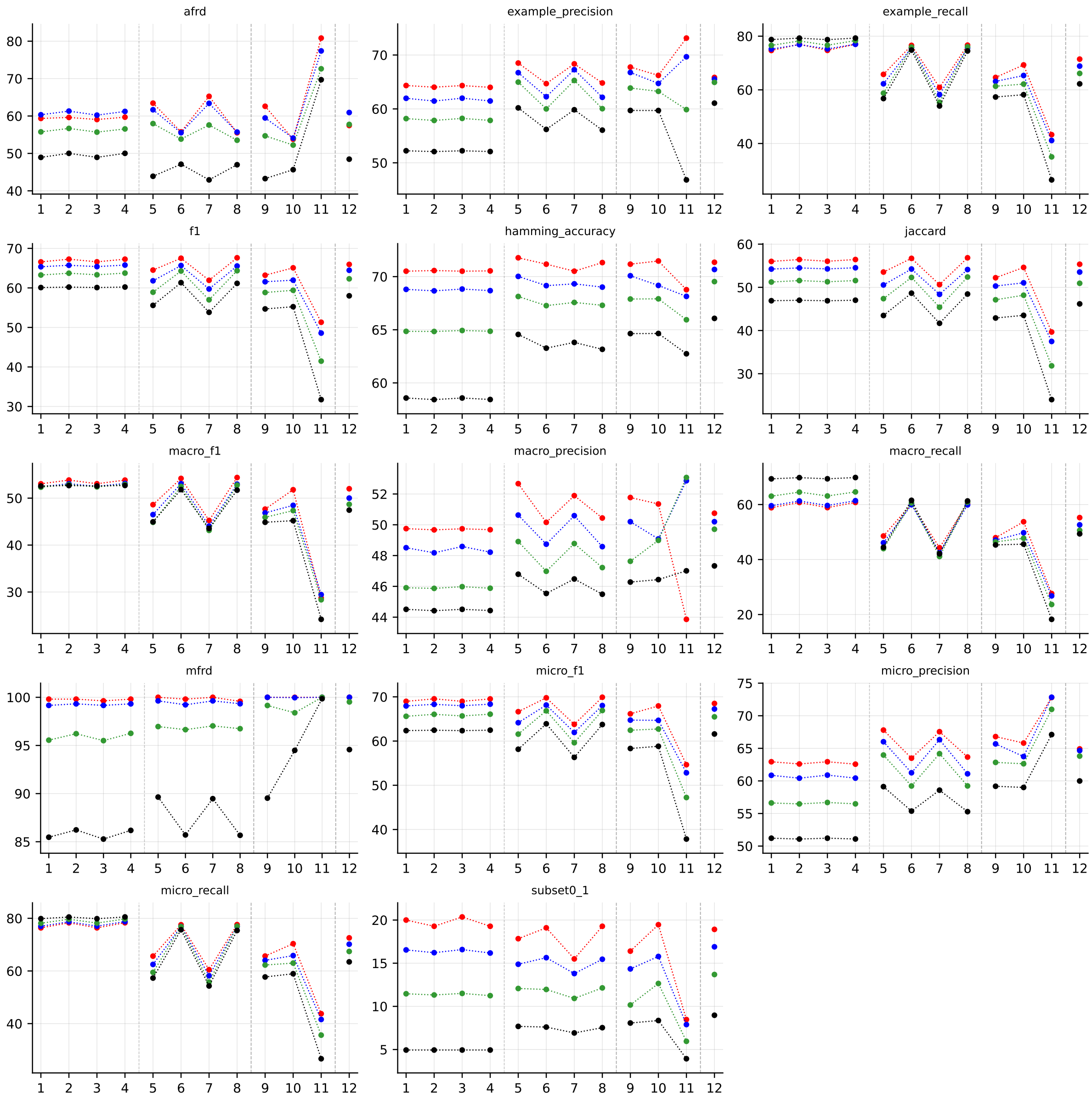


Table 46: Standard MLC predictions (BinaryVector) on CHD_49 (RF base learner).

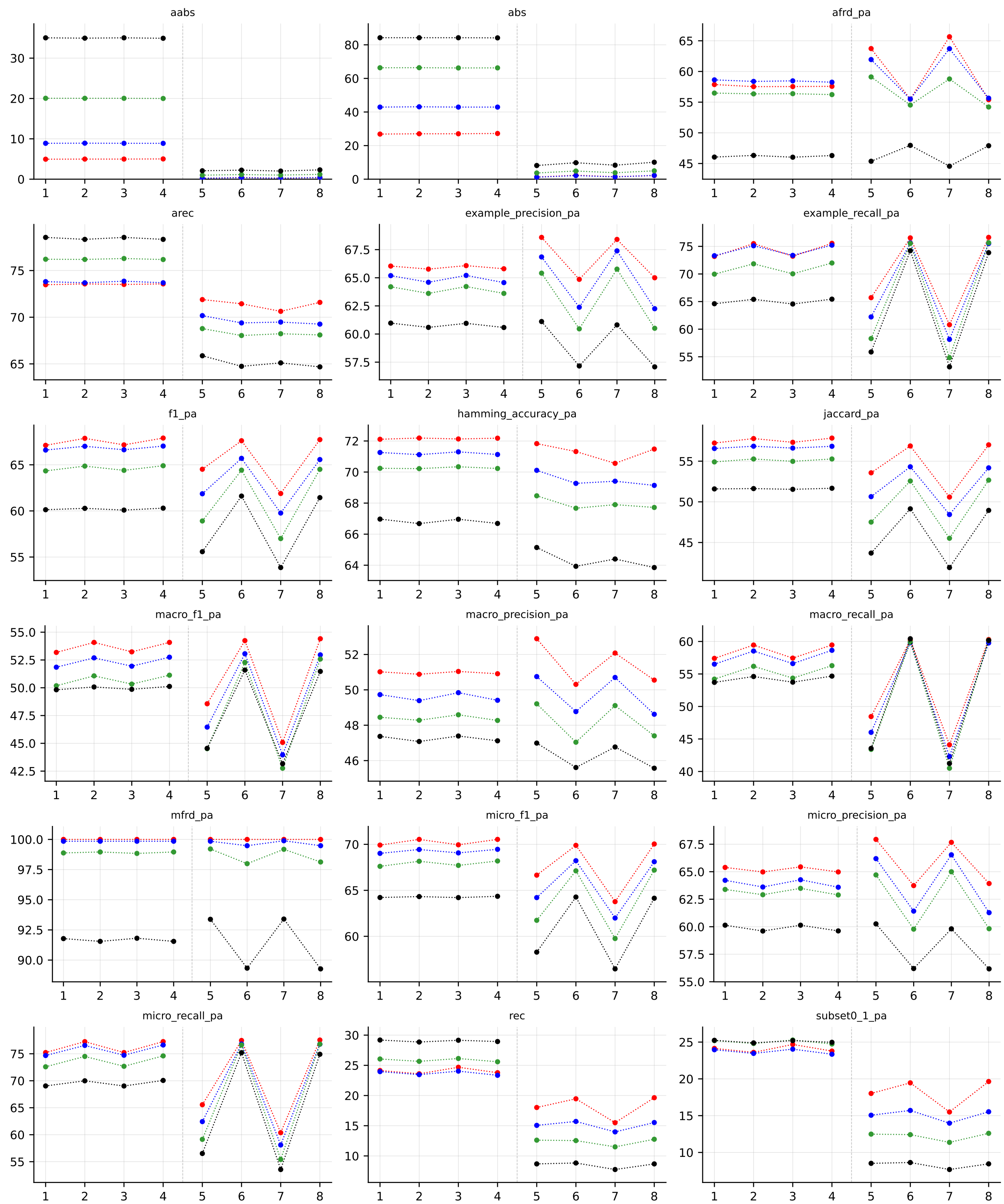


Table 47: Partial abstention (PartialAbstention) on `CHD_49` (RF base learner).

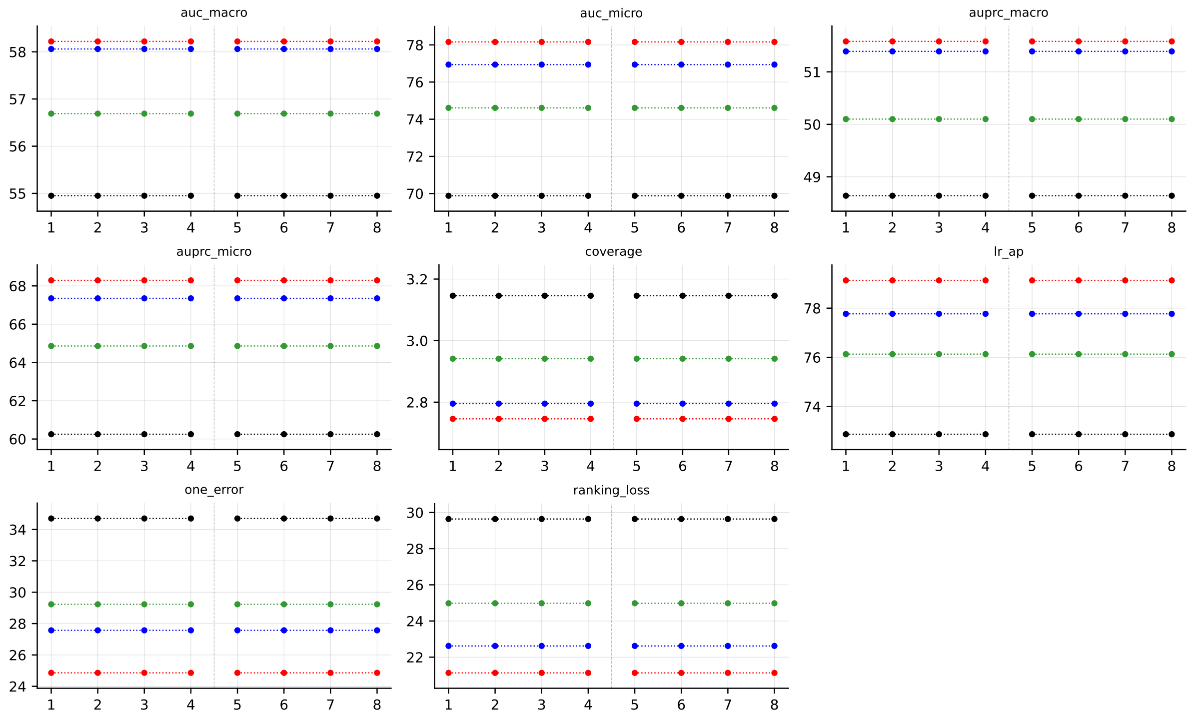


Table 48: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `CHD_49` (RF base learner).

B.8 Water-quality ($K = 14$)

RF • `Water-quality` • BinaryVector — Standard MLC predictions

RF • `Water-quality` • PartialAbstention — Partial abstention

RF • `Water-quality` • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

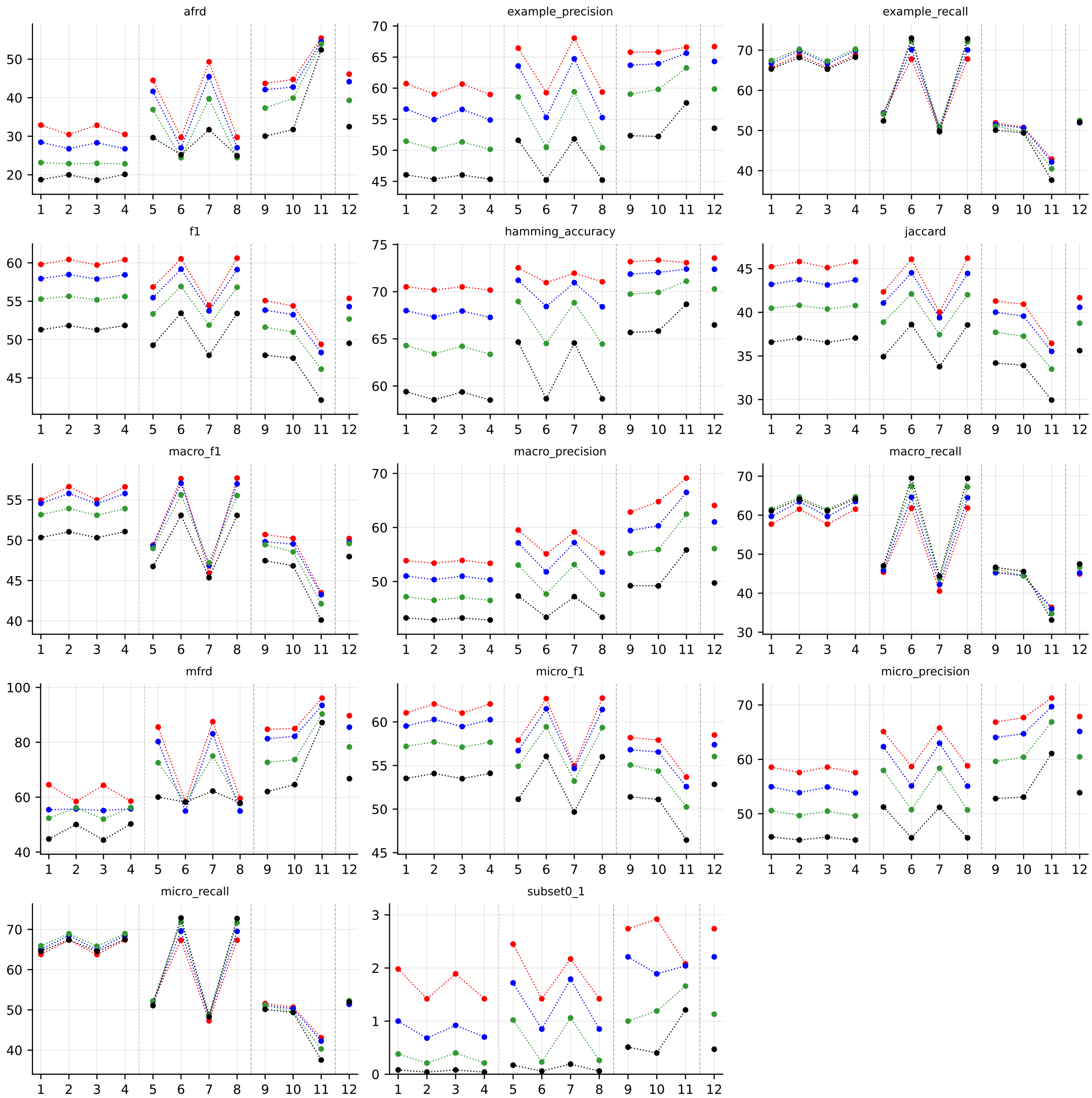


Table 49: Standard MLC predictions (BinaryVector) on `Water-quality` (RF base learner).

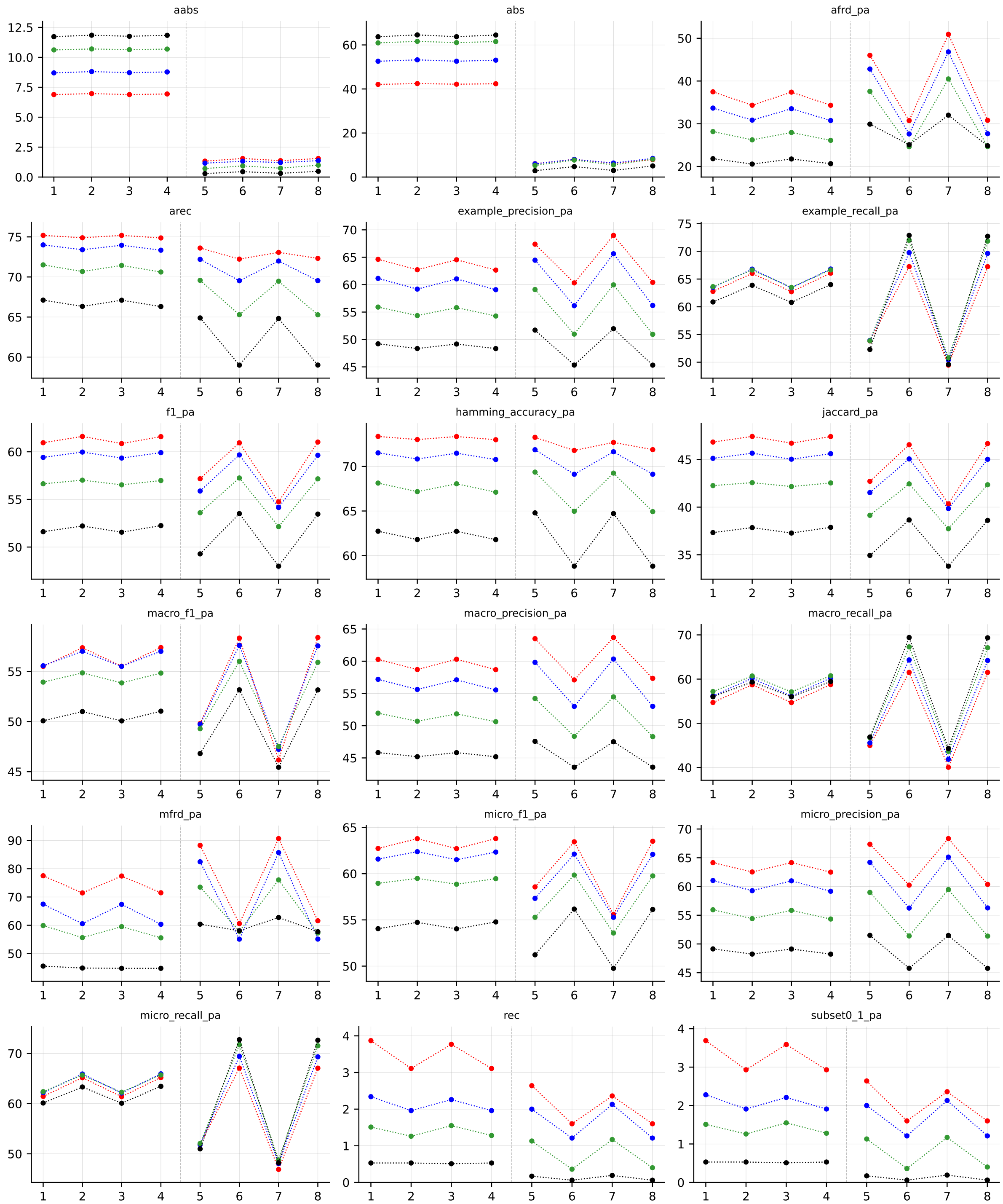


Table 50: Partial abstention (PartialAbstention) on `Water-quality` (RF base learner).

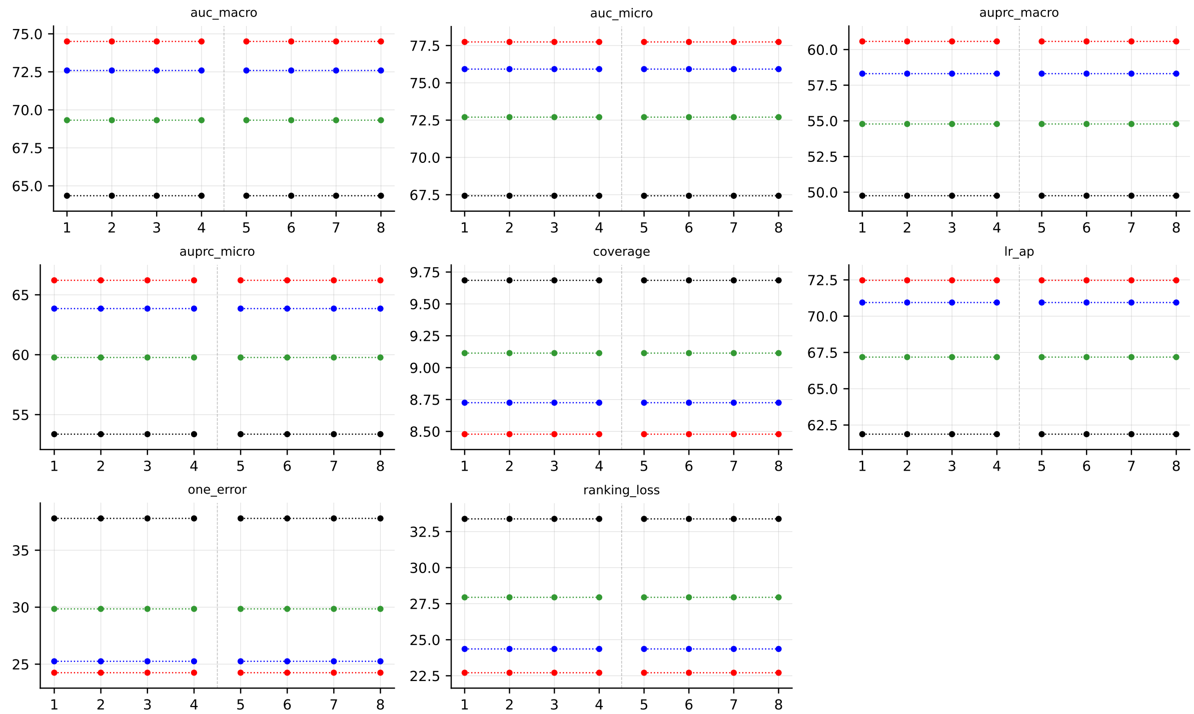


Table 51: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `Water-quality` (RF base learner).

B.9 enron ($K = 53$)

RF • `enron` • BinaryVector — Standard MLC predictions

RF • `enron` • PartialAbstention — Partial abstention

RF • `enron` • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

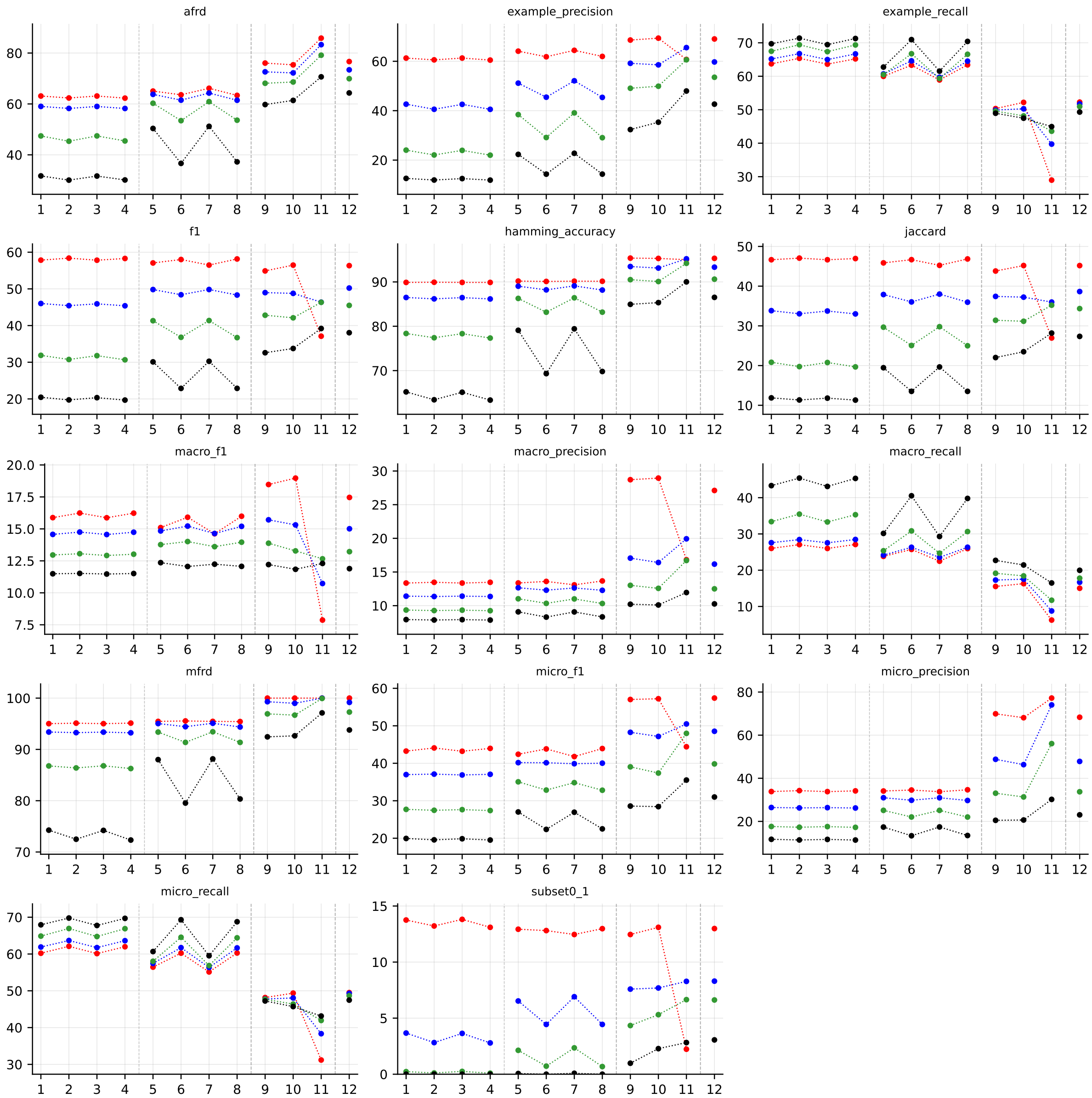


Table 52: Standard MLC predictions (BinaryVector) on `enron` (RF base learner).

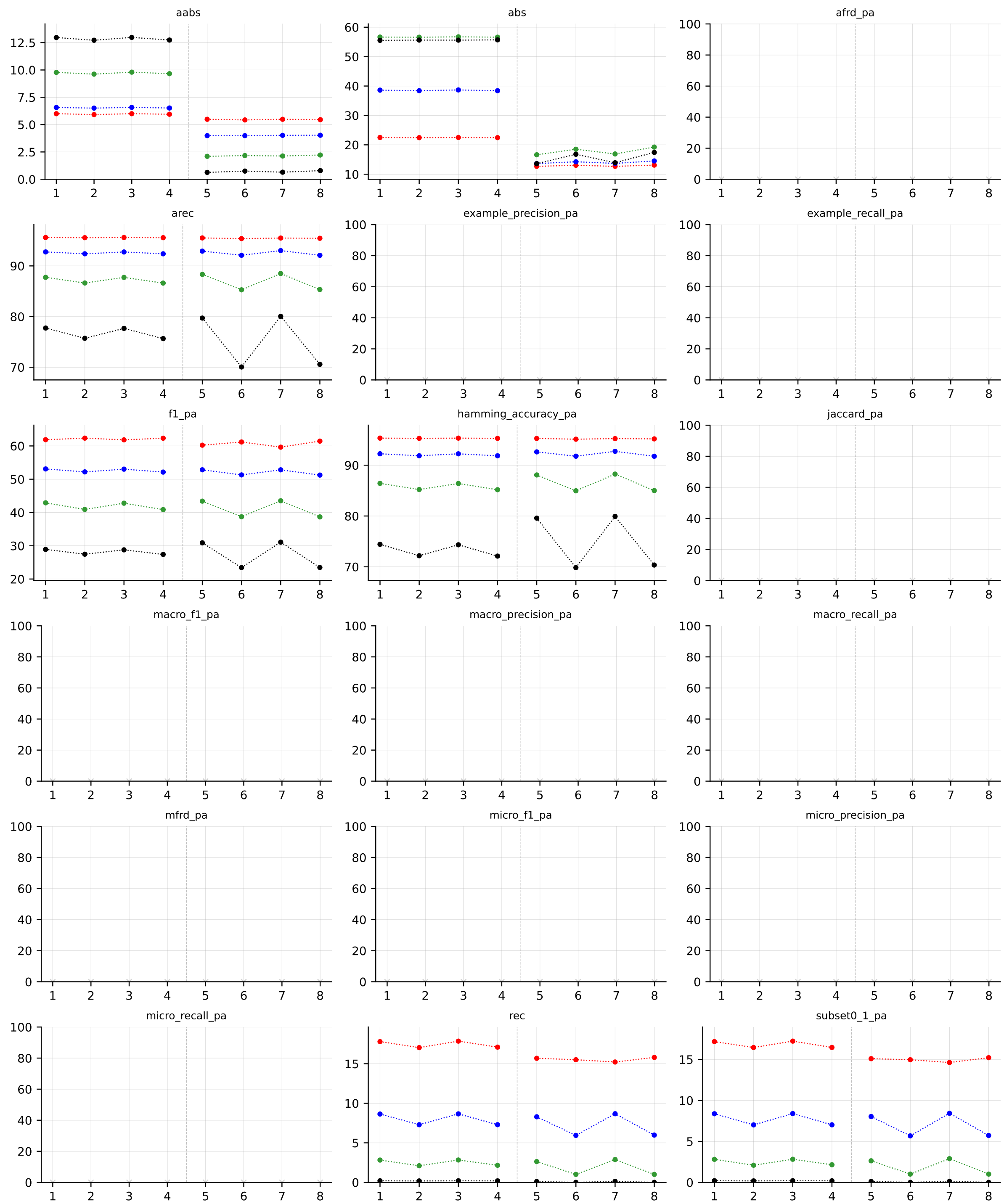


Table 53: Partial abstention (PartialAbstention) on `enron` (RF base learner).

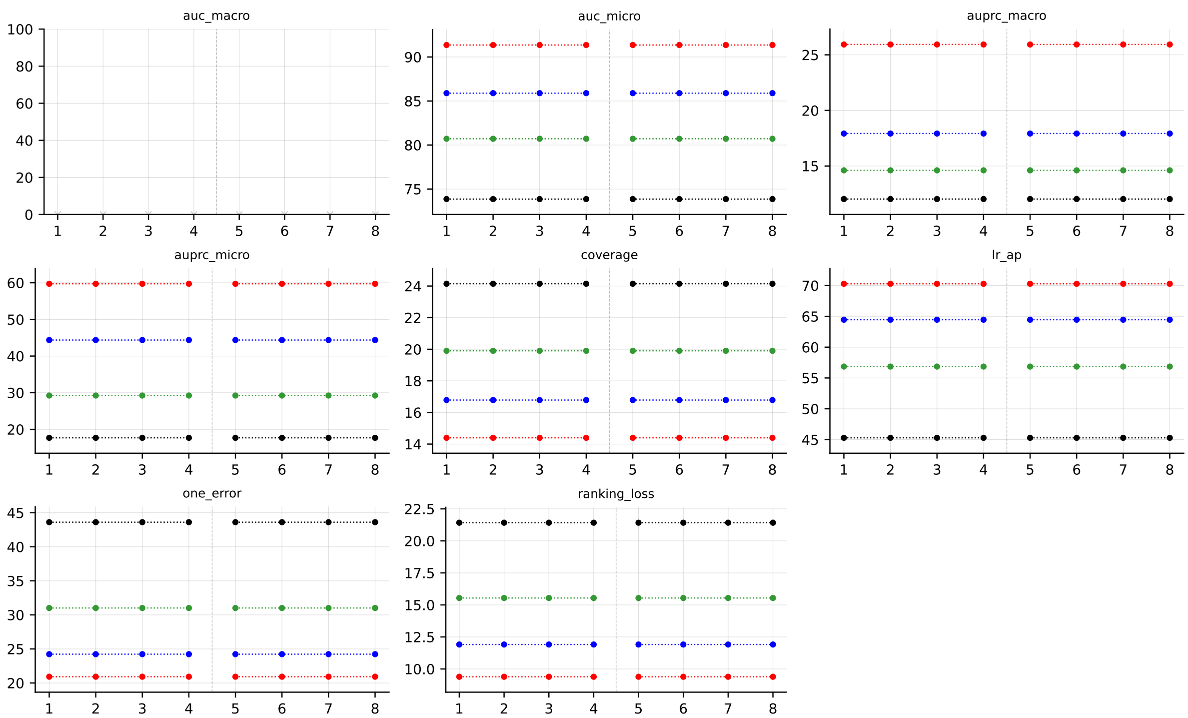


Table 54: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `enron` (RF base learner).

C LGBM aggregated results

LGBM • Aggregate • BinaryVector — Standard MLC predictions

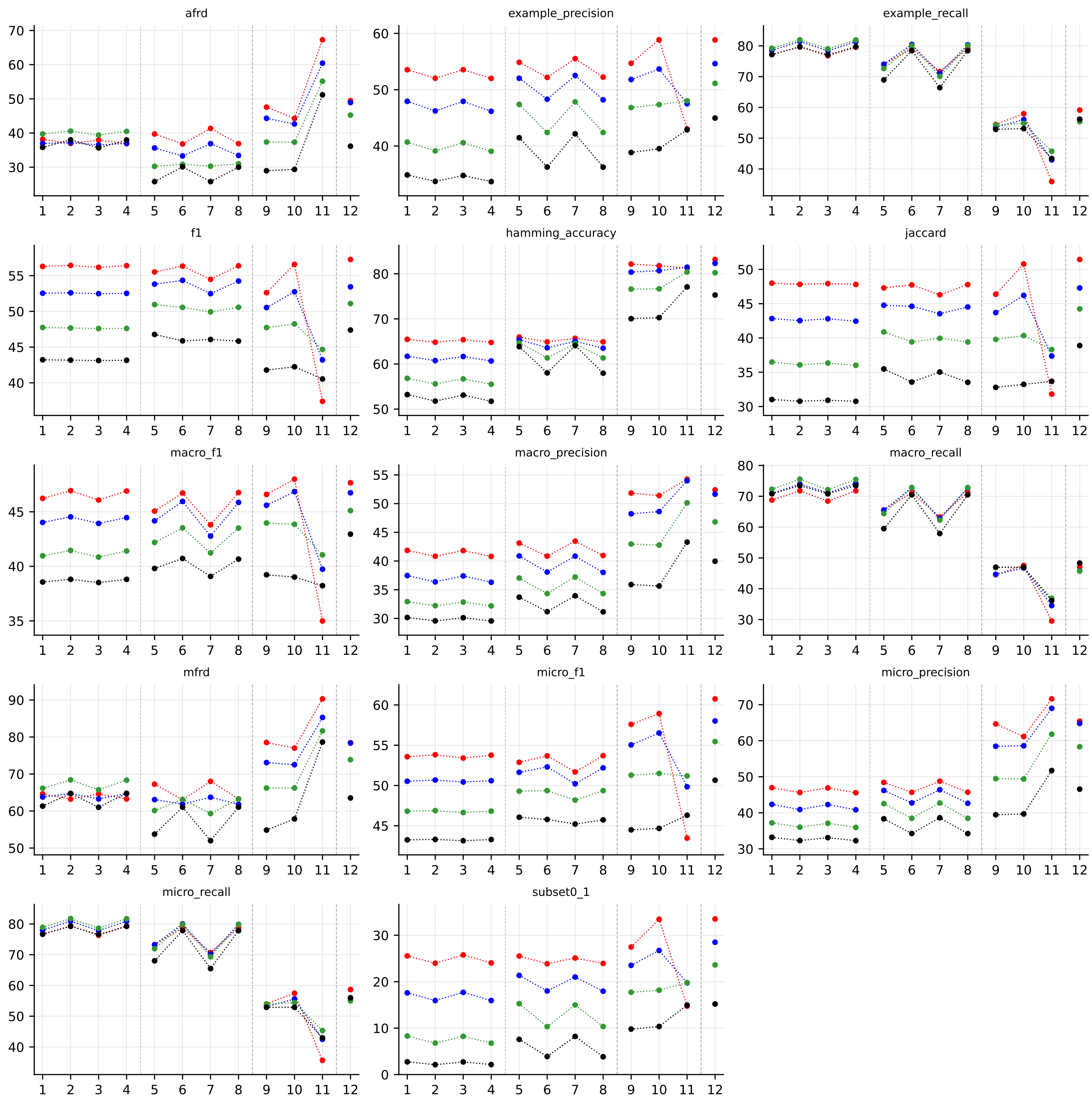


Table 55: Standard MLC predictions (BinaryVector), LGBM base learner, averaged across datasets (8 of 9: enron LGBM still training). Method indices and line colors follow the encoding paragraph above.

LGBM • Aggregate • PartialAbstention — Partial abstention

LGBM • Aggregate • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

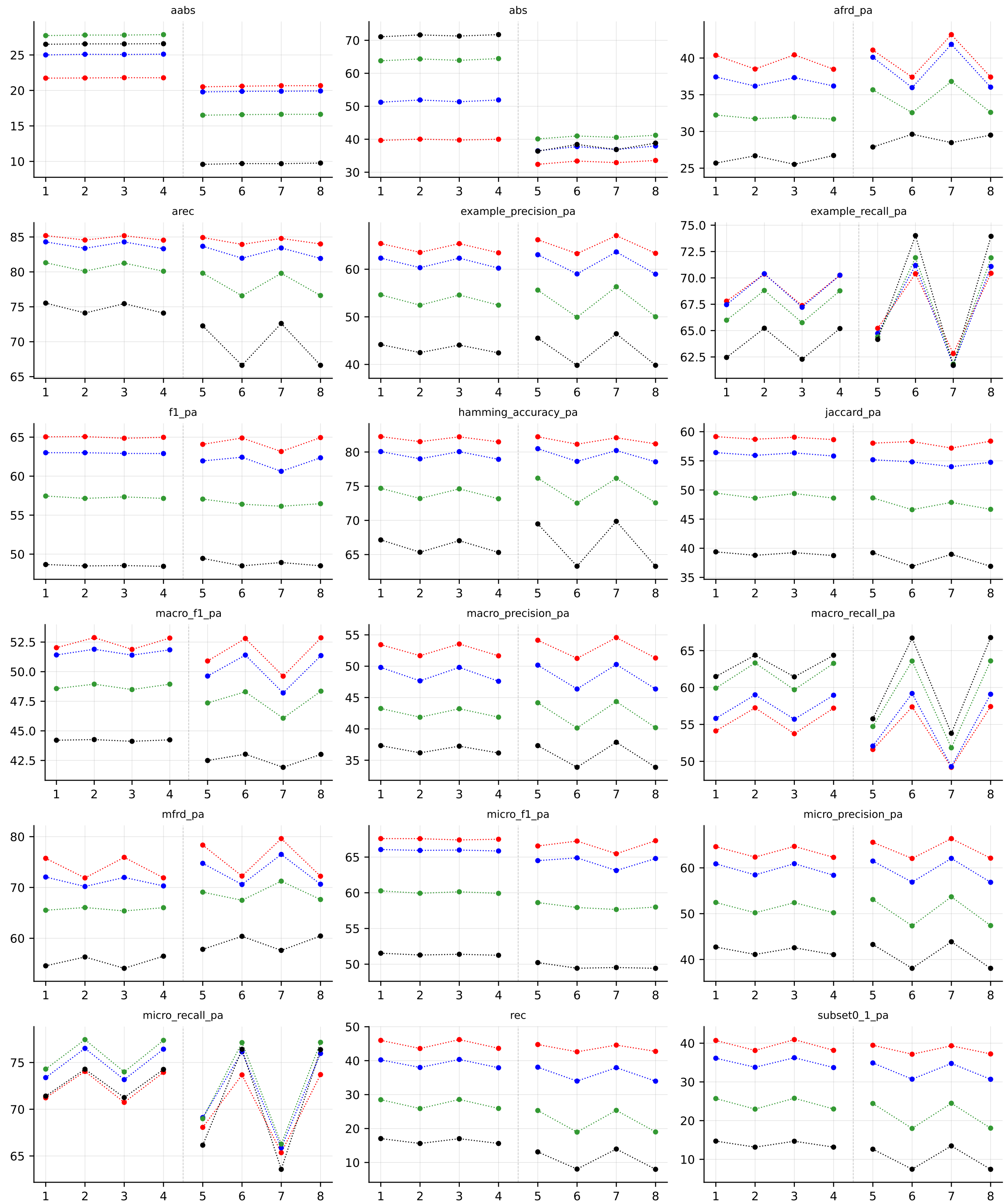


Table 56: Partial abstention (PartialAbstention), LGBM base learner, averaged across datasets (8 of 9: enron LGBM still training). Method indices and line colors follow the encoding paragraph above.

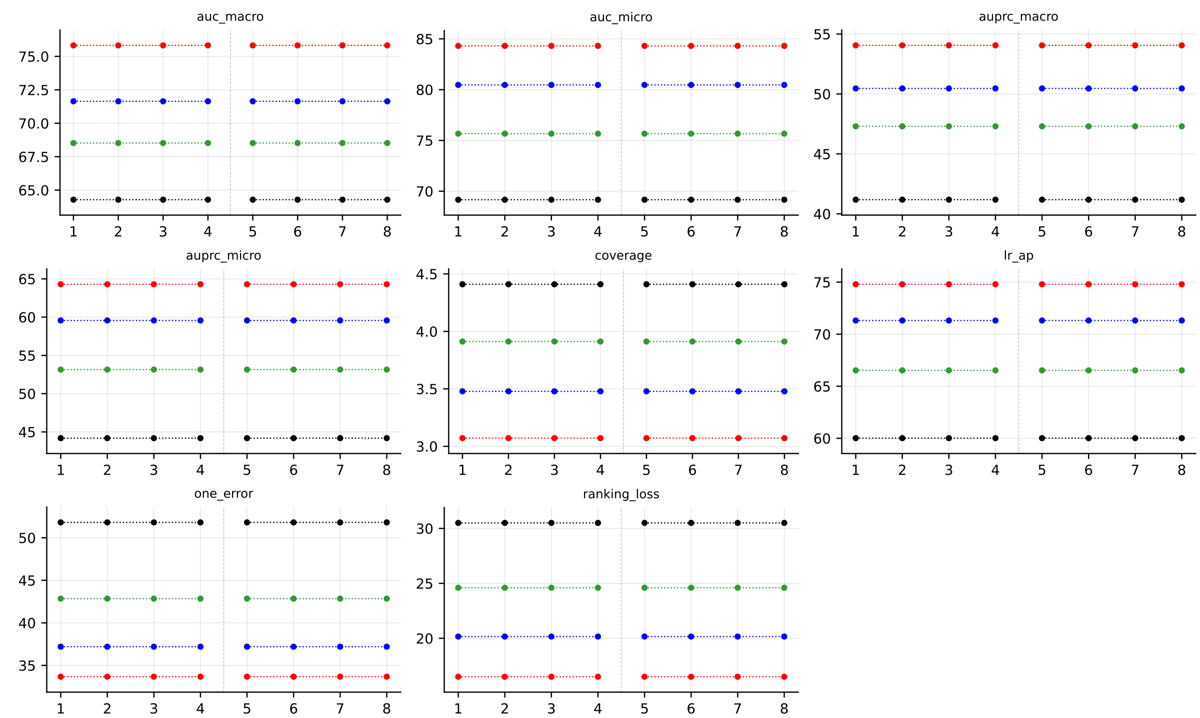


Table 57: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector), LGBM base learner, averaged across datasets (8 of 9: enron LGBM still training). Method indices and line colors follow the encoding paragraph above.